Article

Gravity-Light

Universal Quantum Foam Hypothesis

Ukshin Q. Rexhepi

Independent Researcher
Tübingen, Germany

https://doi.org/10.13140/RG.2.2.33455.91042

Abstract

This paper formulates the Universal Quantum Foam Hypothesis (UQSH) as an ontological field framework. Its starting point is the Qu-ground Q, whose physically readable field projection Φ appears as Qu-foam. The motion of the Qu-foam field, $B(\Phi)$, forms the common basis of light, energy, and gravitational field action:

$$Q \to \Phi \to B(\Phi), \qquad B(\Phi) \Rightarrow \{L, E, \mathfrak{G}\}.$$

Qu-foam is not understood as a classical ether, nor as an additional substance in space. It serves as a boundary concept for the underlying reality in which physical field states arise, couple, saturate, relax, and reorganize. In this view, radiation, matter, gravitation, energy, time, and observation do not appear as separate fundamental substances, but as different readings of the same field motion.

Light is not understood as a flying object, but as free, spherically propagating field motion. Energy describes the motion content of this dynamics, while gravitation describes its regime-effective curvature and tension side. Matter arises as bound field motion, and inertia as the resistance of this binding against forced reorganization.

The paper interprets redshift as the dedensification of propagating light bubbles, the dark-matter effect as baryonically anchored field tension, and the dark-energy effect as a large-scale dynamics of relaxation and dedensification. It further discusses quanta, spectrum, uncertainty, Anektron and Anun, magnetism, chemistry, space, the speed of light, matter formation, and the Planck boundary as a regime-dependent limit of visibility.

Hypothesis Status and Scale-Relative Working Level

The following sections should be understood as a partially formalized UQSH hypothesis. The UQSH does not replace established measurements or the successful mathematical relations of physics. Quantities such as the speed of light, spectral lines, the mass-energy relation, gravitational approximations, rotation curves, lensing signals, or quantized coupling events remain starting points against which every UQSH interpretation must be tested.

The aim of this work is therefore not to discard confirmed formalisms, but to extend their ontological

reading. Within the UQSH framework, established equations can be understood as limiting cases, regime projections, or effective coupling forms of a deeper field organization. Closeness to known measured quantities is explicitly intended: a UQSH interpretation is only viable if it remains connectable to existing mathematical and physical descriptions.

Within this work, the scale-relative basic assumption of the UQSH is taken as the supporting working level: physically readable field reality organizes itself on different scales according to recurring principles of saturation, coupling, coherence, inertia, reorganization, and boundary formation. This recurrence, however, does not mean a literal equation of different objects.

Terms such as electron-like, proton-like, neutron-like, atom-like, or element-analog therefore denote scale-relative functional roles within the same field logic. A galaxy is not claimed to be an actual atom. In the superregime, however, it may possess an atom-like function if central densification, peripheral coupling, neutral stabilization, and transition modes form the same mode of organization.

The UQSH therefore treats atoms, galaxies, and cosmic structures as different regimes of a recurring field mode of operation. What matters is not external sameness, but the recurrence of roles: center, binding space, shell, neutralization, coupling, saturation, reorganization, and transition.

All statements in this section should therefore be understood as working hypotheses. They gain strength not through the mere assertion of an analogy, but only through connectability to measured data, through formal precision, and through testing whether the same organizational roles actually recur on different scales.

Contents

1 Introduction **7**

2 Radiation as Light Bubbles **10**

2.1 Free Radiation as a Spherically Propagating Light Bubble 11

2.2 Superposition of Many Light Bubbles and Cumulative Field Curvature 13

2.3 Local Detection and Wave–Particle Duality 15

2.4 Quanta as Minimal Coupling Events 15

2.5 Spectrum as Internal Modulation Curvature 16

2.6 Spectrum as a Recursive Structure 17

2.7 Observation as Coupling 17

2.8 CMB and Visibility Limit 18

2.9 Densification, Reemission, and Matter Formation 19

2.10 Redshift as Field-Mechanical Dedensification 20

2.11 Dedensification as Distribution of Frequency Strength 21

3 Double Gravitational Effectiveness **22**

3.1 Energy as Field Motion, Frequency Strength, and Reorganization 22

3.2 Energy Distribution and Reorganization 23

4 Qu-Ground, Field Projection, and Motion: UQSH Basic Equation **24**

5 Distinction from Ether and Wheeler's Quantum Foam **32**

6 Dark-Matter Effect as Organized Field Response **37**

7 Dark Energy, Redshift, and Void Relaxation **38**

8 Saturation, Black Holes, and the End of Singularity **39**

8.1 Complete Relaxation as an Unreachable Limit State 41

8.2 Supermassive Black Holes as Long-Term Grown Anun Knots 41

9 Saturation, Inertia, and Regime Transition: Anektron and Anun **42**

9.1 Soft Anun-Anektron Transition Band and Temporal Resolution 45

9.2 Name Formation: Anektron and Anun 46

10 Electrons and Positrons **47**

11 Scale-Relative Atom-Galaxy Analogy **50**

11.1 Proton-Neutron Transitions as Role Changes of Field Organization 51

11.2 Quarks as Internal Intermediate Modes 53

11.3 Galactic Warps as Coupling to Filament Motion 53

12 Maximum Gravitation, Temperature, and Quasar Activity **54**

13 Uncertainty as Stability and Growth Margin **55**

14 Space as Relative Field Order **56**

15 Speed of Light, Scale-Relative Timing, and Time Dilation **57**

15.1 Spacetime as Shared Field Readability. 59

16 Time as a local clock field **59**

17 Proper Time, Inertia, and the Mobility of Bound Field Nodes **61**

17.1 Proper Time as the Internal Clocking of Bound Field Organization 61

17.2 Kinematic and Gravitational Time Dilation 62

17.3 Bound Field Nodes and Inertia 62

17.4 The Test-Body Limit and Mutual Field Reorganization 63

17.5 Anuns and the Mobility of Maximally Bound Field Nodes 63

17.6 Astrophysical Mobility and Reorganization Traces 64

17.7 Galactic Rotation as Directed Field Reorganization 64

17.8 Satellite and Dwarf Galaxies as Partly Integrated Field Populations 65

17.9 Macroscopic Saturation and Integrated Inhibition 65

17.10 Scale-Relative Saturation and Magnetic Field Order 66

17.11 Regime-Bound Saturation and Continued Boundary Dynamics 68

18 Entanglement and the Continuity of the Field Medium **70**

19 Observer, Temporal Linearity, and Scale-Relative Reality **71**

20 Galaxies as Spectral Field Knots **74**

20.1 What Was Measured in the Tests 75

20.2 SPARC, q, and Baryonic Organization 76

20.3 WISE, GALEX, and SDSS 76

20.4 Multiscale Galaxy Spectral Templates 76

21 Matter Formation **77**

21.1 Regime Coupling and Subregime-Like Matter Modes 81

21.2 From Electron-Positron Pairs to Hydrogen and Helium 82

21.3 Dwarf Galaxies as Secondary Matter and Field Knots 84

21.4 Quantum Numbers as Stable Field Markers 86

22 Chemistry as Flexible Field Coupling **86**

22.1 Periodic Table and Scale-Relative Organizational Analogy 88

23 Large-Scale Structures and light bubbles **90**

23.1 Imperfect Symmetry as a Prerequisite for Structure 91

24 Gravitational Wave as Gravitational Light **92**

24.1 Gravitational Waves as an Additional Component of Free Field Tension 95

25 Planck Limit, Vacuum Energy, and Coupling Filter **97**

25.1 Quantum Fluctuations as Short-Lived Potential Spaces 98

26 Electromagnetism as Twisting Order of the Field **98**

27 The Four Interactions within the UQSH Framework **103**

27.1 Gravitation 103

27.2 Electromagnetism 103

27.3 Strong Interaction 104

27.4 Weak Interaction 104

28 Recurring Mode of Field Operation and the Possibility of Enduring Life Formation **105**

29 c as the First Constant of Field Motion. **107**

30 Formalization Program **110**

30.1 Initial Derivation of the Regime Width between Anektron and Anun 111

31 Astrophysical Mass Gap as a Saturation Zone **114**

31.1 Yang–Mills Mass Gap in Light of the UQSH 115

31.2 Coherence, Hysteresis, and Observation Filter 115

31.3 Formal Supplement: Coherence, Hysteresis, and Observation Filter 115

32 Falsifiable Core Points **117**

32.1 No Ontologically Real Particles 117
32.2 No Gravitation without Field Organization 118
32.3 No Arbitrary Decoupling of Matter and Field 118
32.4 No Variable Speed of Light as a Local Fundamental Quantity 118
32.5 No Infinite Saturation 118
32.6 No Purely Random Regime Formation 119
32.7 Concise Falsification Core 119

33 Experimental Test of Wave-Particle Dualism **119**

34 Prediction on the Emergence of Life in the Universe **121**

35 Outlook: Field Coupling, Location, and Nonclassical Motion **123**

36 Comparison with Established Models **125**

37 Possible Misunderstandings and Limits **125**

38 Conclusion **126**
38.1 Concluding Remark on the Qu-Ground 127

1. Introduction

This paper formulates the Universal Quantum Foam Hypothesis (UQSH) as an ontological field framework. Its starting point is not an individual particle, not light, not matter, and not gravitation, but Qu-foam itself. In what follows, this is referred to as the *Qu-ground* Q. It denotes the deepest ground from which physical field reality, motion, curvature, coupling, matter, space, time, and observation become readable in the first place.

In order to distinguish this ground level from physically accessible field reality, the notation Φ is later introduced for the field projection of the Qu-ground. The term *projection* does not mean that the physical world is merely virtual, apparent, or less real. It does not refer to a mere image without reality, nor to a simulation. Rather, Φ denotes that physically real field form in which the Qu-ground becomes readable for a regime as field medium, motion, tension, curvature, and organization.

How Q itself is connected to this physical field reality cannot, according to the present state of this work, be measured directly. What is measurable is not the Qu-ground as such, but its regime-dependent effects: radiation, matter, gravitation, spectrum, coupling, saturation, dedensification, and reorganization.

The search for particles behind the dark-matter effect therefore takes on a different meaning within the UQSH framework. It is not wrong, but it may be directed by an ontological expectation that is too narrow. What is being sought may not be an additional substance in space, but a previously overlooked part of physical reality: the field medium itself. In the standard picture, this medium appears only indirectly as space, geometry, or background. In the UQSH, by contrast, it is the active carrier of matter, radiation, gravitation, and observation. The dark-matter effect would then not be the signal of a hidden particle, but the measurable effect of the very field that first determines what can appear as particle, mass, light, or gravitation.

In established physics, light is successfully described as an electromagnetic wave, a photon, or a quantized field excitation. These descriptions are precise and very well confirmed experimentally. Yet a deeper ontological question remains open: what actually propagates when light travels from a source to an observer? Is it a stream of particles, a wave of a field, a change in spacetime, or a combination of different descriptions?

The UQSH answers this question field-mechanically: light is not a substance flying through empty space. It is an ongoing reorganization of the field medium itself. Put more simply, light is a form of motion of the universe. A useful image is a wave spreading in circles across calm water after a stone has been thrown into it. The stone releases a tension, the water reorganizes itself, and the disturbance continues as a wave. Within the UQSH framework, something comparable occurs in the field medium: a local excitation produces a light bubble that propagates not only circularly on a surface, but spherically into the surrounding field space. If a tension were released in the middle of the water, the natural propagation there too would be spatial and spherical.

The term *gravitational light* denotes precisely this deeper field interpretation in the present work. It does not mean that ordinary light is simply gravitation in the classical sense. It means that light is understood as a free, spherically propagating light bubble of the same field medium whose organized curvature also appears as gravitation. Light therefore carries not only electromagnetically readable modulations such as frequency, color, or spectrum. It is also real field motion: tension, timing, reorganization, and, in the limiting case, also curvature action of the same gravity-capable field.

Radiation of different strength therefore does not describe a fundamentally different kind of motion, but different degrees of tension and timing within the same Qu-foam motion. Radio waves, visible light, X-rays, and gamma rays differ within the UQSH framework mainly in how densely, strongly, and rapidly they reorganize the field.

Here too, the image of water and stone is helpful. A small stone produces a gentle wave order with widely spaced waves; a large or fast-impacting stone produces denser, stronger, and more energetic waves. The water remains the same medium, but the released tension, timing, and wave strength differ. In the same way, within the UQSH framework, Qu-foam remains the same field medium, while different forms of radiation form light bubble of different strength within it.

The higher the frequency and energy density, the more tightly timed and more strongly tension-bearing the front becomes. The lower the frequency, the broader, softer, and more dedensified the same basic motion of the field medium appears.

In this picture, space is not a passive background. It is the dynamic medium in which field states absorb tension, pass it on, and reorder themselves. Radiation arises when a local bound or excited state releases its tension as a propagating front into the surrounding field. The speed of light therefore does not primarily describe the velocity of a transported object, but the maximum reorganization rate of the field. It is the limiting rate at which a change in tension can be transferred from one field region to the next.

It is important to distinguish between the carrying light bubble and its spectral structure. The front itself is not simply color or a spectral line. The spectrum arises as an internal modulation of this front through interaction with bound field states such as atoms, molecules, plasma, stars, or more complex structures. What we perceive as light, color, or spectrum is therefore not a complete view of the front, but the modulation that can couple to our observational regime.

This idea is unfamiliar at first, but it connects to everyday observation. We do not see things because their complete field reality enters our eye. We see the modulations of a radiation front that has been absorbed, changed, and re-emitted or scattered by matter. A red object is not red because it emits an isolated red substance, but because it absorbs certain modulation components and stably couples others back into our visual regime. This simple observation is generalized here: every spectral observation is a coupling to a modulation layer of a carrying front.

For this purpose, the UQSH uses a scale-relative language. Our directly observable regime forms only a specific organizational range of the field and serves as the first basis of comparison. The immediately underlying connecting scale is called the *subregime*, and the immediately higher one the *superregime*. These terms do not refer to separate worlds or additional spatial dimensions, but to recursively effective layers of organization of the same Qu-foam field.

Each regime level can act on neighboring levels and be influenced by them, but only in the coupling form possible in each case. A structure therefore does not become visible independently of its regime, but only when its motion, tension, and organization connect to the resolution and coupling capacity of the observing regime.

Put intuitively: in order to perceive a structure of the superregime directly as its own organizational world, an observer would have to be scaled up so far that galaxies or galaxy clusters appeared to him like atomic or subatomic structures. Conversely, an observer moving toward the subregime would have to be scaled down so far that atoms or elementary field knots appeared to him like extended galactic organizations. These images are not meant as literal changes in the size of a body, but as an

indication of the scale-relative readability of the same field organization.

The term *dimension* is deliberately avoided in this context. Subregime and superregime do not denote additional manifolds or separate spaces. They describe intrinsic organizational and coupling levels of the Qu-foam field itself. Scale relativity therefore does not mean that new dimensions are added, but that the same field reality becomes bound, resolved, and temporally readable in different ways depending on the regime.

A structure may therefore appear differently depending on the observational and coupling regime: in our regime it may appear as a spectrum, in a subregime as an independent field knot, and in a superregime as a modulation of a larger carrying light bubble. Subregime and superregime thus do not denote absolute locations, but relative connecting scales of field organization: the underlying and the higher regime with which a given observational regime is directly and recursively coupled.

This work brings together motifs that had already appeared separately in earlier UQSH works: Light in Light, field-mechanical redshift, the CMB as a coupling band, matter as bound field organization, the dark-matter effect as field tension, black holes as saturation knots, magnetism, chemistry, and regimes as scale-relative readability. What is new is the common placement of these motifs within an ontological field framework in which light, energy, gravitation, matter, time, and observation are understood as different readings of a deeper Qu-foam dynamics.

Conceptual Overview

Qu-ground / Qu-foam Designation for the universal, continuous fundamental field of the UQSH. The term *Qu-foam* denotes the dynamic, multi-scale, foam-like organized fundamental structure from which radiation, matter, gravitation, tension, relaxation, and observation emerge as state forms.

The expression *field medium* or *field* is used additionally in this work in order to describe the physical function of Qu-foam in a generally understandable way: it is the carrying medium in which field motion, coupling, saturation, dedensification, and reorganization take place. *Field medium* is therefore not a second designation for a different object, but a descriptive functional language for Qu-foam.

What is meant is not empty space and not a substance in the classical sense, but the dynamic fundamental structure from which all physical properties arise. The concept of Qu-foam stands conceptually in the tradition of Wheeler's quantum-geometrical considerations [19], but is interpreted here further in Sections 4 and 5 as a continuous fundamental structure specific to the UQSH.

Light Bubble Propagating field motion that arises from local excitation, relief, or reorganization of Qu-foam. In the main text, radiation is developed as a spherically propagating light bubble.

Modulation curvature Internal imprint or fine structure of a light bubble. It denotes the regime-compatible order that can later become readable as spectrum, frequency structure, or path information.

Regime A scale-relative organizational domain of the field with its own coupling capacity, its own stability, its own saturation, and its own observability. A regime is not a separate world, but a specific readable layer of the same field medium.

Subregime A lower-lying organizational domain of the field relative to our observational regime. Structures of a subregime may not be directly visible to our regime, but may become effective as internal modes, fluctuations, coupling thresholds, or bound fine structures.

Superregime A higher-lying organizational domain relative to our observational regime. A structure that appears in our regime as a large-scale knot may be read in a superregime as a smaller modulation, substructure, or coupling unit.

Anektron Hypothetical minimally saturated field mode of a regime. The Anektron does not denote the electron itself and does not replace the established concept of the electron. It describes the first independent coupling unit in which field motion is bound and stabilized to such an extent that it can appear as a measurable matter-capable field form.

Anun Maximally saturated field form of a regime. The Anun denotes the upper limit state of local field saturation, at which further deepening is no longer possible as an increase in local curvature. Within the UQSH framework, black holes are the most plausible astrophysical candidates for Anun states.

Dedensification Field-mechanical process in which a light bubble spreads over larger domains and thereby loses local tension density, timing density, and regime-compatible storage density. Observationally, dedensification may appear as weakening, redshift, or loss of fine modulation structure.

Saturation State in which a field region can no longer carry additional field motion as further local deepening or densification. Saturation does not lead to the end of the dynamics, but to reorganization, boundary formation, relief, recoil, reemission, or regime transition.

Coupling filter The regime-dependent selection of what becomes readable at all for an observer or a measurement system. Not every existing field organization couples to every regime. Only what possesses coherence, frequency, density, duration, and geometry that fit the respective coupling filter becomes visible.

Maximum gravitation The maximum locally effective gravitational depth of a regime. It does not mean infinite gravitation and no singularity, but the limit at which additional field tension no longer appears as further local depth. The depth saturates, while range, boundary structure, and relief dynamics may grow.

Gravity-Light The term *Gravity-Light* is used here as the registered preprint title. In terms of content, it corresponds to the German term *Gravitationslicht* or the English term *gravitational light*. The regime-dependent visibility of macroscopic gravitational field waves.

2. Radiation as Light Bubbles

In the Universal Quantum Foam Hypothesis, radiation is not understood as a small object moving through empty space. Rather, it appears as a free mode of motion of the physically effective field reality Φ. A local field reorganization propagates, in the idealized case, isotropically at the limiting rate c and forms a spherically propagating front. This free form of radiation is referred to here as a *Light Bubble*.

In this interpretation, light is therefore not a special case separated from field reality, but its most visible free mode of motion for us. A Light Bubble arises where a previously bound field state releases local tension, reorganizes, or is transferred into a free propagation form. What propagates is therefore not a small finished particle, but a continuous reorganization of the medium itself.

2.1. Free Radiation as a Spherically Propagating Light Bubble

A single Light Bubble carries not only propagation motion, but also an internal order. In the observational regime, this internal order appears as frequency, spectrum, intensity, line structure, and temporal signature. Radiation therefore has two distinguishable, but not separate, aspects: the propagating Light Bubble itself and the internal modulation of this Light Bubble. The modulation is not a second object alongside the front, but its readable internal structure.

It is important that a Light Bubble must not be understood as a filled sphere. In the idealized description, it is a propagating motion skin of the field. At its leading side, the field is tensioned, densified, or reoriented in the direction of expansion; at its trailing side, the previously excited field state relaxes back into the surrounding regime. A Light Bubble is therefore not a sphere filled with substance, but a moving tension-and-relaxation boundary. Its interior is not, in this idealized representation, the carrier of radiation. The carrier of radiation is the dynamic skin itself: the region in which tension, modulation, and subsequent relaxation are continuously passed on.

The following idealized illustration shows the spherical propagation of individual Light Bubbles.



Figure 1: Idealized spherical propagation of Light Bubbles as expanding tension-and-relaxation skins in the UQSH. A Light Bubble is not to be understood as a filled sphere. The effective radiation structure lies in the propagating motion skin: at the front, the field is tensioned or reoriented in the direction of expansion, while the previously excited field state relaxes at the trailing side. In detection, the entire Light Bubble is not captured; only a small locally and temporally coupled part of its extended propagation is measured. The spacing of successive fronts illustrates frequency. Wave-like behavior arises from the ordered sequence, or from the internal modulation, of the propagating Light Bubble.

Intuitively, a Light Bubble can be imagined as an expanding front shell. As propagation continues, the same field motion is distributed over an ever larger surface. As a result, the locally effective tension and coupling density decreases. The Light Bubble does not disappear, but its locally measurable effect per coupling region becomes weaker.

This de-densification affects several observable properties at once. The intensity decreases because a detector couples only to a smaller effective part of the extended front than near the source. The locally readable energy and coupling content decreases because the same field motion is distributed over a larger effective region. At the same time, the internal modulation is read more broadly: the distances between successive fronts or modulation phases appear larger, and the observed frequency decreases. In this UQSH reading, redshift is therefore not a destruction of radiation order, but a scale-relative stretching of the internal modulation of a propagating Light Bubble.

The local field-curving effect also decreases with de-densification. The decisive factor is not the total front as an extended structure, but the local tension difference that a coupling region actually experiences. A dense Light Bubble carries a stronger local field deformation; a strongly de-densified

Light Bubble produces only a weaker local change of the field state. Its effect therefore does not necessarily disappear, but can pass into a weaker or differently coupled subregime.

In this view, the spectrum is not the Light Bubble itself, but the ordered internal modulation with which it couples to bound field states and is read by them. Spectral lines therefore appear not as properties of a small transported object, but as resonant modulation responses of bound field organizations. This interpretation builds on earlier UQSH work on spectral shift, time dilation, and intensity decay [44**? ?**].

2.2. Superposition of Many Light Bubbles and Cumulative Field Curvature

On macroscopic scales, Light Bubbles do not occur in isolation. Many sources continuously generate spherical Light Bubbles, which cross one another and form a dynamic superposition network. In this interpretation, precisely this continuous superposition is important, because it can contribute to cumulative field tension and therefore to a field-curving effect.

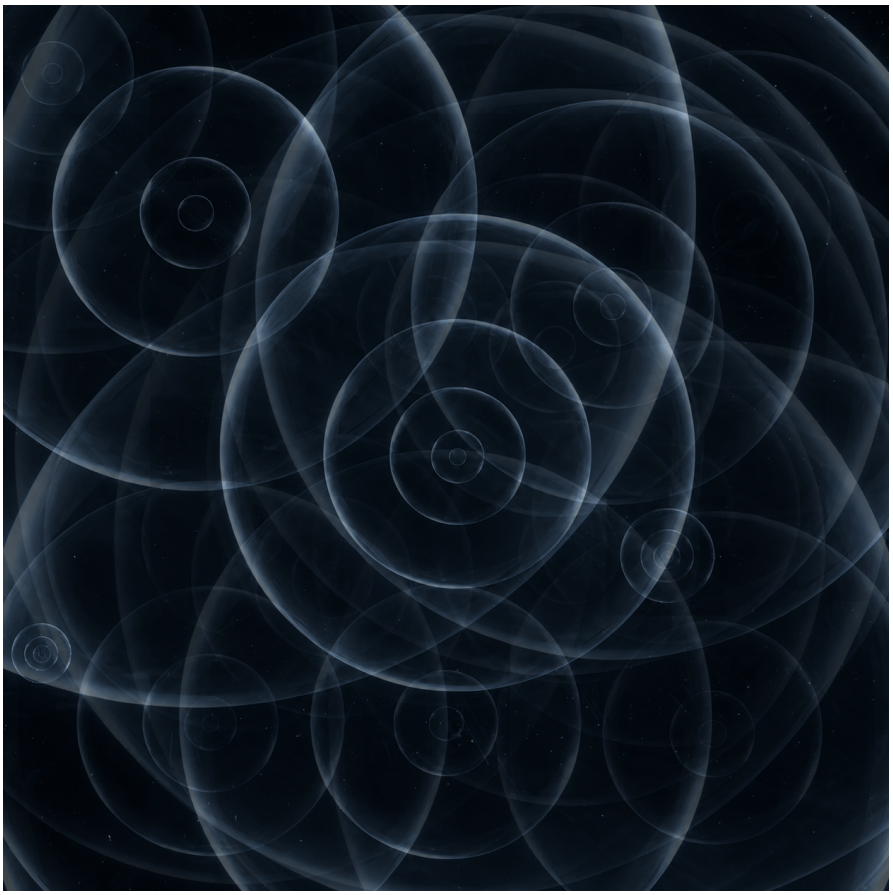


Figure 2: Schematic representation of overlapping spherical Light Bubbles. The figure shows an idealized limiting situation in which Light Bubbles propagate isotropically at the limiting rate c and form a dynamic superposition network. In the UQSH interpretation, the continuous superposition of such Light Bubbles in the quantum regime contributes to cumulative field tension, or to a field-curving and therefore gravitationally effective response. In real systems, the sources of the Light Bubbles are themselves in motion; overlaps therefore do not arise simultaneously or at fixed locations, but in spatially and temporally shifted coupling regions. The representation is therefore not to be understood as a reconstruction of individual real fronts, but as a conceptual visualization of the superposition principle.

In real astrophysical systems, this geometry is substantially more complex. The sources of the Light Bubbles move, so superpositions arise neither simultaneously nor at fixed locations, but in spatially and temporally shifted coupling regions. Moreover, because of the enormous number of overlapping contributions, an individual Light Bubble is generally not isolable. What is observable, therefore, is not the Light Bubbles themselves, but their interactions with matter, for example in the form of emission, absorption, scattering, or reflection.

This interpretation is also relevant for later UQSH readings of the dark-matter effect. Regions of high baryonic activity, strong anchoring, and high radiation density can build cumulative field organization more rapidly than baryon-poor or weakly organized systems. Radiation, baryonic anchoring, saturation, and additional gravitational action are therefore not read as separate processes, but as connected aspects of the same field dynamics.

2.3. Local Detection and Wave–Particle Duality

From this perspective, a new interpretation of wave–particle duality follows. If radiation is a spherically propagating Light Bubble, then what propagates is not point-like. Only the local coupling event in the detector is point-like.

An atom, molecule, or macroscopic detector couples only to that small region of the Light Bubble which reaches its local structure and can trigger a sufficiently strong reorganization there. The observed "photon" is therefore, in this view, not the entire Light Bubble, but the local event in which a section of the extended front becomes effective in a bound material state.

Wave-like behavior thus belongs to propagation: the Light Bubble spreads spatially, superposes, interferes, and carries internal modulations. Particle-like behavior belongs to detection: matter responds locally, threshold-dependently, and discretely. What becomes visible is therefore a point, even though the underlying field organization is extended.

This is particularly intuitive in the double-slit experiment. The Light Bubble acts through both slits as an extended field motion. Behind the slits, an interference pattern of field organization is formed. The screen, however, does not register this pattern as a continuous surface, but as individual local coupling events. Many such discrete hits gradually reproduce the interference pattern.

Formally, the UQSH therefore separates three levels:

$$\text{propagation} = \text{spherical Light Bubble},$$

$$\text{spectrum} = \text{internal modulation of the carrying Light Bubble},$$

$$\text{photonic event} = \text{local, threshold-dependent coupling to matter.}$$

Wave–particle duality thus appears not as an ontological contradiction, but as the consequence of a regime-dependent coupling process: what propagates is field-like and extended; what is measured is local and discrete.

In the UQSH, the photon is not a flying object, but the local coupling event of a propagating Light Bubble.

2.4. Quanta as Minimal Coupling Events

Within the UQSH framework, quanta are not small solid objects that fly independently through space. A quantum denotes the smallest stable coupling portion in which a field motion can be absorbed, emitted, or reorganized within a regime. Quantization therefore does not arise because the field consists of isolated grains, but because bound field states allow only certain stable transitions, thresholds, and coupling forms.

$$\text{Quantum} = \text{minimal stable coupling event of a regime.}$$

The photonic event described above is a special case of such a quantum. It denotes the local, stable coupling of a propagating light bubble to a bound field state.

Within the UQSH, photonic events can be distinguished according to their coupling role:

Emission quantum = local relief of a bound field state,

Absorption quantum = local uptake of a fitting modulation,

Scattering quantum = local deflection or re-imprinting of a front,

Reemission quantum = newly modulated release of previously absorbed field motion.

This distinction does not mean that different kinds of photons are introduced as new particles. It describes different coupling roles of the same photonic event type. A photon is therefore not a small object with a fixed internal content, but the regime-compatible, local, and quantized reading of a propagating field light bubble.

The energy content of a photonic event depends on the timing density of the front. The relation $E = h\nu$ is later interpreted field-mechanically in the section on energy as field motion, as a relation between frequency, tension density, and coupling strength.

Quanta are not field grains, but minimal stable coupling events.

This interpretation connects to the earlier UQSH interpretation of *Light Bubbles*, in which the particle event was described as the local trace of a spatially extended field motion [44].

2.5. Spectrum as Internal Modulation Curvature

Within the UQSH framework, a spectrum is not an external property of a light particle, but an internal modulation of the carrying light bubble. Atoms, molecules, plasma, dust, or other bound field organizations act as resonance structures. They cannot imprint an incoming or emerging front arbitrarily, but only according to their own stable states.

Spectral lines are therefore stable resonance responses of bound field knots. Bound field knots here mean, for example, atoms, molecules, ions, or plasma states, that is, material organizations that allow only certain stable transitions and couplings. When a light bubble interacts with such structures, a characteristic modulation order is imprinted on it.

This modulation order appears to us as a spectral line. It shows which resonance has taken place between the propagating front and the respective material field knot. The front is not to be understood as a thin line or a single ray, but as a spherically growing field surface. Each point of this surface is a local reorganization state of the medium that passes its tension on to neighboring regions. The entire front therefore forms an extended, spherical coupling surface on which the internal modulation

is carried along.

The term “Light in Light” takes up an earlier UQSH motif according to which a spherical field response is itself the carrier of fine internal distortions and modulations [43]. It describes precisely this layering: a larger spherical light bubble carries finer modulations within itself. This finer modulation does not lie outside the front, but is an ordered tension structure within the front surface itself. It is no less real than the carrying front.

It corresponds to a smaller curvature within a larger curvature, an ordered tension within the larger propagating tension. As the front surface grows outward, this internal order remains imprinted on the front as a relational structure and is carried again with each local transfer of the field state.

Over large distances, the modulation of a light bubble can remain partly preserved while its overall timing scale shifts. This shift is described in the following section as field-mechanical dedensification. For spectral analysis, the decisive point here is initially only this: the modulation remains readable as a relative order as long as it is carried coherently enough through propagation.

A spectrum is therefore not an unchanging imprint of the source. It is the readable residual order of a propagating field light bubble that has been modulated along the way and partially dedensified. It contains traces of the source, of the field regions it has passed through, and of the observational regime that finally reads this order.

This view also changes the meaning of astronomical spectra. When a galaxy has a spectrum, we do not see its complete field organization, but the electromagnetically readable modulation layer of its stars, gas clouds, dust components, active cores, and propagation paths. The deeper gravitational field configuration must additionally be inferred through dynamics, rotation curves, lensing, and large-scale field structure.

2.6. Spectrum as a Recursive Structure

The modulation level is itself nested again. A line may appear in our regime as a simple transition, but in the subregime as a complex organization of finer field states. Likewise, an entire galaxy may appear in the superregime as the spectral modulation of an even larger field front. Spectrum thus becomes a scale-relative concept. It denotes not only atomic lines, but more generally the readable structure of a carrier front.

This recursive view can be formulated briefly:

$$\text{Carrier front}_n = \text{Modulation}_{n+1}, \qquad \text{Modulation}_n = \text{Carrier front}_{n-1}.$$

What carries in one regime is itself carried in a larger regime. What modulates in one regime can be its own carrier in a smaller regime. This symmetry is decisive for the UQSH, because it explains why the same terms can be applied to atoms, galaxies, the CMB, and large-scale structures without equating them.

2.7. Observation as Coupling

In the UQSH, observation is not the passive imaging of a fully visible world. It is a resonance coupling between a propagating field structure and a bound observer system. We do not see the entire light

bubble. We see only the modulation layer that can couple stably to our biological, atomic, and technical structures.

Visible light is therefore not a preferred universal state, but the coupling window of our material regime. Color is not an absolute property of radiation. It arises from the way in which our eye and nervous system read certain modulation ranges of the front. Other living beings, other instruments, or hypothetical observers in other regimes would read other layers of the same field reality.

Instruments do not change the basic logic either. They merely extend our coupling window. Infrared, ultraviolet, X-ray, or gamma radiation are not completely different carriers, but different tension densities and modulation ranges of the same field propagation. An X-ray telescope reads a different coupling layer than an optical telescope, but both read regime-compatible modulations of a field front.

A simple example is the seeing of the environment. We rarely see the direct original solar front. Most of the time, we see radiation that has been reflected, absorbed, scattered, and newly modulated by matter on Earth. If one looks directly into the Sun, a much denser front couples in and overloads the biological system. Starlight from great distances, by contrast, reaches us dedensified enough to be stably readable.

These examples show that observation is always selection. It does not select consciously, but through coupling capacity. A detector measures only what can pass into its material state. Everything else may be real and yet remain invisible. For the UQSH, this is not a weakness of observation, but a basic condition of every observation.

It follows that horizons must not be understood only geometrically. Within the UQSH framework, a horizon is always also a coupling horizon. This consequence is developed in the following section with reference to the cosmic microwave background and to the visibility limit of our observational regime.

2.8. CMB and Visibility Limit

In the standard model, the cosmic microwave background is precisely characterized by the Planck measurements [3]. The UQSH does not dispute this finding, but interprets its coupling meaning differently. In this context, it is not the end of the universe and not a fixed wall behind which nothing lies. Rather, it describes the lower coupling limit of our observational regime: the domain in which strongly dedensified light bubbles can no longer be resolved as individual sources or clearly separated spectra, but still interact statistically with our regime.

What we measure as the CMB would then be the smoothed residual order of many dedensified boundary dynamics from different directions, distances, and propagation paths. Their commonality does not necessarily lie in an identical origin, but in their similar coupling state upon arrival in our regime. The carrying field motions are so strongly dedensified that individual sources, line structures, and fine modulations no longer remain separately readable. What remains is a nearly isotropic background band of weak, statistically superimposed field tension.

The visibility horizon is therefore not an absolute edge of the universe, but a coupling horizon. How far we can see depends not only on distance, but also on how much modulation order a light bubble has retained by the time it arrives in our regime. Dense, highly timed fronts may remain readable as separate signals over larger distances, while already strongly dedensified fronts tend rather to pass into a smoothed background coupling.

The more precise field-mechanical interpretation of this dedensification, including the shift from high

to low frequency, is treated later in the section on energy, frequency strength, and reorganization. For the CMB, only this is decisive here: it does not mark an absolute wall, but the domain in which many strongly dedensified field fronts couple to our regime only statistically as a background band.

An observer at another cosmic location, e.g. one billion light-years away, would therefore not have a CMB horizon identical in all details. Such an observer would measure different field paths, different local structures, different degrees of dedensification, and different anisotropies. The CMB would be statistically comparable because it describes the same general coupling domain of strongly dedensified field fronts, but its concrete fine structure would be position-dependent, path-dependent, and regime-dependent.

The CMB therefore appears not as the final boundary of reality, but as the limit of our present spectral readability. It shows that domain in which separate sources and fine modulations largely pass, for our regime, into a common background coupling.

The denser a front starts, the longer it remains readable as an ordered signal despite dedensification.

2.9. Densification, Reemission, and Matter Formation

Dedensification is only one direction of field reorganization. The opposite direction is densification. In the UQSH, every bound structure has its own field tension. This does not appear only in extreme objects such as neutron stars, magnetars, or black holes, but already in ordinary stars, planetary bodies, plasma fields, and atomic field knots. The difference does not lie in whether field tension is present, but in how strongly, densely, coherently, and sustainably it is organized.

When broadly distributed or low-frequency field motion encounters such a field knot, it can be absorbed, densified, modulated, or reemitted there. How this occurs depends on the local tension order of the knot. More weakly tensioned field knots tend to release absorbed field motion in longer-wavelength ranges. More strongly tensioned field knots can time the same motion more densely and transfer it into higher frequency ranges.

The observed spectral color is therefore not only an indication of temperature. It can also be read as an indication of how a field knot organizes the motion it has absorbed and releases it again. A small red star has a lower surface timing and tension density and therefore emits predominantly in the red and infrared range. Sun-like stars couple more strongly in the visible range. Hot blue stars and compact objects reach further toward UV, X-ray, or gamma radiation. High-energy outbursts can also occur in magnetically active small stars, for example as flares; there, however, they do not belong to the quiet basic emission, but to special relief events.

Field knots thus act as reorganization centers. They absorb field motion, reorder it according to their own carrying capacity, and release it again under relief conditions. In this view, high-energy emission does not appear as isolated energy generation, but as the reemission of previously focused, stored, or bound field tension. The general conservation and reorganization of this field motion is treated later in the section on energy.

It is important here: not every more weakly tensioned region automatically converts high-frequency radiation into lower frequencies. Without sufficient coupling, absorption, scattering, or resonance, a dense front can pass through a region largely unchanged. Densification and reemission therefore

require actual coupling.

A field knot reemits according to its own tension order; high-frequency radiation is reorganized only where it actually couples.

Matter formation is the limiting case of this densification. In this case, a densified field motion is not merely focused temporarily, but closed in such a way that it can persist as a bound organization. Densification alone is not sufficient for this. A fitting coupling geometry is required: phase, orientation, energetic level, functional role, and local carrying capacity must fit together at the same time. In earlier UQSH works, this was described as triadic coupling or geometrically selective stabilization, and it is clarified again later here in the discussion of matter formation.

The role of deeper regimes remains essential here. During densification, subregime-like modulation components are not simply compressed homogeneously. Some components may be co-densified coherently, while others may evade into deeper coupling levels, remain preserved as the internal structure of the new knot, or return to free field motion. This fine structure must be formalized in the future. The basic mechanism, however, remains closed: field motion is not destroyed, but redistributed between free propagation, local absorption, densification, reemission, and bound organization.

$$\text{free light bubble} \rightarrow \text{absorption by field knot} \rightarrow \text{densification according to local field tension}$$
$$\rightarrow \begin{cases} \text{reemission}, \\ \text{bound field knot}. \end{cases}$$

The spectral color of a reemission can therefore be understood as a visible indication of the tension strength and timing order of the emitting field knot:

$$\text{lower field tension} \rightarrow \text{red / infrared},$$

$$\text{medium field tension} \rightarrow \text{visible light},$$

$$\text{high field tension} \rightarrow \text{blue / UV / X-ray / gamma}.$$

2.10. Redshift as Field-Mechanical Dedensification

The observed cosmological redshift is not disputed within the UQSH framework. Distant sources appear redshifted, and this finding remains central. The UQSH, however, shifts its interpretation: redshift is not read primarily as the metric expansion of a passive space, but as the field-mechanical dedensification of propagating light bubbles within the dynamic Qu-foam. What appears in the standard model as a stretching of the spatial scale is interpreted here as a decrease in the local organizational density of the radiation front.

Such a light bubble propagates spherically and distributes its field motion over a growing surface:

$$A(r) = 4\pi r^2.$$

As a result, the local tension density decreases. The modulation carried along is supported more broadly, the timing density decreases, and the frequency readable for our regime shifts into lower ranges. This very field-mechanical dedensification appears observationally as redshift.

$$\text{propagating light bubble} \rightarrow \text{dedensification} \rightarrow \text{larger timing scale} \rightarrow \text{redshift}.$$

Redshift is therefore not only a property of the source and not exclusively a property of an expanding space. It is the integrated coupling trace of the entire propagation history. Source, travel time, traversed voids, filaments, halos, local field tension, scattering, modulation smoothing, and observational regime together affect the frequency order that is ultimately measurable.

Especially in large-scale relaxed regions, such as voids, dedensification can become more pronounced because there is less baryonic anchoring, less field saturation, and less coherent re-imprinting. In dense filaments, galaxy clusters, or strongly tensioned halo regions, by contrast, dedensification may be inhibited, superimposed, or partially newly modulated. The observed redshift within the UQSH framework is therefore not only a distance marker, but also a trace of the field state through which the front has passed.

The radiation does not disappear in this process, but is transferred into a broader, more weakly timed coupling form that is longer-wavelength for our regime.

Redshift is not disputed; it is interpreted field-mechanically as the dedensification of propagating light bubbles.

2.11. Dedensification as Distribution of Frequency Strength

As described in the section on redshift, a free light bubble distributes its field motion during propagation over a growing spherical surface. For the interpretation of energy, the decisive point here is this: the same field motion is distributed over larger regions of the field medium. The local tension density decreases, the timing is carried more broadly, and the frequency strength readable for our observational regime decreases.

Dedensification therefore does not mean that energy disappears. It means that the local density of field motion decreases. An originally very high-frequency light bubble can thereby gradually shift its regime-compatible readability into lower frequency ranges. In illustrative form, this corresponds to a sequence such as:

$$\gamma \rightarrow \text{X-ray} \rightarrow \text{ultraviolet} \rightarrow \text{visible light} \rightarrow \text{infrared} \rightarrow \text{microwave}.$$

This sequence is not to be understood as a rigid course of every single radiation front, but as an example of the direction of field-mechanical dedensification: high timing density is transferred over large distances into broader, more weakly timed modulation that is longer-wavelength for our regime.

A gamma front possesses a very high frequency strength. Its field motion is tightly timed, locally dense, and strongly reorganizing. During propagation, this density can decrease through geometrical distribution, path coupling, scattering, interaction with matter knots, traversal of relaxed field regions, and internal modulation smoothing. The front remains field motion, but its locally readable organization becomes broader. For an observational regime, this appears as weakening, redshift, or transition into lower frequency ranges.

Dedensification therefore initially describes the shift of the local timing and frequency order. The question of why no energy disappears in this process, but field motion is only reorganized, is treated in the following paragraph.

3. Double Gravitational Effectiveness

If spectral modulation itself is a form of tension curvature, then it is not merely information. It acts field-mechanically. A propagating front thereby possesses a double effect. First, as a carrying front, it spans a large-scale domain of action. Second, it carries internal modulation curvatures that generate their own couplings and local field responses on finer scales.

A dedensified front therefore does not become ineffective when it falls out of our direct visibility window. Its visible modulation may disappear, while its carrying field tension continues to alter tension relations in the field. Conversely, a higher-regime front can take up new atomic, molecular, or plasma-like modulations when entering our regime and thereby appear as an ordinary electromagnetic spectrum.

Within the UQSH framework, spectrum thus receives a double meaning. On the one hand, it is information about the readable modulation order of a front. On the other hand, every such modulation refers to real field tension that cannot be separated from gravitational field response.

Radiation = carrying light bubble + intrinsic modulation curvature.

Observed spectrum = regime-compatible reading of this modulation curvature.

Gravitational effect = field response to carrier front and modulation curvature together.

This view also explains why electromagnetic spectra alone need not contain the complete field configuration of a galaxy. Light shows the readable modulation layer. The gravitational field organization must additionally be inferred through rotation curves, lensing, dynamics, baryonic structure, and large-scale environment.

3.1. Energy as Field Motion, Frequency Strength, and Reorganization

Within the UQSH framework, energy receives a simple common denominator: energy is field motion. It is not an independent substance added to the field, but the effective activity of the field itself. Wherever the field oscillates, tensions, binds, curves, carries, relaxes, or reorganizes itself, energy appears.

This definition connects phenomena that are otherwise treated separately. Light is freely propagating field motion. Matter is bound and stabilized field motion. Gravitation is organized field tension and curved field motion. Mechanical motion is the displacement or reordering of a field pattern. Temperature is the statistically readable unrest of many small field motions. Inertia is the resistance of bound field motion against forced reorganization.

The energy content of a field motion appears in its strength, density, and timing. In freely propagating radiation, this timing appears as frequency. The well-known relation

$$E = h\nu$$

is read field-mechanically within the UQSH framework: the frequency ν describes the timing density of the propagating light bubble. The higher the frequency, the more densely and strongly the local reorganization of the field is timed; the lower the frequency, the more dedensified and weaker the field motion readable for our regime is.

Frequency is therefore not only an external wave property, but a measure of the energy content of free field motion. High-frequency radiation carries more energy because it engages the field at shorter intervals and with higher reorganization density. Low-frequency radiation carries less energy because its light bubble is broader, more relaxed, and less densely timed.

For bound states, the same relation appears in another form. Matter does not possess energy primarily as a freely running frequency, but as bound, circulating, and stabilized field motion. Its frequency strength is then not directly visible as light frequency, but organized as internal timing, binding energy, mass, inertia, and coupling capacity.

Energy is therefore not one of these phenomena alone, but their common origin. It is what underlies all state forms: the field in motion. What differs is only the degree of binding, direction, coherence, saturation, timing, and coupling form of this motion.

$$\text{Energy} = \text{field motion}.$$

$$\text{Energy content} \sim \text{strength, density, and timing of field motion}.$$

$$\text{free field motion} \rightarrow \text{frequency} \rightarrow \text{radiation energy},$$

$$\text{bound field motion} \rightarrow \text{internal timing} \rightarrow \text{mass, inertia, and binding energy}.$$

3.2. Energy Distribution and Reorganization

If energy is field motion, then within the UQSH framework it cannot simply disappear. It can, however, change its form, lose its local readability, dedensify, bind, scatter, relax, or pass into other coupling regimes. A free light bubble does not lose its field character in this process; it distributes its motion over larger field regions. What disappears is not the energy itself, but a particular order in which it was recognizable to an observational regime.

Conversely, distributed field motion can be refocused and bound again under suitable conditions. When it encounters tensioned field regions, matter nodes, plasma, magnetic fields, filaments, stars, accretion zones, or Anun environments, it can be densified again, modulated, or transferred into bound organization. The same field motion then no longer appears as free radiation, but as heat, acceleration, chemical bonding, matter formation, gravitation, or high-frequency re-emission.

In this view, conservation of energy therefore does not mean that a specific form of appearance remains unchanged. It means that the underlying field motion does not fall out of the field. It is reorganized between different state forms:

$$\begin{aligned}\text{free field motion} &\rightleftharpoons \text{bound field motion} \rightleftharpoons \text{tensioned field organization} \\ &\rightleftharpoons \text{relaxed field motion.}\end{aligned}$$

In established language, this corresponds approximately to transitions between radiation, matter or binding energy, fields, and thermal or diffuse background energy:

$$\begin{aligned}\text{radiation} &\rightleftharpoons \text{matter / binding energy} \rightleftharpoons \text{field energy} \\ &\rightleftharpoons \text{heat / diffuse background energy.}\end{aligned}$$

For a single observational regime, this redistribution may appear as loss. Radiation that dedensifies from gamma readability into infrared or microwave readability no longer has the same effect for certain couplings. It can no longer trigger processes that require high timing density. Field-mechanically, however, it has not been destroyed, but has passed into a different form of distribution and coupling.

Dedensification and densification can therefore be understood as two directions of the same conservation process. Dedensification distributes field motion over larger regions and lowers its local frequency strength. Densification brings distributed field motion back into stronger local organization. Neither process describes the creation or destruction of energy, but the reorganization of the same field motion.

Within this framework, entropy can be understood as a measure of the distribution and reorganizational freedom of field motion. A more strongly bound state has high local order, but less free capacity for reorganization. A dedensified state has lower local order, but a wider distribution in the field. An increase in entropy therefore does not mean the destruction of order, but the redistribution of locally bound organization into more broadly distributed field motion.

Within the UQSH framework, conservation of energy is the conservation of field motion through reorganization.

4. Qu-Ground, Field Projection, and Motion: UQSH Basic Equation

Within the UQSH framework, light, matter, gravitation, or spacetime do not stand at the beginning, but a deeper fundamental reality, which in this work is addressed with the term *Qu-foam*. This term, however, must not be too quickly understood as an ordinary physical medium. It denotes neither a classical substance in space nor an additional field alongside the known fields. Rather, it is a boundary and umbrella term for the connection between the preceding Qu-ground Q and the physically readable field projection Φ.

Here, the expression Qu is pronounced as (tschu / chu)-foam. Qu comes from an Albanian dialect and means, roughly, "to stand up" or "to rise." In the quantum domain as well, the idea of emergence plays a role in the formation of light bubbles, which likewise finds a place in this Qu designation. Within the UQSH, this term is used symbolically in order to describe the arising or emergence of organized physical structure from a continuous underlying medium. The term "foam" does not describe literal physical foam bubbles, but a continuously superimposing, dynamic, and interconnected organizational structure of the medium itself. Foam bubbles merely symbolize the overlap, local firmness, and connectedness of the Qu-foam.

The Qu-foam concept is thereby characterized by emergence from its foundation, which is designated here as Q. It is not a single field within the universe and also not a physical measurable quantity in the usual sense. According to the present understanding of this work, this Qu-ground possesses neither a definable zero, nor a definable unit, nor a place or body in space and time. Concepts such as space, time, scale, number, field state, and observation already presuppose physical readability. Q therefore does not denote an object within physics, but the preceding foundation from which physical readability first becomes possible at all.

The expression of a non-definable zero means here that Q cannot be set, like a physical field value, to an initial, empty, or reference state. Zero, unit, and scale already belong to the physically readable level of Φ.

Since Q itself does not appear directly as a single measurable quantity, its physically readable field reality is designated, in generally understandable terms, as Φ. Φ is thus not a second object separated from Q, but the field projection of the Qu-ground into a specific regime of readability. In this projection, the Qu-foam appears as a field medium that can carry motion, tension, curvature, densification, dedensification, and organization. Thus, Q or Φ is not a mechanical material. It is also not simply an ether in new clothing. The Qu-foam describes the fundamental reality from which physical readability emerges and in which this readability appears as field, motion, tension, coupling, saturation, and relaxation. It therefore stands at the boundary between ontology and physics: on the one hand, it refers to the Qu-ground Q; on the other hand, it appears in the physically readable field projection Φ.

The term *projection* is expressly not to be understood in the sense of an optical image, a projector image, a virtual environment, or mere appearance. What is meant is no simulation and no less real level. Rather, Φ denotes the real physical field form.

A direct physical measurement connection between Q and Φ is not claimed in this work. According to the present state of knowledge, not even the difference between Q as Qu-ground and Φ as physical field reality can be fully formalized according to the standards of our universe. Every measurement already presupposes space, time, coupling, scale, and observational regime and thus lies on the level of Φ, not on the level of Q itself. What can therefore be formalized initially are only the field states, motions, curvatures, couplings, and reorganizations within Φ.

Qu-Ground, Field Projection, and Possible Other Field Realities. The distinction between Q and Φ is fundamental for the UQSH. Thus, Φ is not to be equated with the totality of possible realities, but only with the field form physically measurable and coupling-capable for us.

This distinction prevents all conceivable forms of multidimensionality, existence, or metaphysical foundation from having to be constructed out of Φ itself. If one were to posit Φ as the only and ultimate

foundation, possible other forms of reality would have to be described as additional dimensions, hidden manifolds, or extensions within the same field projection. This would immediately raise questions of connectivity, coupling, and physical accessibility.

Within the UQSH framework, this is not necessary. Q is not bound to exactly one physical projection known to us. From Q, in principle, not only our field projection Φ could emerge, but also other field types or forms of reality with similar, different, or non-coupling-capable properties for us.

$$Q \to \Phi$$

therefore describes only the projection relevant for our regime. More generally, one could write:

$$Q \to \{\Phi_1, \Phi_2, \Phi_3, \ldots\},$$

where

$$\Phi_1 = \Phi = \text{our physically readable field projection.}$$

The possible further projections $\Phi_2, \Phi_3, \ldots$ are not to be understood here as confirmed physical fields, but as an ontological possibility of the framework. Without coupling to our regime, they cannot be treated as objects of our physics.

Thus, Q can be understood as a deeper foundation that does not exclude questions about existence, metaphysical foundation, or also about what is traditionally associated with concepts such as "God." The UQSH, however, does not turn this into a physical claim. It merely separates cleanly between the physically measurable field projection Φ and a deeper foundation Q, which is not fully exhausted by our physical concepts.

For the hypothesis itself, the decisive point remains: the observable physical properties arise within Φ. Light, energy, gravitation, matter, time, saturation, and dark-matter effects are readings and organizational forms of our field projection.

$$\Phi \to B(\Phi) \Rightarrow \{L, E, \mathfrak{G}, M, \tau, \ldots\}.$$

Without the assumption of a Qu-ground, Φ could still be used as a working concept for a physical field medium. For the complete UQSH, however, Q is important because it explains why Φ need not be understood as the absolutely final reality. Q keeps the framework open to the possibility that our field reality is only one specific physically readable projection of a deeper foundation.

In short:

$$Q = \text{preceding foundation of possible field projections,}$$

$$\Phi = \text{our physically measurable field projection,}$$

$$B(\Phi) = \text{motion and reorganization of this projection,}$$

$$\text{Physics} = \text{that which becomes measurable and coupling-capable within } \Phi.$$

Qu-Foam as a Boundary Concept between Q and Φ. The term *Qu-foam* is not understood here as an additional substance, not as a classical substance, and not as a further dimension. Rather, it denotes a boundary and umbrella term for the interplay of Qu-ground Q and physically readable field projection Φ.

Formally, the distinction

$$Q \to \Phi \to B(\Phi)$$

is preserved. Q denotes the preceding foundation, Φ the field projection physically readable for our regime, and $B(\Phi)$ the motion and reorganization of this projection. The term Qu-foam, however, describes the connection in which these levels appear not as separate worlds, but as different descriptive sides of the same fundamental reality.

Thus, the Qu-foam is neither simply identical with Q nor simply identical with Φ. It denotes the boundary language between the two:

$$\text{Qu-foam} = \text{boundary concept between } Q \text{ and } \Phi.$$

Or, more fully:

$$\text{Qu-foam} = \text{the interplay of ontological foundation and physical field appearance.}$$

This formulation avoids two misunderstandings. First, the Qu-foam is not made into a movable medium within space. Second, Φ is not declared to be the absolutely final reality. Instead, the Qu-foam denotes the deeper overall order in which this projection emerges from the Qu-ground and appears as moving field reality. What is meant is the connection in which the metaphysical Qu-ground and the physical field projection can be thought together without completely separating them from one another.

$$Q = \text{Qu-ground,}$$

$$\Phi = \text{physically readable field projection,}$$

$$B(\Phi) = \text{motion and reorganization of the field projection,}$$

$$\text{Qu-foam} = \text{boundary designation for the fundamental reality that connects } Q \text{ and } \Phi.$$

From this perspective, motion is not a property of a substance in space, but the way in which the field

projection Φ appears physically. Free motion possesses no proper time and no mass in the sense of a bound internal state. Only bound, stabilized, and saturation-capable motion within Φ appears as matter, inertia, and proper time.

$$B_{\text{free}}(\Phi) \Rightarrow (M = 0, \quad \tau = 0)\,,$$

$$B_{\text{bound}}(\Phi) \Rightarrow (M, \quad \text{inertia}, \quad \tau)\,.$$

The Qu-foam is therefore a particularly suitable term wherever the UQSH describes not only measurable physics, but also touches on the question of how a physical field reality can emerge at all from a preceding foundation. It stands for the connectedness between Q and Φ, without interpreting this connectedness as an additional spatial dimension or as a classical medium.

The first UQSH basic postulate is therefore:

$$\boxed{Q \to \Phi} \tag{1}$$

In words:

$$\boxed{Q \text{ is the Qu-ground; } \Phi \text{ is its physically readable field reality.}} \tag{2}$$

In this view, physics does not begin with finished objects, forces, or particles. It begins where Q becomes physically readable as Φ. What we describe as field, space, light, energy, gravitation, or matter are therefore not independent fundamental substances, but states, motions, and organizational forms of this field reality.

The decisive next step is motion. Without motion there is no dynamics, no energy, no propagation, no curvature, no reorganization, and no observation. The motion of the field projection is denoted by $B(\Phi)$. It describes the original dynamics from which different physical readings emerge for an observational regime.

This yields the dynamic basic form:

$$\boxed{Q \to \Phi \to B(\Phi)} \tag{3}$$

Here, $B(\Phi)$ is not an additional substance alongside the field projection, but its reality of motion. For the field projection itself, motion is initially not separated into light, energy, gravitation, matter, or time. These distinctions arise only when motion becomes bound, curved, timed, densified, dedensified, twisted, saturated, or observable.

The first physical readings of this single motion are, within the UQSH framework:

$$\boxed{B(\Phi) \Rightarrow \{L,\, E,\, \mathfrak{G}\}} \tag{4}$$

Here:

$$\begin{aligned} L &= \text{light / free radiation,} \\ E &= \text{energy / motion content,} \\ \mathfrak{G} &= \text{gravitational field effect.} \end{aligned}$$

The direction of the arrow does not mean a temporal sequence in the ordinary sense. These three fundamental properties of physics are thus contained in one motion and are therefore the first readability of the same field motion. For our observational regime, they appear as different physical quantities. For the field projection itself, however, they are not separate things, but different aspects of the same motion.

Light is the freely propagating form of motion of Φ. It appears as radiation when field motion does not remain locally bound, but carries itself onward as a front. Energy is the motion content of this dynamics, that is, its timing, frequency, intensity, or capacity to do work. Gravitation is the curvature and tension effect of the same motion insofar as it becomes field-effective and measurable for a regime.

In simple form:

$$B(\Phi)^{\text{free}} \rightarrow L, \tag{5}$$

$$\nu(B(\Phi)) \rightarrow E, \tag{6}$$

$$\kappa(B(\Phi)) \rightarrow \mathfrak{G}. \tag{7}$$

Here, $B(\Phi)^{\text{free}}$ denotes freely propagating field motion. It is read as light or radiation. The quantity $\nu(B(\Phi))$ denotes the timing or frequency order of the motion and becomes readable as energy content. The quantity $\kappa(B(\Phi))$ denotes the curvature or tension side of the motion and becomes readable as gravitational field effect.

From these first readings, further organizational forms arise. As soon as field motion no longer propagates freely, but is bound, stabilized, directed, or transferred into recurring internal dynamics, matter, inertia, magnetism, measure of time, chemistry, and complex structure arise:

$$\boxed{\{L,\ E,\ \mathfrak{G}\} \Rightarrow \{M,\ T,\ \text{Mag},\ \tau,\ C,\ S\}} \tag{8}$$

Here:

$$\begin{aligned} M &= \text{matter}, & T &= \text{inertia}, & \text{Mag} &= \text{magnetism}, \\ \tau &= \text{measure of time}, & C &= \text{chemistry}, & S &= \text{structure}. \end{aligned}$$

Matter arises as bound field-motion dynamics. It is not slowed-down motion of the field, but an internally organized, bound motion whose free propagation has been transferred into circulation, binding, saturation, and local stability. From outside, this bound dynamics appears as mass and persistent form.

Inertia arises from the same binding. It is the resistance of bound field motion against forced reorganization. A material field knot already carries internal motion and order. Every external acceleration therefore demands a reordering of this bound dynamics. This resistance to reorganization appears as inertia.

Magnetism stands closer to the first readings than many later structures, but is nevertheless not a first basic reading of motion itself. It arises only when field motion forms a directed twisting and coherent

near-order. Magnetism is therefore the directed twisting order of Φ, not simply motion as such:

$$\text{Mag} \rightarrow \text{directed twisting of } B(\Phi). \tag{9}$$

Time likewise does not belong to the first level. Motion is more original than time. Time arises only where motion becomes readable as sequence, timing, change, or reorganization. Within the UQSH framework, time is therefore not an independent container in which the field happens, but a measure of the reorganization of field states:

$$\tau \rightarrow \text{readability of the reorganization of } B(\Phi). \tag{10}$$

This gives the complete basic order:

$$\boxed{\begin{aligned} &Q \rightarrow \Phi \rightarrow B(\Phi) \\ &\quad \Rightarrow \{L,\ E,\ \mathfrak{G}\} \\ &\quad \Rightarrow \{M,\ T,\ \text{Mag},\ \tau,\ C,\ S\}\,. \end{aligned}} \tag{11}$$

This equation is not to be understood as an ordinary causal equation in the sense of a temporal chain of emergence. It describes an ontological order: Q is the Qu-ground, Φ its physically readable field reality, $B(\Phi)$ its motion, and the known physical quantities are readings, bindings, or organizations of this motion.

The shortest UQSH core sentence is therefore:

$$\boxed{\text{The Qu-ground is } Q\text{; physics is the readability of its field reality } \Phi.} \tag{12}$$

Or even more briefly:

$$\boxed{Q \text{ is the foundation; } \Phi \text{ is the readable field.}} \tag{13}$$

This avoids prematurely positing individual phenomena such as light, gravitation, or matter as ultimate fundamental quantities. They are not unimportant, but first or later readabilities of the same deeper field reality. For the observer, motion differentiates into light, energy, gravitation, matter, inertia, magnetism, time, and chemistry. For the Qu-ground itself, prior to every physical distinction, there is the possibility of becoming readable as field reality in motion and organization.

This motion, however, is not of the same kind in every case. Free motion possesses no proper time and no mass in the sense of a bound internal clock process. It is open reorganization, not a body that ages during its motion.

$$B_{\text{free}}(\Phi) \Rightarrow (M = 0, \quad \tau = 0)$$

for ideal free field motion.

For this reason, no time can pass for light. Within the UQSH framework, light is not a small object moving through a field. It is the pure free motion of the field projection itself. It possesses no bound internal clock, no locally closed circulation, and no material proper structure that could carry proper time.

$$L = B_{\text{free}}(\Phi),$$

$$\tau_L = 0.$$

Nevertheless, light is not ineffective. It possesses frequency, energy, and a tension side. It carries no proper time, but it carries a timing order that can be measured by an observational regime.

$$\nu_L \neq 0,$$

$$E_L = h\nu_L,$$

$$K(B_{\text{free}}(\Phi)) \neq 0.$$

The apparent contradiction dissolves when proper time and frequency are distinguished. Proper time presupposes a bound internal structure. Frequency, by contrast, describes the measurable timing order of a free or bound field motion relative to an observational regime. Only with a second following motion does a timing-frequency measurement become possible.

$$\text{proper time} = \text{internal time of a bound field pattern},$$

$$\text{frequency} = \text{measurable timing order of multiple field motions}.$$

Time is therefore not introduced within the UQSH framework as an independent dimension. Time is not an external stage on which motion takes place. Time arises only where motion becomes bound, repeatable, stabilized, and readable within a regime as a sequence of states.

$$\boxed{\text{Time is not the stage of motion, but its bound readability.}}$$

Mass, inertia, and proper time therefore arise only where motion within Φ becomes bound, stabilized, saturation-capable, and readable within a regime:

$$B_{\text{bound}}(\Phi) \Rightarrow (M,\ \text{inertia},\ \tau)\,.$$

Matter is therefore not the opposite of motion. It is bound, stabilized, and regime-readable motion. The electron or Anektron appears in this context as the minimally stable bound field organization of our regime. It is not an isolated fundamental thing, but a stable field knot whose internal motion is organized in such a way that it couples to our regime as mass, inertia, and charge.

$$e^- \sim B_{\text{bound}}(\Phi) + K_{\text{stabilized}}(B(\Phi)).$$

This formulation does not mean that the electron is composed of small material components. It means

that the electron is understood as a bound organizational form of field motion. Its stability arises from the fact that motion does not freely continue onward, but remains preserved in a closed, tensioned, and regime-coupled form.

5. Distinction from Ether and Wheeler's Quantum Foam

The UQSH uses the concept of a field, but does not mean a classical ether by this. The historical luminiferous ether was conceived as a substance-like carrier medium that fills space and through which light propagates. This gave rise to the image of an invisible material in space. The UQSH does not adopt precisely this image.

Historical ether experiments, in particular interferometer experiments such as the Michelson–Morley experiment, searched for a direction-dependent change in the travel time of light caused by the motion of the Earth relative to a resting luminiferous ether. An interference difference was expected when light traveled once with and once transverse to this hypothetical ether wind. From the perspective of the UQSH, the largely negative result is not surprising: if Φ is not a movable substance and not a mechanical carrier medium, then there can also be no ether wind against which an observer moves.

What was measured, therefore, was not the non-existence of a physical field reality, but the absence of a classical, substance-like light carrier moving relative to the observer. This is precisely where the UQSH differs from the historical ether: it expects no measurable drift of the field as a material flow, but local state changes, tensions, and reorganizations within a stationary field reality.

The UQSH shares with the historical ether only a few very general features: the field is space-filling, not visible as ordinary matter, and fundamental for the propagation of physical effects. The decisive difference, however, lies in the fact that the ether was conceived as a carrier medium *in* space. The UQSH field Φ, by contrast, is the physical field reality itself. It is not a substance in space, but the stationary foundation whose local states become physically effective as motion, tension, curvature, coupling, and relaxation.

The UQSH is also not identical with Wheeler's quantum foam. Wheeler's quantum foam primarily denotes the idea that spacetime fluctuates strongly on extremely small scales. The UQSH uses the term Qu-foam differently. It does not mean merely microscopic unrest of spacetime, but a scale-spanning field reality.

In this sense, the Qu-foam is to be understood scale-relatively. It denotes not only a structure at the smallest length, but the way in which field organization propagates across different scales. In the microscopic domain, local reorganizations may figuratively appear like small bubbles, foam zones, or fluctuating field cells. In the macroscopic domain, the same principle no longer appears as a microscopic bubble image, but as the large-scale propagation of field organization, for example as a gravitational wave, as a supernova-driven field excitation, or as a large-scale tension and relaxation structure around astrophysical systems.

The term Qu-foam therefore does not describe a single fixed size scale. It describes a scale-relative form of motion and organization of field reality:

microscopic: local foam-like reorganizations,

macroscopic: large-scale propagation of field tension and field relaxation.

The "bubbles" of the Qu-foam are therefore not to be understood as small material spheres. They are an image for local state changes of the field, in which the overlaps of the spherically propagating light bubbles act like a dense foam on every scale. On larger scales, the same basic principles appear as waves, fronts, tension zones, or relaxation processes.

In short:

luminiferous ether = conceived carrier medium for light in space,

Wheeler's quantum foam = microscopically fluctuating spacetime structure,

UQSH Qu-foam = scale-relative reorganization of a stationary field reality.

The UQSH is therefore not a return to the classical ether. Nor does it simply adopt Wheeler's quantum foam. It formulates its own field interpretation: the field is stationary, but not ineffective. What is physically effective are its local and scale-spanning reorganizations, tensions, and relaxations.

Apparent Paradox of the UQSH Field. At first glance, the field reality Φ appears contradictory: it is supposed to be invisible, permeable, stiff, stationary, dynamic, relaxing, and at the same time not locally stretchable without limit. This contradiction arises, however, only when Φ is understood like an ordinary substance. This is precisely not what is meant in the UQSH. Φ is not a material that flows through space. It is a stationary field reality within which local states can arise, change, and relax again. Stationary therefore does not mean complete rigidity. It means only that the field as a substance does not migrate from place to place. What is dynamic are the states of the field.

The apparently opposing properties describe different aspects of the same field reality. Φ is invisible because it is not an object within the field. It is permeable because no material friction occurs as in a fluid. It is stiff because state changes do not occur arbitrarily, but are limited by a maximum reorganization rate, the speed of light. It is stationary because it is not transported as a material flow. It is dynamic because its local states can be reorganized. It relaxes because tension does not remain locally without limit. It is not locally stretchable without limit because saturation, relief, and redistribution set in.

This local boundedness does not mean, however, that field reality as a whole would simply be bounded like a finite material body. The UQSH distinguishes between local saturation and scale-spanning dedensification. What we observe is only the field tension that has reached us furthest or is still coupled to our regime. At the limit of observability, this tension may already be strongly dedensified and close to decoupling into a deeper or differently coupled regime.

The dedensification of field motion need not be restricted to a single spatial extent. It can continue across further scales. Since field motion and field tension within the UQSH framework are not lost, but only reorganized, bound, dedensified, or relaxed, it is natural to understand the total field reality at least as possibly unbounded. This "infinity," however, would not be understood only as spatial infinity

into the distance, but also as scale-spanning unboundedness: field organization can not only propagate further, but also dedensify into ever deeper or finer regimes and, upon saturation, redistribute into higher regimes.

This gives rise to a second, deeper paradox. Locally, Φ is not infinitely stretchable, because every concrete field tension is capable of saturation. Across scales, however, the reorganization of the same field motion may nevertheless possibly continue without bound. The limitation then concerns not the existence of field reality itself, but the coupling capacity of a specific regime.

The apparent paradox disappears as soon as one does not ask how a substance can simultaneously be solid, permeable, and dynamic, but how a stationary field reality can carry changeable states:

The field does not flow.

The states of the field change.

Not every field tension remains locally preserved.

It can dedensify, relax, and pass into other scale regimes.

From this perspective, light is not the motion of a body through a medium. Light is a free reorganization of the field state. Matter is a bound reorganization of the field state. Gravitation is the tension and curvature side of such field organization. Relaxation is the tendency to reduce tension, distribute it, or transfer it into other organizational forms and scale regimes.

Thus, the UQSH remains close enough to the idea of a space-filling foundation to make older ether intuitions understandable. At the same time, it avoids the decisive error of the ether: it does not turn the field into a movable substance. Φ is not a material substance in space, but a stationary field reality whose local states are finitely load-bearing, while its scale-spanning reorganization may possibly continue without bound.

$$\boxed{\text{Physics is finite; the universe can be infinite.}} \tag{14}$$

On the Dimension of the Field. The concept of dimension must be used with caution within the UQSH framework. The scale structure of the field described here does not mean an additional spatial dimension and no externally added manifold. Rather, what is meant is an intrinsic organizational and coupling depth of the Qu-foam field itself.

The field reality Φ can carry light bubbles, modulations, bound field organizations, and large-scale structures on different scales, without these levels having to be understood as separate spaces. They interpenetrate one another because they do not exist as material layers in space, but as different coupling and organizational forms of the same field.

Illustratively, one may imagine this as many transparent, mutually shifted tension networks. Each network possesses its own knots, pulling directions, and tension patterns. The networks, however, do not lie on top of one another as fixed material layers, but interpenetrate because they represent

different scale and coupling states of the same field reality. A structure may be strongly effective in one regime, while remaining almost invisible to another regime.

$$\text{Regime} = \text{readable organizational layer of } \Phi,$$

$$\text{Subregime} = \text{more deeply coupled organizational layer},$$

$$\text{Superregime} = \text{more highly coupled organizational layer}.$$

An interaction between regimes does not occur automatically through mere spatial interpenetration. It becomes physically relevant only when a coupling threshold is reached. If, in one regime, for example, celestial bodies, galaxies, or clusters meet, enormous field tensions and energy transfers may arise that are observable for this regime. For a subregime, however, the same processes do not necessarily appear as an event. There, they may appear merely as a slow or non-resolvable state change of the field, because temporal resolution, coupling form, and energy threshold are not the same.

$$\text{event in regime } n \neq \text{event in regime } n-1.$$

$$\text{Visibility} = \text{coupling} + \text{energy threshold} + \text{temporal resolution}.$$

It is therefore more precise to speak not of additional dimensions, but of scale-relative coupling regimes. The UQSH describes no further spatial axes, but a field reality whose organization is nested across many scales. Each scale possesses its own possibilities of energy transfer, tension propagation, modulation, and saturation. These possibilities superimpose without necessarily obstructing one another like material bodies.

In this sense, the Qu-foam field has no dimensions in the sense of additional spaces, but an intrinsic regime depth:

$$\boxed{\text{Regime depth} = \text{scale-relative coupling and organizational capacity of } \Phi.}$$

If one nevertheless speaks of a "dimension," then only in an extended, non-geometrical sense:

$$\boxed{\text{intrinsic field dimension} \neq \text{additional spatial dimension}.}$$

$$\boxed{\begin{array}{c}\text{intrinsic field dimension} = \\ \text{nesting of scales, couplings, tensions, and state resolutions}\end{array}}$$

The UQSH thus remains distinct from classical multidimensionality models. It does not claim that the field unfolds only through additional manifolds. Rather, the scale-relative nesting is an internal property of the field itself. What we perceive as space, time, energy transfer, and physical structure is in each case only that layer of this field organization which couples to our regime.

Regime, Subregime, and Superregime. In what follows, a *regime* denotes a specific organizational and coupling domain of the Qu-foam field. Our directly observable physical regime is designated as the starting level R_0. The immediately underlying organizational layer is called the *subregime* and is denoted by R_{-1}. The immediately overlying organizational layer is called the *superregime* and is denoted by R_{+1}.

This designation can be continued recursively:

$$R_0 = \text{our observable regime},$$

$$R_{-1} = \text{subregime}, \qquad R_{-2} = \text{subsubregime}, \qquad R_{-3} = \text{subsubsubregime},$$

$$R_{+1} = \text{superregime}, \qquad R_{+2} = \text{supersuperregime}, \qquad R_{+3} = \text{supersupersuperregime}.$$

In general:

$$R_n = \text{scale-relative organizational and coupling level of the field.}$$

Here, n describes no additional spatial dimension and no separate world. The index denotes only the relative position of an organizational layer with respect to our observational regime. Negative values ($n < 0$) denote deeper, more finely coupled regimes. Positive values ($n > 0$) denote higher, more coarsely coupled or more large-scale organized regimes.

$$n < 0 \Rightarrow \text{subregime},$$

$$n > 0 \Rightarrow \text{superregime}.$$

A structure may appear in one regime as small, stable, and elementary, while from the perspective of a deeper regime it would be readable as an extended or even extremely tensioned field organization. Conversely, a large-scale structure of our regime may, from the perspective of a higher regime, assume the role of a smaller bound organizational form. The terms subregime and superregime therefore describe not size alone, but relative coupling, resolution, and stability relations.

Illustratively speaking: an electron appears in our regime as a minimally stable field knot. From the perspective of a subregime, the same organization could appear as an extensive or strongly tensioned field structure. A galaxy or galaxy cluster appears large-scale to us; from the perspective of a superregime, by contrast, such a structure could assume the role of a smaller bound field knot. These statements are not to be understood as a literal identity of electrons, galaxies, and black holes, but as scale-relative homology of their organizational roles.

What is compared are not objects, but roles:

coupling, saturation, tension, binding, relaxation, stability.

Interactions between neighboring regimes may occur, but only where a coupling threshold is exceeded. An event in R_0 therefore does not have to appear immediately as an event in R_{-1} or R_{+1}. What is decisive in each case are the energy concentration, coupling capacity, and temporal resolution of the regime concerned.

$$\text{Visibility in } R_n = \text{coupling} + \text{energy threshold} + \text{temporal resolution}.$$

6. Dark-Matter Effect as Organized Field Response

Within the UQSH framework, the dark-matter effect is not understood as an indication of an additional non-baryonic matter component, but as the dynamic response of a continuous field medium to baryonic organization, radiation excitation, and local saturation. Visible matter acts here not only as a classical gravitational source, but as an anchor structure of the field. Radiation appears as a propagating light bubble that continuously excites the field and generates local reorganization work. The observed gravitational excess arises from the baryonically guided field tension that does not fully relax.

Formally, the observed acceleration can initially be described through an effective potential:

$$\mathbf{g}_{\text{obs}}(\mathbf{x}, t) = -\nabla\Phi_{\text{eff}}(\mathbf{x}, t).$$

This potential is composed of the baryonic contribution and an additional field-organizational response:

$$\Phi_{\text{eff}} = \Phi_{\text{bar}} + \Phi_{\text{field}}.$$

The field contribution describes no new matter density, but a saturation-limited response of the medium to radiation and baryonic organization:

$$\Phi_{\text{rad}}(\mathbf{x}, t) = \int d^3x' \, \frac{Q_{\text{rad}}\left(\mathbf{x}', t - \frac{R}{c}\right)}{4\pi R^2} S_\phi(\mathbf{x}, t) O_{\text{bar}}(\mathbf{x}, t), \qquad R = |\mathbf{x} - \mathbf{x}'|.$$

Here, Q_{rad} describes the local strength of the radiation-induced field excitation. The factor $1/(4\pi R^2)$ captures the spherical propagation of the light bubbles. S_ϕ describes the saturation limit of the field response, while O_{bar} describes the organizing effect of baryonic structures.

The observable excess in rotation curves can correspondingly be written as the difference between observed and baryonically expected dynamics:

$$C(r) = v_{\text{obs}}^2(r) - v_{\text{bar}}^2(r).$$

This term does not directly measure an invisible mass, but the additional dynamic effect of the organized field response. The dynamic amplification factor

$$D(r) = \frac{g_{\text{obs}}(r)}{g_{\text{bar}}(r)}$$

describes how strongly the observed acceleration deviates from the purely baryonic expectation.

In earlier works it was shown that this excess is not distributed arbitrarily, but possesses preferred

regimes. The normalized structural form

$$F(x) \propto x^p e^{-x/q}$$

describes whether the field response is organized in a more locally concentrated or diffuse manner over larger radii. Small values of q correspond to a localized peak response, while larger values of q correspond to an extended, diffuse field organization. The parameter q is therefore read as a dynamically activated transition scale, not as a freely arbitrary halo parameter.

The UQSH thereby connects three levels: baryonic anchoring, radiation-driven field excitation, and saturation-limited reorganization. The dark-matter effect arises where these three levels act coherently together. It is not an additional substance in space, but the regime-dependent visible gravitation of a deeper field organization.

Within the UQSH framework, dark matter is not a missing substance, but organized field tension.

Baryonic matter anchors the field; radiation excites it; saturation limits the response.

A more detailed derivation of this effective field formulation was developed in [42]; the empirical regime structure of rotation curves was discussed in [41].

7. Dark Energy, Redshift, and Void Relaxation

Within the UQSH framework, dark energy must be treated in a differentiated way. Most of what is interpreted in the standard model as evidence for accelerated expansion, and therefore as dark energy, lies in the observed redshift of distant radiation sources. This redshift is not disputed within the UQSH framework, but interpreted differently: not primarily as the metric expansion of a passive space, but as the field-mechanical dedensification of propagating light bubbles.

This shifts the role of dark energy. It appears first not as an additional global force or substance, but as the interpretation of a large-scale dedensification dynamics of the Qu-foam. Over long propagation paths, a light bubble loses local timing density, modulation sharpness, and regime-compatible storage density. What is read in the standard model as the stretching of space appears here as relaxation, distribution, and a change in timing of the field motion itself.

The relaxation tendency of the field is, in principle, present everywhere. Every field state carries within it the possibility of reducing tension, smoothing modulation, broadening timing, and transferring bound dynamics into freer field motion. This relaxation, however, is not equally visible everywhere. In galaxies, filaments, clusters, halo regions, or near strongly saturated field nodes, field motion is more often bound, scattered, reemitted, focused, or newly modulated. There, dedensification is not abolished, but it is overlaid by baryonic anchoring, plasma, radiation fields, magnetization, and field saturation.

Voids remain important in this view, but they are not the actual main cause of the dark-energy interpretation. Rather, they are the regions in which the normal dedensification of propagating light bubbles is interrupted especially little. In voids, dense baryonic anchors, strong reemission centers,

compact field nodes, and pronounced saturation zones are largely absent. As a result, a light bubble can relax more freely over large distances, broaden its timing, and lose modulation density.

Dark energy and redshift are thereby connected, but not identified with one another. Redshift is the observable trace of dedensification. Within the UQSH framework, dark energy is the cosmological interpretation of this large-scale dedensification and relaxation dynamics. Voids describe the field regions in which this dynamics can proceed especially undisturbed. They contribute to normal dedensification, but they do not explain it by themselves.

Formally, this view can be written as a path function of dedensification:

$$E_{\text{path}} = E_{\text{geom}}\, E_{\text{relax}}\, E_{\text{bind}}^{-1}.$$

Here, E_{geom} describes the geometric distribution of the tension front over growing spherical surfaces. E_{relax} describes the intrinsic relaxation of the field motion during propagation. E_{bind} describes the inhibiting, binding, or re-imprinting effect of bound structures such as galaxies, filaments, halos, plasma, matter nodes, or saturation zones. Voids correspond to regions with small E_{bind}, so that normal dedensification is less interrupted there.

This also makes clear why the UQSH does not require an additional dark-energy substance. The observed redshift remains the central finding. Its cause, however, is not sought exclusively in the global expansion of a passive space, but in the field-mechanical history of the radiation itself: in its spherical propagation, its relaxation, its passage through differently bound field regions, and its gradual dedensification.

In this reading, dark energy does not describe a force acting uniformly everywhere, but an interpretation of the large-scale behavior of propagating field motion. Where the field is strongly bound, dedensification is overlaid by coupling, reemission, and saturation. Where the field is weakly bound, especially in voids, the same dedensification can appear more freely. The difference therefore lies not in a second form of energy, but in the local coupling environment of the Qu-foam.

Within the UQSH framework, dark energy is read primarily as an interpretation of redshift through field-mechanical dedensification; voids favor this dedensification, but are not its main cause.

Dark-matter effect ↔ baryonically anchored field tension,

Dark-energy effect ↔ large-scale dedensification of propagating light bubbles,

Voids ↔ weakly bound regions of facilitated relaxation.

8. Saturation, Black Holes, and the End of Singularity

If gravitation is the effect of organized field curvature, then it cannot grow into depth without limit. The Qu-foam possesses a finite carrying capacity within a given regime. If a local saturation limit

is reached, additional field motion can no longer be carried as further local deepening of the same regime curvature. The field then does not respond with infinite curvature, but with reorganization.

For black holes, it is above all the local side of saturation that is decisive here: the maximally effective gravitational depth of our regime is reached without a mathematical singularity having to arise. Further field motion then no longer appears as additional local depth, but as greater range, boundary structure, relief, or regime connection.

In this view, black holes appear not as singularities, but as saturated field knots of maximum local regime curvature. In the terminology introduced later, this upper saturation form corresponds to the Anun function of a regime. Their depth does not grow infinitely; what can grow is the domain in which the field is organized at this saturation limit.

The depth saturates, the range can grow.

This sentence summarizes the mechanism. A small black hole and a supermassive black hole do not differ in that one would be finite and the other infinite. Both locally reach the same limiting character of field saturation within our regime. The larger black hole possesses a larger saturation range, a larger boundary structure, and a larger coupling surface. It is not more infinite, but more extended.

This also dissolves the question of why recursive field curvatures of all regimes do not lead to megagravitation. Although every organized field curvature acts gravitationally, not every curvature couples completely into the same observational regime. Each level possesses its own saturation, its own carrying capacity, and its own coupling filters. The observed gravitation is therefore not the raw sum of all curvatures, but the saturated and regime-filtered projection of those components that become effective for our regime.

$$G_{\text{obs}}^{(n)} \neq \sum_{k=-\infty}^{+\infty} G^{(k)}$$

but approximately

$$G_{\text{obs}}^{(n)} = \mathcal{F}_n \left[\sum_k S_k\left(K^{(k)}\right) \right],$$

where $K^{(k)}$ describes the field curvature of a regime, S_k its saturation function, and $\mathcal{F}_n$ the coupling filter of the observing regime. This equation is not meant here as a finished theory, but as a structural formula: gravitation is saturated, regime-filtered field response.

Beyond this limit, the field remains dynamic. For our regime, however, this dynamics is no longer readable as additional local deepening, but as boundary structure, inertia, relief, and possible connection coupling to other regimes.

This also explains why black holes do not simply fall into one another immediately during mergers. When two Anun states approach each other, it is not two isolated mass points that meet, but two extended domains of nearly maximum field tension. Their outer saturation zones begin to overlap even before the actual contact. The space between them thereby itself becomes a highly loaded field domain. The field cannot absorb this additional load arbitrarily as further deepening. It must reorganize the tension to a compatible measure. This gives rise to deceleration, orbital motion, oscillating approach,

lateral relief, radiation, and finally a common saturation envelope. In this view, the apparent long "dancing around" of two black holes is not merely a delay of a gravitational fall, but the field-mechanical process by which the medium distributes, smooths, and transfers the maximum load between two Anun states into a sustainable common order.

Where depth is saturated, approach must first be made compatible as field tension.

The prolonged and wave-radiating merger process therefore fits the UQSH interpretation: additional tension is not translated into depth without limit, but into reorganization, boundary structure, and new field order.

8.1. Complete Relaxation as an Unreachable Limit State

Just as an absolute final saturation of the field is not reachable, a fully relaxed Qu-foam state cannot really arise either. A totally relaxed field would be a state without residual tension, without internal timing, without reorganization capacity, and without minimal coupling. Such a state would be the inverse singularity to the infinite density of a black hole: not infinite densification, but complete dynamic dissolution. In both cases, field mechanics would end.

The UQSH therefore excludes both extremes. Neither maximum compression nor complete relaxation can occur as real final states. The field always remains in an intermediate domain of tension, residual dynamics, and reorganization capacity. Even in the largest voids, the Qu-foam is therefore not absolutely calm, but remains dynamic.

In this sense, the cosmic background radiation can be read as an indication of a minimal residual timing that remains coupling-capable. It does not mark a fully relaxed zero state, but a lower coupling domain of our observational regime. Even in the most strongly relaxed regions, the field therefore remains dynamic and, in principle, capable of reorganization.

Complete relaxation would be just as singular as complete saturation.

8.2. Supermassive Black Holes as Long-Term Grown Anun Knots

Within the UQSH framework, supermassive black holes appear not as suddenly formed singular mass points, but as long-term grown Anun knots. Their extreme size therefore does not have to be explained from a one-time primordial initial state. It is more plausible that they arose over long periods through matter uptake, galactic interactions, and above all through mergers with other black holes.

This interpretation becomes especially important when known physics has difficulty allowing the rapid formation of very massive black holes within short cosmological times. Within the UQSH framework, the visible distance does not automatically have to be understood as an absolute beginning horizon if redshift is also read as dedensification of propagating field motion. An older or larger universe would accordingly provide more time for generations of galaxy formation, accretion, and merger.

Matter uptake contributes to the growth of black holes, generates accretion disks, radiation, jets, and local field reorganization. The largest mass increases, however, are more likely to be expected through mergers. When two supermassive black holes couple, not only mass is added. Two already highly

saturated Anun zones pass into a common larger saturation order. The result is a more extended Anun knot with greater range, larger coupling surface, and altered boundary dynamics.

Hawking radiation does not contradict this view either. A black hole can, in principle, lose energy through the relaxation tendency of the field, but this process is extremely slow compared with accretion, interactions, and possible mergers. Before such a knot disappears through radiation, it will with high probability take up matter again, couple, structure its environment, or merge with another saturated knot.

In this sense, black holes are not absolutely unchanging objects, but functionally extraordinarily long-lived structure-givers. They collect field tension, stabilize galactic centers, drive jets, distribute energy, generate densification and relief spaces, and can thereby favor new star formation, gas structures, filaments, or compact knots. They therefore act not only as endpoints of bound matter, but as regenerative centers of an ongoing field organization.

Within the UQSH framework, supermassive black holes are thus no exception to the field logic, but one of its most long-term manifestations: saturation, merger, relief, and renewed structure formation form a cycle in which Anun knots not only store, but repeatedly reorder the cosmic field. That is why they are functionally immortal life-givers.

9. Saturation, Inertia, and Regime Transition: Anektron and Anun

The previous considerations lead to a more precise interpretation of the concept of saturation within the UQSH. Saturation does not simply denote a high energy density and also not an absolute standstill of the field. It describes the state in which a field region can no longer carry additional tension motion onward in the previous open form. The dynamics is not destroyed in this process, but transferred into another organizational form: into binding, inertia, stored tension, boundary structure, reemission, or into a cross-regime coupling form.

Saturation is therefore not rest of motion, but a change of coupling form. As long as a light bubble propagates freely within a regime, its local transfer occurs at the limiting rate of that regime. In our observational regime, this limiting rate appears as the speed of light c. If, however, the front encounters a region of very high field tension, for example in the environment of a black hole, it increasingly loses the capacity to be continued as a free front of the same regime. It is not c that becomes smaller, but the share of the dynamics that remains readable for our regime as free c-propagation decreases.

This distinction is essential. The invariance of the speed of light is not violated thereby. c describes the local maximum reorganization rate of free, regime-internal light bubbles. The apparent slowing near maximum field saturation does not concern this local limiting rate, but the readability of the open propagation form. The front is increasingly transferred into a bound, inert, and saturated field state. For an external observer, this appears as time dilation, slowing, or horizon formation. Field-mechanically, it means that open dynamics is translated into inert field organization.

It is not the speed of light that slows down;
free field motion is transferred into inertia.

In this context, inertia is not an additional attribute of matter, but the expression of stored field motion in bound form. An inert state resists rapid reorganization because its internal field tension is already

stabilized and bound. Inertia is therefore the coupling resistance of a saturated or partially saturated field state against further reorganization. Mass appears in our regime as the measurable form of such bound field tension.

From this perspective, a regime transition does not have to be understood as a break of c. Each regime possesses its own resolution, its own coupling threshold, and its own form of time. What is readable within our regime as fast, free dynamics may be too fine, too fast, and not directly resolvable for a coarser superregime. Our electron or photon dynamics would there not be a stable individual structure, but an unresolved submodulation. Only when such dynamics is transferred through saturation into a more inert, coarser, and more stable field organization can it become readable for a higher regime as a coherent coupling form.

Saturation therefore acts as a scale translator. It does not slow the field absolutely, but transforms an open, rapidly timed dynamics into a bound, more coarsely readable, and more inert field order. What appears for our regime as an upper limiting state can become, for a higher regime, the first sufficiently stable coupling form. This creates the possibility that the end of one regime at the same time forms the beginning of the coupling of the next higher regime.

In order to distinguish these two limiting forms, two terms are introduced in what follows: *Anektron* and *Anun*.

An *Anektron* denotes, in the UQSH, a hypothetical minimally saturated field mode. It is not simply a new name for the Standard Model electron. The electron remains the experimentally confirmed observed state with charge, mass, and spin. The Anektron, by contrast, denotes the postulated field-mechanical foundation behind such an appearance: a minimally saturated field mode that can act as the first autonomous regime-forming coupling unit. The existence of Anektrons is therefore a prediction of the UQSH. It does not claim that today's measurements already clearly distinguish different electron classes, but that the electron appearance could rest on a deeper class of minimally saturated field modes.

An *Anun* denotes the maximally saturated field form of a regime. In the observational regime, the clearest candidate for this is the black hole. In this view, a black hole is not a singularity, but a maximally saturated field state in which additional field motion is no longer carried as further local deepening. Instead, range, boundary structure, coupling surface, and relief channels grow. The Anun is therefore not merely an endpoint, but an upper saturation anchor of the regime architecture.

Anektron = minimal saturation; Anun = maximum saturation.

Between Anektron and Anun lies the actual interior domain of a regime. In this interior domain, material, atomic, molecular, stellar, galactic, and satellite-like field formations arise. Not every coupling-capable structure is an Anektron. Coupling capacity alone is not sufficient. A dwarf galaxy may possess a measurable field effect and assume a similar coupling role in the galactic field space as an electron does in the atomic field space. It is, however, not necessarily a minimally saturated regime-transition mode, but a secondary coupling-capable field formation within an already established potential space.

This distinction is decisive. The measurement regime initially does not recognize the internal object type, but the stable total field effect. Whether a measured coupling effect arises from stars, gas, a central black hole, a dark field tension, or a mixed form is initially secondary for a coarse observational

regime. What is measured first is the effective field organization. Only higher resolution and additional modeling allow the internal assignment. Precisely for this reason, dwarf galaxies, satellite systems, or other substructures in the galactic field space can appear functionally analogous to electron states without themselves being Anektrons.

The role of the Anun, by contrast, is more fundamental. Without an Anun, no permanently stable large-scale potential spaces arise. An Anun collects, centers, and saturates field tension so strongly that the environment is forced into ordered reorganization paths. In the UQSH, black holes are therefore not only endpoints of collapsed matter, but generators and stabilizers of large-scale field architecture. Their relief zones, jet channels, boundary tensions, shock fronts, density gradients, and magnetic orders can enable new field formations. In this view, galaxies arise not despite their central saturated field states, but through their structuring effect.

The formation of dwarf galaxies can thereby be understood as a secondary reorganization process. They arise preferentially where sufficient coupling energy, field tension, and relief dynamics are present, for example at the edges of jet channels, in filament axes, in shear surfaces, or in galactic boundary zones structured by the activity of an Anun. A dwarf galaxy is therefore neither an upper nor a lower regime boundary form, but a coupling-capable field formation that emerges from the interior domain of a regime stabilized by Anun action.

This gives a closed scale-relative chain of emergence:

$$
\begin{aligned}
\text{Anektron} &\rightarrow \text{matter formation} \rightarrow \text{electrons, nuclei, atoms} \\
&\rightarrow \text{molecules} \rightarrow \text{stars and complex matter} \\
&\rightarrow \text{black holes as Anun} \rightarrow \text{jet and boundary-field relief} \\
&\rightarrow \text{galaxies and dwarf galaxies} \rightarrow \text{Anun as Anektron of the superregime.}
\end{aligned}
$$

This chain does not mean that every electron is directly an Anektron or that every dwarf galaxy is a cosmic electron. Rather, it describes a functional order. The Anektron denotes the minimally saturated regime-transition mode. The electron is the observable standard form in which such minimal coupling could appear in our regime. Dwarf galaxies are secondary, coupling-capable field formations within a galactic potential space. The Anun is the maximum saturation form that limits a regime upward, stabilizes it, and at the same time can translate it into the next regime.

The central thought is therefore:

The Anektron opens the coupling of a regime;
the Anun closes it and in the superregime
again assumes the Anektron function.

This also makes understandable why electrons in the Standard Model appear so extraordinarily stable, universal, and hardly destructible. They could represent the observable surface of a deeper minimally saturated field-mode class. Likewise, it becomes understandable why black holes should not be regarded only as destructive endpoints. They are maximally saturated field forms that constitute the upper stability of a regime, generate its interior architecture, and enable new coupling-capable structures through relief.

The UQSH thereby obtains a clear regime architecture:

Regime = field domain between minimal saturation and maximum saturation.

The minimal saturation appears as Anektron. The maximum saturation appears as Anun. Between the two, the observable diversity of the respective scale unfolds: radiation, matter, atoms, molecules, stars, galaxies, satellites, and dwarf galaxies. The end of a regime is thereby not the end of field motion, but its translation into the first coupling-capable form of the next regime.

9.1. Soft Anun-Anektron Transition Band and Temporal Resolution

Within the UQSH framework, the minimal black-hole mass does not automatically mark a fully developed Anektron form of the next higher regime. It first denotes the lower threshold from which a field state of our regime reaches the Anun function. From this point onward, the local curvature depth is saturated. Further field motion does not deepen the state without limit, but increases range, boundary structure, coupling surface, and inertia.

This does not give rise to a hard point-like regime transition, but to a soft transition band. A minimal Anun may initially be only weakly, transiently, or not yet coherently readable for the superregime. With growing mass and range, however, the saturated field form becomes increasingly coarser, more inert, and more stable. As a result, it can gradually enter the observational and coupling spectrum of the next higher regime.

This readability depends not only on the mass or energy of the Anun, but also on the temporal resolution of the observing regime. A process that runs over long periods within our regime may appear to a superregime as a short or soft transition. What appears to us as the slow growth of a black hole through accretion, merger, and field saturation could be read in the superregime as the formation of a new stable coupling unit.

The regime bridge thereby becomes temporally relative. The lower threshold of the transition is given by the minimal Anun formation,

$$E^{(n)}_{\mathrm{U,min}} \sim M_{\mathrm{BH,min}} c^2 .$$

The coherent readability in the next higher regime, however, is reached only in an extended transition domain:

$$E^{(n+1)}_{\mathrm{A}} \in \left[E^{(n)}_{\mathrm{U,min}},\, E^{(n)}_{\mathrm{U,coh}} \right],$$

where $E^{(n)}_{\mathrm{U,coh}}$ denotes that Anun energy from which the saturated field form becomes stably readable for the superregime as a coherent Anektron function.

This transition can occur dynamically very quickly if the minimal Anun rapidly binds further field motion. In our regime, this may appear as strong accretion growth, merger, jet activity, or increasing boundary-field relief. In the superregime, the same process would not be visible as the sum of individual rapid processes, but as an increase in the coherence of a new coupling unit.

Temporal resolution is therefore decisive. A regime does not read every subordinate dynamics individually, but only that development which becomes coherent within its own measure of time. An Anun of our regime can thus initially lie below superregime readability, but then, through rapid growth, pass into a stable Anektron function of the superregime.

In short: the threshold of regime transition is given by minimal saturation, but the observability of the

new Anektron arises softly, time-dependently, and coherence-dependently.

Saturation sets the threshold; temporal coherence makes the transition observable.

An Anun probably does not become fully the Anektron of the next higher regime at the same moment. It grows into its observability.

9.2. Name Formation: Anektron and Anun

The terms *Anektron* and *Anun* are introduced in this work as independent UQSH terms. They are intended to name two complementary saturation forms of a scale-relative regime: the minimally saturated lower coupling form and the maximally saturated upper limiting form.

The common component *An* refers on the one hand to Albert Einstein, formed from the first letter of *Albert* and the last letter of *Einstein*. On the other hand, *An* stands not only for the beginning of the first name and the end of the last name, but symbolically for beginning and end at the same time. This double meaning fits directly with the UQSH, because a regime transition has exactly this structure: what appears for one regime as the end of its maximum saturation can become the beginning of a new coupling capacity for the next higher superregime.

The term *Anektron* connects this *An* with a phonetic proximity to the electron. This is not intended to replace the established electron concept of the Standard Model. The physical role of the Anektron was already defined above as the minimal saturation form of a regime. The choice of name is intended merely to indicate that the observed electron appearance within the UQSH framework could point to a deeper minimally saturated coupling function.

The term *Anun* denotes the complementary upper limiting form. It too contains *An*, supplemented by the *u* for universality and UQSH. The final *n* can at the same time be read as a mathematical regime or dimension index. The physical role of the Anun was already defined above as the maximum saturation form of a regime. The choice of name connects this limiting function with the motif of closure, transition, and new beginning function in the next higher regime.

The phonetic proximity of *Anun* to ancient Near Eastern terms such as *Anunna* or *Anunnaki* is not understood here as a historical derivation. It is accidental and arose independently. It can, however, be read as symbolic resonance. In these ancient cosmic images, upper and lower world domains, an original carrying water or a cosmic primordial ocean, as well as ordering powers that mediate between these domains, appear. The UQSH does not adopt these ideas as justification, but the imagistic proximity is remarkable: here, too, the issue is a carrying fundamental medium, superposed or nested regime domains, saturation limits, and transitions through which new stable structures emerge from a deeper field order.

This linguistic proximity is not intended to mystify the term. It does, however, make it more illustrative. *Anun* therefore carries several levels at once: the honoring of Einstein, the motif of beginning and end, the *u* of universality and of the UQSH structure, the mathematical *n* as regime index, and a distant poetic echo of ancient creation images in which world order emerges from a carrying cosmic ground.

10. Electrons and Positrons

In the Standard Model, electron and positron appear as particles with the same mass and opposite electric charge. If both meet under suitable conditions, they can annihilate and transfer their energy into radiation. This description is empirically successful and is not replaced within the UQSH framework. The UQSH, however, asks about the field-mechanical meaning of this symmetry: what does it mean that two states possess the same stability, but opposite coupling direction?

Within the UQSH framework, matter is not understood as an independent substance, but as bound, stabilized, and regime-coupled field motion. A material structure is a field state whose motion does not freely propagate onward, but is held in a stable internal organization.

$$M = B_{\text{bound}}(\Phi).$$

More precisely, matter can be described as bound field motion with stabilized tension order:

$$M = B_{\text{bound}}(\Phi) + K_{\text{stabilized}}(B(\Phi)).$$

In this view, an electron is not a small material body in space, but a minimally stable bound field organization of our regime. It possesses mass, inertia, and charge because its internal field motion is not free, but remains organized in a closed, tensioned, and coupling-capable form.

The positron is correspondingly not interpreted as a foreign substance. It is the complementary coupling form of the same minimally stable field organization. Both states possess the same stability class, but opposite coupling orientation.

$$e^- \sim B^{(+)}_{\text{bound}}(\Phi),$$

$$e^+ \sim B^{(-)}_{\text{bound}}(\Phi).$$

Here, the sign denotes not an additional substance, but the direction of coupling. Electron and positron are therefore, within the UQSH framework, two mirror-image orientations of a minimally stable field mode.

$$\sigma \in \{+1, -1\}.$$

$$B^{(\sigma)}_{\text{min}}(\Phi) = \text{minimally stable field mode with coupling orientation } \sigma.$$

Thus, one can write:

$$e^- \leftrightarrow B^{(+)}_{\text{min}}(\Phi), \qquad e^+ \leftrightarrow B^{(-)}_{\text{min}}(\Phi).$$

In this reading, the concepts matter and antimatter do not denote two separate kinds of reality. They describe two complementary coupling orientations within the same field projection. Antimatter is

therefore not “negative matter” and also not a substance from another world. It is the same bound field organization in the opposite coupling direction.

Matter = bound field motion with one coupling orientation,

Antimatter = complementarily bound field motion with opposite coupling orientation.

This view explains why matter and antimatter do not simply remain stable next to one another upon direct encounter. If two complementary field modes meet coherently, their oppositely directed coupling orders can mutually cancel the bound structure. What disappears in this process is not the field motion itself. What disappears is the bound form of this motion.

$$B^{(+)}_{\text{bound}}(\Phi) + B^{(-)}_{\text{bound}}(\Phi) \rightarrow B_{\text{free}}(\Phi).$$

Field-mechanically, annihilation therefore does not mean destruction of reality, but unbinding. The previously bound field motion loses its stable closed coupling form and passes into free radiation.

$$e^- + e^+ \rightarrow \gamma + \gamma$$

is read within the UQSH framework as:

$$B^{(+)}_{\text{min}}(\Phi) + B^{(-)}_{\text{min}}(\Phi) \rightarrow B_{\text{free}}(\Phi) + B_{\text{free}}(\Phi).$$

The resulting radiation is thus not a newly created substance, but the released field motion that had previously been stored in the two complementary bound modes. In this language, conservation of energy means that the motion and tension content of the bound field organization does not disappear, but passes into free field propagation.

$$E_{\text{bound}} \rightarrow E_{\text{free}}.$$

From this perspective, the so-called disappearance of antimatter also becomes readable differently. If matter and antimatter had remained exactly symmetrical and permanently equally frequent as stable, separate field organizations, large-scale amounts of both coupling directions would have to be observable. In fact, however, our observational regime appears predominantly matter-dominated.

Within the UQSH framework, this may mean that our regime possesses a stable preferred coupling direction. The counter-orientation is possible and is also observed, but it is less permanently embedded in normal field space. It occurs when local energy and coupling conditions allow its formation, but it usually remains stable only as long as it is not coherently coupled with its complementary counter-form.

Regime $\Rightarrow$ preferred coupling orientation.

Antimatter therefore does not disappear because it would be less real, but because complementary

coupling states in a matter-dominated regime very easily pass into unbinding. As soon as a positronic mode encounters the fitting material counter-structure, the bound organization is cancelled and reemitted as free radiation.

In this interpretation, the asymmetry between matter and antimatter does not first have to be understood as an inequality of two substances. It can be read as regime asymmetry: our field regime stabilizes one coupling direction on large scales better than the other.

$$N_M \gg N_{\bar{M}}$$

would then not mean:

Antimatter was never possible,

but:

one coupling orientation was preferentially stabilized on large scales.

The UQSH thereby obtains a field-mechanical interpretation of the matter-antimatter problem without presupposing a one-time beginning of the universe. The decisive question is therefore not why more matter than antimatter should have arisen in an original event. Within the UQSH framework, matter arises continuously wherever light bubbles are coupled, densified, and geometrically closed in such a way that they can remain stable as bound field organization.

The actual question is therefore why our regime permanently stabilizes one coupling orientation more easily than its complementary counter-orientation.

Striking here is the special role of high-frequency radiation, especially gamma radiation. During the unbinding of complementary field modes, not just arbitrary radiation is released, but often an extremely densely timed form of free field motion. Within the UQSH framework, this indicates that gamma radiation lies close to the saturation and relief dynamics of the Qu-foam. It appears where bound field tension is abruptly returned into free propagation or where matter formation and matter unbinding are tied to especially dense coupling states.

Gamma radiation thereby receives a double meaning. On the one hand, it can appear as a high-frequency relief form of bound field tension. On the other hand, it can play a special role in matter formation, because its high timing density is suited to driving local light bubbles into the domain of geometric closure and saturation. Matter then does not arise from radiation as from a substance, but from the binding, densification, and stabilization of field motion. In this process, gamma radiation marks an especially dense boundary domain between free and bound field organization.

Within the UQSH framework, matter and antimatter are not two kinds of substance, but complementary coupling orientations of bound field motion.

Annihilation is not the destruction of field motion, but the unbinding of complementary field modes into free radiation.

This interpretation remains compatible with the established description: electron and positron possess

the same mass, opposite charge, and can generate radiation during annihilation. The UQSH merely shifts the ontological reading. The electron is the observable material coupling form of a minimally stable field mode. The positron is the complementary orientation of the same stability class. Their encounter releases the bound field motion, rather than destroying a substance in the actual sense.

11. Scale-Relative Atom-Galaxy Analogy

Under the scale-relative working hypothesis of the UQSH, a galaxy appears as an atom-like field organization of a higher structural scale. What is meant is a functional correspondence of the organizational form: a central field knot generates a potential space in which visible coupling zones, neutral stabilization regions, and peripheral substructures together form an element-analogous order. For our observational regime, only those components become visible whose coherence, mass, duration, and coupling geometry are sufficient. What is decisive is therefore not the mere number of existing structures, but their effective coupling capacity.

For a galactic system, an effective satellite or electron analogy can be introduced:

$$N_{\mathrm{eff}}^{\mathrm{sat}} = \sum_i w_i,$$

where w_i describes the coupling strength, coherence, and stability of the respective dwarf galaxy or satellite structure. Satellites that couple as independent field knots with the central galactic potential space contribute substantially to this effective number. Weakly bound, transient, torn-apart, or only partially coherent structures accordingly enter only with weighting.

A galaxy such as the Milky Way thereby possesses a specific scale-relative element function. This does not result solely from the visible total mass and not solely from the number of known dwarf galaxies. It arises from the combination of central field organization, baryonic coherence, satellite binding, halo tension, and not directly visible field persistence. The counting of known satellites therefore forms the first observable approximation of the electron-like shell structure.

The proton-analogous side of a galactic system lies in the coherent, baryonically visible high-tension regions. These include the galactic core, the bulge, dense stellar and gas regions, as well as stable spiral-arm segments. These regions act proton-like because they couple openly outward, carry visible structure, and help determine the central organizational strength of the system. They form the visibly structuring core and inner order of the galaxy.

The neutron-analogous side lies in neutrally stabilizing field configurations. On the galactic scale, such contributions arise from two interconnected components. First, there are baryonic structures that are really present, but do not stand out as independent, strongly luminous, or clearly delimited coupling units. Second, there are tension fields that are not directly baryonically visible: halo persistence, diffuse field organization, voids with connecting tension, shear surfaces, filament connections, and non-coupling-capable transition zones.

In this reading, voids are neutron-like as soon as they assume a neutrally stabilizing role in the total field. They then do not act as the mere absence of matter, but as buffering, connecting, or tension-distributing field components. They separate and connect open baryonic coupling zones, mediate between spiral-arm segments, hold field tension in the halo, and contribute to the overall stability of the galactic knot.

This yields the structural assignment:

electron-analogous ↔ coupling-capable dwarf galaxies and satellites,

proton-analogous ↔ visible coherent baryonic high-tension regions,

neutron-analogous ↔ neutrally stabilizing, only partially or not directly visible field organization.

The element analogy of a galaxy is therefore reconstructed from two observational levels. The first level is the already introduced effective number of coupling-capable satellites $N_{\text{eff}}^{\text{sat}}$. It corresponds to the electron-like shell structure. The second level is the internal distribution of visible baryonic coherence and neutral field persistence:

$$Z_{\text{eff}}^{\text{gal}} \sim \mathcal{K}\left[\rho_{\text{bar}}^{\text{coh}}, \Phi_{\text{core}}, \Phi_{\text{arms}}\right],$$

$$N_{\text{neutral}}^{\text{gal}} \sim \mathcal{K}\left[\rho_{\text{bar}}^{\text{non-coupl}}, K_{\text{halo}}, K_{\text{void}}, K_{\text{fil}}\right].$$

Here, $Z_{\text{eff}}^{\text{gal}}$ denotes the effective visible core and structural coupling of a galaxy. $N_{\text{neutral}}^{\text{gal}}$ denotes the neutrally stabilizing field organization that does not couple, or couples only indirectly, to our observational regime.

A galaxy thereby appears as an enlarged atom-like field organization. The reason lies not in identical particle numbers, but in the recurring mode of operation of the field: central densification generates a potential space; visible coherent structures couple outward; neutral field components stabilize; peripheral substructures occupy preferred coupling domains. Because the field works on all scales according to the same organizational principles, galaxies of different size and form arise with recurring classes, transitions, and stability patterns.

The central question is therefore: what effective scale-relative element function does a galactic field knot possess if only those core, shell, and neutral components are counted that couple coherently enough for our regime?

What is counted is not everything present, but what is coherently coupling-capable.

11.1. Proton-Neutron Transitions as Role Changes of Field Organization

Within the scale-relative UQSH framework, the distinction between proton-like and neutron-like components is not a rigid classification of objects. Protonic and neutronic states describe changeable coupling roles of bound field knots. Already in the microphysical domain, neutron-proton transitions show that a field knot can change its internal role. What is described in the Standard Model as beta decay or electron capture appears here as selective reorganization of the internal tension distribution:

$$n \rightarrow p + e^{-} + \bar{\nu}_e,$$

$$p + e^{-} \rightarrow n + \nu_e.$$

Within the UQSH framework, free beta decay means that a previously neutron-like, neutrally stabilized field order breaks open and forms a more open coupling asymmetry. The residual state becomes more

proton-like: it couples more strongly outward, becomes more directed, and appears more clearly as an independent structure for an observational regime. The emitted electron and antineutrino component is read as an exported counter-coupling through which the new open order is stabilized:

$$\text{neutrally stabilizing field order} \longrightarrow \text{openly coupling field knot} + \text{exported counter-coupling}.$$

The reverse direction, electron capture, describes a neutralization. A proton-like state takes up a fitting compensating counter-coupling. Its open coupling asymmetry is thereby balanced, and the state passes into a more neutron-like, more inwardly closed, and neutrally stabilizing field order:

$$\begin{aligned}\text{openly coupling field knot} + \text{compensating counter-coupling} \\ \longrightarrow \text{neutrally stabilizing field order}.\end{aligned}$$

In this reading, the weak interaction describes the selective channel through which such boundary-stable field configurations can change their coupling role.

Scale-relatively, the same role logic applies on the galactic level. A baryonic structure appears proton-like as long as it is coherent enough to act visibly, luminously, and dynamically as its own coupling unit. If it becomes more diffuse, loses coherence, or if its open coupling is compensated by environmental tension, it assumes a neutron-like function: it stabilizes, buffers, and connects without strongly standing out as an independent visible structure.

Conversely, an initially neutral field organization becomes proton-like when it builds up sufficient coherence, density, and coupling asymmetry. A diffuse tension region, a halo component, a filament zone, a gas-rich transition region, or a non-coupling-capable piece of a spiral arm can, under suitable conditions, tip into a visible baryonic structure: into star formation, gas densification, a satellite knot, or a young dwarf galaxy. A previously neutron-like region becomes proton-like when it gains coherence, density, and visible coupling.

One example is a dwarf galaxy or a gas-rich subknot that enters a spiral arm or a galactic density wave. If it encounters there a region that was previously only weakly coupling-capable, it increases its density, coherence, and degree of organization. The region becomes more visible for our regime and acts more proton-like. Conversely, a formerly visible, proton-like region can recede into a neutron-like stabilization role through dispersion, relaxation, gas loss, or compensation of its open coupling.

Proton and neutron analogy are therefore two roles within the same field mechanics:

$$\text{proton-like} = \text{openly coupling, visibly structuring,}$$

$$\text{neutron-like} = \text{neutrally stabilizing, buffering, connecting.}$$

The transition between the two roles corresponds to a change in coupling asymmetry:

$$\text{neutron-like field order} \rightleftharpoons \text{proton-like field order} + \text{free coupling modes}.$$

On the galactic scale, this is a slow reorganization of visibility, coherence, and field tension. Regions that appear as dark, neutral, or not directly coupling-capable field components pass into visible structures under relief or densification conditions. Visible baryonic structures recede into a neutron-like stabilization role through dispersion, relaxation, or coupling balance.

Proton-like and neutron-like denote changeable
coupling roles: open visible structure on the one hand,
neutral stabilization and buffering on the other.

11.2. Quarks as Internal Intermediate Modes

Within the scale-relative UQSH framework, quarks are not understood as an isolated fundamental substance, but as internal intermediate modes of strongly saturated field knots. A proton-like or neutron-like state therefore does not consist of small solid particles, but of a closed organization of several internal tension directions.

These intermediate modes possess no independent stable external coupling. They become sustainable only within the entire field knot. This is why quarks appear confined in the standard picture: not because they are held fast by an external force as in a cage, but because a single internal tension direction without the remaining overall organization does not form its own stable field form.

Protons and neutrons are therefore complete nuclear field knots. Quarks are their internal orientation, tension, and coupling modes. The proton-like or neutron-like role arises only from the overall balance of these intermediate modes:

$$\begin{aligned}\text{internal intermediate modes} &\rightarrow \text{closed nuclear field organization}\\ &\rightarrow \text{proton-like or neutron-like external role.}\end{aligned}$$

Scale-relatively, this can be transferred to galactic structures. Within a spiral arm, a bulge, or a dense star-forming region, local vortices, shears, gas densifications, magnetic-field directions, and not directly visible tension components can be read as quark-like intermediate modes. By themselves, they do not yet form complete proton-like or neutron-like field knots; only their shared organization generates a visible, coherent coupling form.

Quarks or quark-like modes thus form the internal transition level between diffuse field tension and complete knot formation:

$$\text{diffuse field tension} \rightarrow \text{quark-like intermediate modes} \rightarrow \text{proton-/neutron-like field knots.}$$

11.3. Galactic Warps as Coupling to Filament Motion

Within the UQSH framework, galactic warps can be interpreted as visible traces of a coupling between the intrinsic rotation of a galaxy and the large-scale motion of the Cosmic Web. A galaxy is not only an isolated rotating field knot. It is also embedded in filaments, nodes, voids, and tension paths of the superregime. Its disk therefore carries its own internal rotational order, while the surrounding filament imposes an additional large-scale direction of pull, push, or twisting.

In this view, a warp arises where the galactic intrinsic rotation does not fully coincide with the external field motion of the filament. The inner disk remains stabilized by its own rotation, mass distribution, and coupling, while the outer, more weakly bound regions respond more sensitively to the large-scale tension direction of the web. As a result, especially the boundary regions of the galaxy are bent out of the original disk plane.

The warp structure would therefore not be a purely local disturbance, but a regime transition between two orders of motion: the bound intrinsic motion of the galaxy and the superordinate field motion of the filament. The inner galaxy follows its own rotational order more strongly, while the outer disk, gas components, and diffuse boundary structures couple more strongly to the direction of flight or tension direction of the Cosmic Web.

Within the UQSH framework, it would therefore be expected that warps are not randomly oriented. Their axes and deflections should preferably correlate with the local filament direction, with large-scale velocity fields, or with the direction of motion of the galaxy within the Cosmic Web. Such effects should be especially strong in galaxies whose outer gas and stellar disks are widely extended and only weakly bound, since these regions can be twisted more easily by superordinate field tension.

This yields a testable prediction: galactic warps should statistically be aligned more often along the direction of flight, pull, or push of the surrounding Cosmic Web than would be expected for random orientation. A combination of HI warp maps, galaxy rotation, filament reconstructions, and large-scale peculiar-velocity fields could investigate this coupling.

Galactic warps should not be purely randomly oriented, but should preferably correlate with the local filament direction, the large-scale direction of motion, or the tension axis of the Cosmic Web.

12. Maximum Gravitation, Temperature, and Quasar Activity

On this basis, the question of temperature and quasars can be interpreted more precisely. An Anun denotes the maximally locally effective gravitational depth of a regime, but not a final standstill of field motion. If additional field tension is no longer sustainable as further local deepening, it is transferred into internal tension, inertia, boundary structure, relief, recoil, and regime coupling.

The saturation of an Anun is therefore a progressive limiting process. The more strongly the field is locally loaded, the more strongly its tendency grows not to deepen additional tension further, but to redistribute it at the boundary, along relief axes, and into the environment. The Anun thereby does not become a singularity, but an active redistribution center.

This view connects directly with the interpretation of the temperature of black holes. The effective temperature of an inactive black hole can be understood as a measure of the still open internal mobility of the field. The more strongly an Anun is saturated, the more free dynamics is bound into internal tension organization. The saturated core therefore appears not hotter, but colder: not because energy is absent, but because open oscillation, free reorganization, and thermally readable mobility are increasingly inhibited. This interpretation does not replace the standard basis of the thermal radiation of black holes, which goes back to Hawking's calculation of particle creation at black holes [4].

With increasing size of an Anun, therefore, what grows is not an infinite gravitational depth, but the range of the saturated tension domain. The tension distributes itself over larger boundary zones, coupling surfaces, and transition domains. Maximum gravitation thereby does not become harder in the sense of a singularity, but more far-reaching and field-mechanically softer: the field does not fully yield to the pressure, but translates it into extended saturation zones, relief channels, and reorganization structures.

The largest Anun therefore show especially pronounced boundary and recoil effects: what appears at the center as coldness, inertia, and blocked reorganization can become visible externally as extreme

activity. In this view, quasars can be understood as Anun systems operating close to the maximum load-bearing limit of their regime. The inner saturated core remains extremely cold in the sense of low open reorganization capacity, while the environment appears highly active. The observed heat and radiation do not belong to the internal temperature state of the Anun itself, but to the outer relief and reemission zone: accretion disks, jets, shock fronts, and magnetized plasmas are the regions in which bound or focused field tension is transferred back into running radiation and new matter organization.

A quasar can thereby remain long-lived. It is stabilized by the ongoing coupling between maximally saturated Anun, surrounding field, and relief structures. The system partly sustains itself because it relieves field tension, generates reemission, and at the same time prevents complete relaxation of the saturated region. The possible matter-forming consequences of such relief zones are treated in the section *Matter Formation*.

In this sense, the largest black holes are not simply the "most extreme singularities," but the largest extended zones of maximum gravitation within our regime. Their internal dynamics is strongly bound and cold, while their boundary regions belong to the brightest and most productive reorganization zones of the universe.

Maximum gravitation does not stop dynamics; it binds it internally and enforces relief at the boundary.

13. Uncertainty as Stability and Growth Margin

Within the UQSH framework, uncertainty possesses not only epistemological, but ontological significance. It describes not merely a lack in our measurement, but a necessary margin of the field. A completely sharp, completely determined field state would have no remaining internal degree of freedom. It could not evade, could not relax, could not couple, and could not form a new order. Precisely the remaining uncertainty therefore prevents the rigid final state.

This view connects uncertainty directly with stability. A stable field organization is not stable because it is absolutely fixed. It is stable because it remains open within certain limits. It possesses enough binding to carry form and at the same time enough residual motion to redistribute tension, absorb disturbances, and enter into couplings. Stability therefore does not arise through total determinacy, but through controlled openness.

Uncertainty is thereby connected with the two unreachable limiting states of the UQSH: complete saturation and complete relaxation. Both extremes would be singular because they would deprive the field of any margin for further reorganization. The real working domain of the field lies between them: in limited order, limited uncertainty, and ongoing reorganization capacity.

Uncertainty is thereby also the condition for growth to remain possible. Within the UQSH framework, growth does not merely mean an increase in size, but the capacity of a field state to take up new tensions, expand its boundary structure, form couplings, and integrate into larger organizations. Without uncertainty, there would be no boundary formation, no transition domains, no regime connections, and no formation of new stability.

This applies especially to Anektron and Anun, though in different ways. In the Anektron, uncertainty appears as the necessary openness of the minimal coupling form. It can act stably only if it is not completely point-like, closed, and absolutely determined. Its residual openness enables internal

dynamics, coupling capacity, and interaction.

In the Anun, by contrast, uncertainty does not appear for our observational regime as slight mobility, but as boundary openness of maximum saturation. An Anun appears to us as extremely inert, time-dilated, and almost maximally bound. What remains open is not its local depth, but its boundary structure, range, relief channels, and connection capacity to a higher regime.

From the perspective of a superregime, the same slowed tension motion can become readable differently. What appears in our regime as an extremely dilated Anun can there be temporally contracted and appear as a new minimal, still uncertainly resolved coupling unit. In this way, the Anun of one regime can assume Anektron function in the superregime.

Thus, uncertainty acts at both boundaries of the regime. Below, it protects the first stable coupling form from complete fixation. Above, it prevents singularity and enables redistribution into the next higher regime. Anektron and Anun are therefore not only saturation boundaries, but also uncertainty boundaries: they show that stable field organization always requires a residual openness.

World formation as a whole is also based on this principle. Matter, chemistry, life, and cosmic structure can arise only because field states neither completely flow apart freely nor are completely rigidly fixed. They remain stable enough to carry form and open enough to enable exchange, self-reorganization, and adaptation.

The world is therefore not stable because everything is perfectly determined. It is stable because the field possesses a controlled uncertainty: enough order to carry form and enough openness to enable change, coupling, and future.

Uncertainty is not an error of order, but its margin for stability and growth.

14. Space as Relative Field Order

General relativity successfully describes gravitation as the geometry of spacetime [18]; the UQSH reads this geometry as the expression of carried field tension.

Within the UQSH framework, space is not an empty container in which things are located. It denotes the relational order of the field medium itself: how field states lie with respect to one another, how tensions are distributed, where couplings become possible, and which paths propagating field motion can take. Space is therefore the geometrical reading of the current field organization.

A completely empty space is not presupposed in this view. Where field exists, tension, residual dynamics, coupling capacity, and reorganization possibility exist. Even apparently empty regions such as voids are not nothing, but weakly bound and more strongly relaxed field orders. They possess less baryonic anchoring and lower saturation, but still carry a field structure through which light bubbles can propagate, dedensify, and be modulated.

This spatial order is relative to the respective regime. A structure that appears in our regime as an extended intermediate space between galaxies can be readable in a superregime as an internal modulation zone of a larger field front. Conversely, a field structure that is microscopic for us can represent its own spatial organizational domain in a subregime. Space is thus not an absolute stage, but the scale-relative geometry of field organization.

Space = relational order of coexisting field states.

The spatial extent of a structure therefore describes not only its geometrical size, but also its coupling range. A field knot possesses space by distributing tension, generating bindings, forming boundaries, and incorporating other field states into its organizational form. Spatiality thereby appears as the visible order of these coupling and tension relations.

Space is therefore field-relative within the UQSH framework: it is the way in which the field orders its states with respect to one another and makes them geometrically readable for a regime.

Space is not the container of the field, but the order of its couplings.

15. Speed of Light, Scale-Relative Timing, and Time Dilation

Within the UQSH framework, the speed of light and time are not quantities of the same kind. The speed of light c denotes the maximum reorganization rate of free field motion. It is the limiting motion of the field itself: the most open form in which the field can pass its local order on to neighboring field domains.

c = maximum reorganization rate of free field motion.

Time must be distinguished from this. It is not an independent background quantity and not a second constant alongside c, but the emergent measure of the sequence of field reorganizations. Wherever field states change, couple, relax, bind, or dedensify, a temporal order arises.

Time = order of field reorganization.

Without field change, there would be no physically readable time. Thus, c is more fundamental than the respective time form of a regime. Time can be dilated, slowed, or read scale-relatively; the limiting rate of free field motion remains preserved.

This distinction is especially important for scale relativity. Each regime possesses its own internal timing. A process that appears extremely dilated in our regime can proceed normally in its own regime. Conversely, a process that lasts long for a deeper regime can appear almost instantaneous for a higher regime. This does not mean that the speed of light would be different in the regimes. It means that the temporal readability of field reorganization is scale-relative.

scale-relative time $\neq$ variable speed of light.

Scale-relative time rather means different timing of bound field organization. A local observer always measures light with the same speed, because his matter, his clock, his measuring rods, and the local tension front belong to the same field regime. For him,

c = constant.

An external observer, however, can see that the time form of another regime is extremely dilated. The processes there appear slower, not because c has become smaller, but because the bound field organization performs fewer free reorganization steps per external comparison.

In strongly tensioned or saturated field regions, this difference becomes especially clear. The more strongly field motion is bound, the less freely it can reorganize itself. A larger share of the motion is held in tension, curvature, inertia, or saturation. For external observers, the passage of time there appears slowed. For the affected local observer, however, the observer's own timing remains consistent.

$$\text{stronger field binding} \rightarrow \text{dilated local timing} \rightarrow \text{time dilation.}$$

Time dilation therefore does not arise from a local reduction of the speed of light. It arises from the different reorganization order of bound field states. Light remains the limiting motion of the field. Time is the way in which bound systems count this motion in their own internal order.

At the Anun, this difference reaches its strongest form. Field motion is maximally bound and appears to external observers as extremely slowed or almost frozen. Nevertheless, c is not abolished. The limiting dynamics of the field remains present, but no longer appears as freely outwardly continuable propagation, but as internally bound, inert, and boundary-indeterminate field organization.

$$\text{Anun} \rightarrow \text{maximum field binding} \rightarrow \text{extreme timing dilation} \rightarrow \text{no violation of } c.$$

This yields a clear ordering of the concepts. Space describes the coupling order of the field. Time describes the reorganization order. The speed of light describes the limiting rate of free field motion. Space and time are therefore regime-dependent readings of the field, while c represents the constant limiting dynamics against which these readings become measurable.

$$\text{Space} = \text{coupling order of the field,}$$

$$\text{Time} = \text{reorganization order of the field,}$$

$$c = \text{limiting motion of the field itself.}$$

The constancy of the speed of light and the relativity of time therefore do not contradict one another. They belong together. Precisely because c is the constant limiting rate of free field motion, different regimes can time their bound reorganization differently relative to it. Time thereby appears scale-relative, but not as an independent fundamental substance. It is an emergent measure arising from field motion itself.

The speed of light is the limiting motion of the field; time is the emergent measure of its reorganization.

15.1. Spacetime as Shared Field Readability.

Space and time are thereby not discarded, but interpreted field-mechanically. Space describes the coupling order of coexisting field states; time describes their sequence of reorganization. What is geometrically summarized in relativity theory as spacetime appears in the UQSH as the dynamic readability of the field medium: field states possess spatial extent because they distribute couplings and tensions; they possess temporal duration because they reorganize their order step by step.

16. Time as a local clock field

In the UQSH, time is not understood as an independent background in which field motion takes place. It is the readable order of this field motion itself. A field state does not possess time because it exists inside an external container of time, but because its internal order changes, repeats, binds, relaxes, or reorganizes. Time therefore appears wherever field motion becomes measurable as a sequence of stable states.

Thus, time is more closely connected with bound field organization than with free propagation. A free light bubble carries frequency, direction, energy, and modulation, but it has no proper time in the sense of an internal bound clock process. Proper time arises only where field motion is locally stabilized and can carry its own reorganization as a persistent sequence of states.

free field motion	bound field motion
$\Downarrow$	$\Downarrow$
frequency and propagation	proper time and internal taktung

The fundamental relation is therefore:

$$\text{time} = \text{readable taktung of bound field reorganization.}$$

This taktung is never absolutely identical everywhere. Every field region has its own order of tension, density, motion, binding, and coupling. Even apparently calm regions are only locally averaged as calm. Over a sufficiently stable region, a common temporal order can be defined because many small field differences are practically combined into an average takt. Fundamentally, however, time remains locally field-dependent.

$$\tau_{\text{local}} \approx \langle \tau(\mathbf{x}) \rangle_{\text{region}} .$$

This averaging explains why time appears stable in everyday experience. A laboratory, a planet, or a stellar system possesses a sufficiently common field order, so that clocks within this region run comparably. At a more precise level, however, this equality is only approximate. Every change in gravitation, motion, density, radiation, rotation, saturation, or rarefaction also changes the local order of reorganization.

field tension and binding

$$\Downarrow$$

local capacity for reorganization

$$\Downarrow$$

proper time of bound systems

Time dilation is therefore not a rare special case, but the measurable difference between local field taktungs. In strong gravitational fields or at high velocities this difference becomes especially visible. There, a larger share of field motion is involved in tension, binding, direction, or inertia. Correspondingly, less freedom remains for the internal reorganization of a bound system. To an outside observer, its time appears dilated.

stronger binding or directed motion

$$\Downarrow$$

reduced free internal reorganization

$$\Downarrow$$

dilated proper time

This also gives the relativity of time a field-mechanical reading. Relativity theory describes the measurable geometric side of this difference. The UQSH interprets it as an expression of different field tensions and different capacities for reorganization in bound field states. Time is therefore not first absolute and only later disturbed by mass or motion. From the beginning, it depends locally on the field order in which a system is bound.

This field-dependence also applies in a scale-relative sense. A subregime and a superregime do not simply possess the same time at a different size. They possess their own orders of taktung, because their stable field states, coupling thresholds, and reorganization speeds become readable in different ways. A process that appears in our regime as a short-lived fluctuation may possess an ordered internal development in a subregime. Conversely, a process that unfolds over very long periods for us may appear in a superregime as a compact change of state.

$$\tau_{n-1} \neq \tau_n \neq \tau_{n+1}.$$

This inequality does not mean that every regime has a different speed of light in the ordinary sense. It means that bound reorganization is counted differently. The limiting rate of free field motion remains the local limit of propagation within a regime. What changes is the way bound systems translate this motion into proper time, duration, repetition, and development.

$$\text{regime time} = \text{taktung of bound reorganization in } R_n.$$

An event is therefore not of identical temporal form in every regime. What appears as an event in R_n may appear as an extended structure in R_{n-1}, and as a fine modulation of a larger order in R_{n+1}. Time is thus not only relative between moving observers, but also relative to the coupling and resolution level of the field.

$$\text{event in } R_n \neq \text{same temporal form in } R_{n-1} \neq \text{same temporal form in } R_{n+1}.$$

The UQSH does not thereby introduce an additional time dimension. It describes a regime-depth of time: every stable field order possesses its own readable taktung. Translations may exist between regimes, but there is no absolute clock that measures all levels equally and independently of the field state.

$$\begin{gathered}\text{regime } R_{n-1}\\ \Downarrow\\ \text{own bound taktung}\\ \text{regime } R_n\\ \Downarrow\\ \text{own bound taktung}\\ \text{regime } R_{n+1}\\ \Downarrow\\ \text{own bound taktung}\end{gathered}$$

Time is therefore permanently relative in operation. It appears nearly identical only where a field region is sufficiently stable, homogeneous, and jointly coupled to form an average temporal order. This average time is practically real, but not fundamentally absolute. It is the averaged proper order of a local field region.

Time is not an external stage for field motion.
It is the local taktung of bound field reorganization.
Therefore time is always field-dependent and only in stably averaged regions approximately the same.

17. Proper Time, Inertia, and the Mobility of Bound Field Nodes

17.1. Proper Time as the Internal Clocking of Bound Field Organization

In this work, the speed of light c is not read only as the propagation speed of electromagnetic radiation, but as the causal limiting rate at which a physically effective field state Φ can reorganize. A bound field state therefore cannot change its internal order arbitrarily fast. Its internal dynamics remain tied to motion, local field tension, and gravitational organization.

The relativistic relation

$$d\tau = dt\sqrt{1 - \frac{v^2}{c^2}}$$

remains unchanged. The field interpretation does not replace this relation, but gives it a mechanical reading. If a bound state moves externally, part of its available reorganization capacity is tied into the

directed translation of the entire state. Relative to an external observer, less free reorganization rate remains available for its internal clocking. Its proper time therefore appears slowed.

In this view, proper time is the internal clocking of bound field organization. Time dilation does not mean that a substance-like time medium flows more slowly. Rather, it describes that the internal reorganization rate of a bound state appears relatively reduced under motion or strong field binding [24–26].

17.2. Kinematic and Gravitational Time Dilation

In this reading, kinematic and gravitational time dilation have the same basic structure. In both cases, the free internal clocking of a bound field state is reduced relative to a more weakly bound or external observer.

At high velocity, part of the reorganization capacity is tied into the translation of the whole state. In gravitational time dilation, the cause is not external motion, but the local field organization itself. Near massive or strongly bound objects, the field projection Φ is more strongly tensioned, curved, or organized. A bound state in such a region therefore exists in an environment with less freely available reorganization capacity.

Both cases can therefore be read as variants of the same principle:

$$\text{time dilation} = \text{reduced internal reorganization freedom.}$$

This interpretation does not contradict relativity. It preserves its formal limiting structure and understands time dilation as the observable expression of a limited field-internal reorganization rate [27–29].

17.3. Bound Field Nodes and Inertia

A bound field node is a state whose internal dynamics are not freely distributed through the field, but organized into binding, tension, and self-stabilization. The stronger this binding is, the more of its reorganization capacity is already internally bound, and the greater the resistance to changes in its state of motion becomes.

Inertia thus appears as the resistance of bound field organization against reorganization. A body can carry an already existing motion, but any change of this motion requires a rearrangement of the entire bound tension structure. The deeper the field organization, the smaller the acceleration response to a given external influence.

Schematically:

$$\text{stronger binding} \rightarrow \text{less free reorganization capacity} \rightarrow \text{greater inertia.}$$

For an external force, the classical expression

$$a = \frac{F}{M_{\text{inert}}}$$

remains valid. What changes is the interpretation: M_{inert} is read as a measure of bound field organization. Larger or more deeply organized states therefore respond less strongly to the same non-gravitational influence. This applies, for example, to radiation pressure, gas drag, collisions, jets, accretion, or asymmetric mass loss. The universality of free fall is not abandoned by this interpretation; it continues to hold in the test-body limit [29, 30].

17.4. The Test-Body Limit and Mutual Field Reorganization

The interpretation of inertia used here should not be understood to mean that large bodies generally fall differently from small bodies in a given gravitational field. For local test bodies, the equivalence principle remains intact. A test body is a bound field state whose own organization modifies the external field only negligibly.

However, when the involved organizational masses become comparable, the problem is no longer a simple test-body problem. In that case, one body does not fall through an unchanged external field. Rather, two bound field states reorganize each other. The common field center shifts, and both states move relative to their shared field organization.

Free fall remains locally equivalent, but the global field structure is no longer passive. The transition from test-body motion to mutual field reorganization is therefore not a violation of the equivalence principle, but the natural limit of its test-body approximation [29, 30].

17.5. Anuns and the Mobility of Maximally Bound Field Nodes

A special limiting case is a maximally bound field node, referred to here as an Anun. An Anun is not an ordinary body with large mass, but an extremely organized and self-stabilized field state whose internal reorganization freedom is reduced to a minimum.

This does not mean that such a node would be absolutely immobile. The decisive distinction is between internal reorganization and translation of the entire node:

$$\text{internal reorganization} \neq \text{translation of the entire node.}$$

A maximally bound node can move as a whole structure. Its acceleration, however, is strongly inhibited, because any change in its state of motion requires a reorganization of the entire bound field structure. Its defining feature is therefore not a particular velocity, but its extreme resistance to changes in motion.

For bound field nodes, one may write qualitatively:

$$\frac{dv_{\max}}{dM_{\text{org}}} < 0,$$

where M_{org} denotes the bound organizational mass. This quantity does not simply mean Newtonian mass, but the strength, density, and stability of field binding. Here, $v_{\max}$ is not a new universal light-speed limit, but the effectively sustainable translational velocity of a bound node within a given field regime. The relativistic limit $v < c$ remains unchanged.

A possible phenomenological limiting form is

$$v_{\max}(M_{\mathrm{org}}) = \frac{c}{1 + \left(M_{\mathrm{org}}/M_A\right)^p}, \qquad p > 0.$$

This expression is not proposed as a fundamental law. It only summarizes the principle that increasing bound organization raises inertia and reduces mobility.

This also allows one to distinguish minimal and supermassive Anuns. A minimal Anun would be a maximally bound node on the smallest stable organizational scale of its class. A supermassive Anun, or a supermassive black hole, would correspond to a substantially larger, more far-reaching, more strongly bound, and more inert organizational state. Minimal Anuns could therefore be much more easily accelerated than supermassive Anuns, without reaching the light-speed limit. Supermassive Anuns, by contrast, would appear more as inertia-stable ordering anchors of galactic or cosmic field structure.

17.6. Astrophysical Mobility and Reorganization Traces

The observed velocity of a body is not determined by its organizational mass alone. Formation history, environment, collisions, recoil processes, and field gradients also play a role. For this reason, compact and massive objects can also possess high velocities. Anuns, for example, may be accelerated through many-body interactions or gravitational-wave recoil [35, 36].

The important point here is not absolute rest, but the acceleration response. A strongly bound field node can be in motion, but changing this motion requires exceptionally strong reorganization events. In the field reading, such a node leaves behind a wake of tension, compression, and local reorganization as it passes through gas or a circumgalactic medium. An observed stellar trail would then not represent matter formation from nothing, but the baryonically visible projection of field and gas reorganization triggered by the moving node.

Schematically:

strongly bound field node + motion through a medium
$\rightarrow$ tension and compression wake
$\rightarrow$ local stellar or matter organization.

Such cases would be particularly informative, because they could show that compact field nodes are not merely passive gravitational centers, but can generate organizing wake structures during their motion.

17.7. Galactic Rotation as Directed Field Reorganization

In this view, a galaxy is not merely an assembly of independent mass points, but a baryonically anchored rotation and tension system. Stars, gas, dust, and compact nodes are embedded in a shared field topology. The rotational velocity of a star is therefore not only a local orbital property, but also an expression of the systemic field organization.

Galactic rotation requires the ongoing reorganization of the baryonically anchored field structure in the tangential direction. This directed reorganization capacity is not unlimited. It is determined by

baryonic organization, local coupling, saturation, and nonlocal redistribution of the field.

The flattened rotational velocity then appears as a limiting form of collective field reorganization:

$$v_{\mathrm{rot}} \leq v_{\mathrm{rot,max}} \left(O_{\mathrm{bar}}, \Sigma_{\mathrm{local}}, M_{\mathrm{org}}, S_{\mathrm{sat}} \right) .$$

Here, $v_{\mathrm{rot,max}}$ is not a universal constant, but a system-dependent limit of the rotational reorganization capacity of the respective galactic field state. An individual star may temporarily deviate from this through local perturbations or its formation history. Without sufficient external acceleration and without a genuine escape condition, however, it remains embedded in the field context of the galaxy.

Galactic saturation therefore does not mean that the total mass of a galaxy reaches a simple global limit. It concerns the reorganization capacity of the baryonically anchored field structure in a specific dynamical direction. For rotation, this is the tangential, rotation-aligned reorganization. The flat rotation curve can thus be read as the macroscopic signature of a limited but continuously maintained field reorganization [1, 31, 32].

17.8. Satellite and Dwarf Galaxies as Partly Integrated Field Populations

In this view, dwarf galaxies and satellite systems are baryonically visible sub-organizations within a larger field context. Their motion and internal dynamics should therefore not be treated as fully independent of the host galaxy.

As long as a dwarf galaxy has not yet been fully integrated into the field regime of the host galaxy, it can retain its own relative motion and internal organization. Its baryonic mass, gas content, internal dynamics, and tidal disturbance then reflect both its formation history and its degree of embedding in the larger field topology.

This does not imply that all satellites should have the same velocity. Rather, one would expect a structured distribution of relative velocities, masses, and integration stages. Recently embedded satellites may show high relative velocities. Partly integrated systems should have intermediate relative motion. Strongly integrated satellites should be more tightly coupled to the host galaxy's motion regime.

This is empirically testable. For satellite systems of the Milky Way, Andromeda, or other nearby host galaxies, baryonic mass, distance, relative velocity, gas content, stellar populations, and tidal signatures can be studied together. A systematic coupling between population mass, degree of integration, and relative velocity would support this reading. A completely random distribution of these quantities with no recognizable relation to the host galaxy's field organization would be critical [37–40].

17.9. Macroscopic Saturation and Integrated Inhibition

Saturation is not a standstill of field dynamics. It describes the transition in which more and more reorganization capacity becomes bound into binding, stabilization, and self-coupling. The more deeply organized a field state is, the more strongly its inhibition is already integrated into the structure itself.

This becomes especially important on macroscopic scales. A planet, a star, or an Anun is not simply a larger version of a freely moving field packet. Through its depth of reorganization, such a state already contains an internal counter-coupling. This prevents additional field tension or external influence from

being converted directly into excessive acceleration of the overall system. Motion remains possible, but it is filtered, distributed, and inhibited by the bound organization.

In magnetized astrophysical systems, magnetic field components may also take on a direction-setting role. They do not impose rigid order in a mechanical sense, but they structure plasma, charge transport, jets, accretion flows, and rotation axes. As a result, the available reorganization is not distributed arbitrarily and isotropically, but is ordered along preferred directions. Magnetic coupling therefore acts as a macroscopic ordering mechanism within already bound field systems.

Saturation thus does not mean that a field state becomes passive. It means that every further change of its state requires more internal reorganization work and is more strongly channeled by the existing order. In weakly bound states, additional tension can still be absorbed or relaxed relatively freely. In strongly bound states, it is increasingly converted into redistribution, counter-coupling, directional ordering, or large-scale reorganization.

The transition sequence is therefore:

$$\text{free field excitation} \rightarrow \text{bound field node} \rightarrow \text{macroscopically saturated state} \rightarrow \text{Anun}.$$

With increasing saturation, the required reorganization work grows:

$$S_{\text{org}} \uparrow \quad \Rightarrow \quad \Delta\mathcal{R}_{\text{required}} \uparrow .$$

Here, S_{org} denotes the organizational or saturation degree of the field state, and $\Delta\mathcal{R}_{\text{required}}$ denotes the reorganization work required for a further change of state.

Macroscopic saturation thus connects inertia, time dilation, inhibited acceleration, magnetically directed order, and large-scale field structure. It is not the end of dynamics, but the form in which dynamics in deeply bound systems is stabilized and ordered directionally.

The considerations regarding macroscopic saturation discussed so far concern not only the inertia and mobility of bound field nodes. They also lead to two further consequences: First, saturation must exhibit observable traces of ordering and stress relief at every scale. Second, maximum saturation in a given regime does not signify the end of field dynamics, but rather a limit to local readability.

17.10. Scale-Relative Saturation and Magnetic Field Order

Within the UQSH framework, saturation is not restricted to microscopic or compact objects. If the same field logic operates scale-relatively, saturation must also appear on macroscopic and cosmic scales. There, however, it does not necessarily appear as a particle state or as a local boundary node, but as a large-scale field configuration.

In general, saturation means that a field region can no longer absorb additional motion, tension, or densification indefinitely in the same local form. The field then responds through redistribution, directional ordering, boundary formation, relief, reemission, plasma ordering, or transition into a larger organizational form. Saturation is therefore not the end of dynamics, but a change in the way dynamics couples.

On the atomic scale, such saturation appears as stable bound field organization. On the stellar scale, it appears as a balance between gravitation, pressure, radiation, rotation, and magnetic order. On

the galactic scale, it appears as a limited rotational and halo response, as baryonically anchored field tension, or as saturation of the additional dynamical acceleration. On cluster and filament scales, the same logic appears in still larger field configurations: in plasma distribution, shock fronts, large-scale magnetization, lensing structure, gas displacement, boundary zones, and relief axes.

Magnetic order is especially important here. In this work, magnetism is not understood as an independent substance, but as a directed torsional and coupling order of the field motion $B(\Phi)$. Where plasma, rotation, shear, radiation, and baryonic density act together, this torsional order becomes macroscopically coherent. A large-scale magnetic field is then not merely a by-product of matter, but a visible organizational form of the tensioned field.

In a strongly filled or strongly organized region, such as a galaxy cluster, a filament node, or a dense cluster environment, field tension reaches a boundary range in which additional densification is no longer simply absorbed locally. The consequence is not an absolute blockade, but a field-mechanical counterreaction: additional tension is redistributed laterally, redirected into turbulence, shear, magnetization, or relief structures, or it produces boundary regions in which gas and plasma are organized differently.

In this sense, a macroscopically saturated field configuration acts repulsively without requiring a new fundamental repulsive force. The repulsion is not the action of a new substance, but the response of an already strongly tensioned field region to further compression. A saturated region does not absorb additional field motion arbitrarily; it redirects it, distributes it, or forces it into other coupling forms.

Magnetic fields are a preferred visible trace of this redistribution, because they indicate direction, torsion, and tension of the field. Especially in plasmas, magnetic orders generate pressure, guiding paths, and boundary surfaces. They channel gas motions, structure shock fronts, stabilize jets, guide accretion flows, and prevent diffuse matter from densifying in an arbitrarily isotropic way. In the UQSH framework, such effects are macroscopic readings of the same saturation logic: the field does not become indefinitely denser, but organizes its tension spatially and directionally.

Saturation thereby obtains an observable side. A saturated macroscopic field region is not characterized only by high density or strong gravitation, but by a combination of several signatures:

$$\text{saturation}_{\text{macro}} \Rightarrow \begin{cases} \text{high field or plasma density,} \\ \text{coherent magnetic order,} \\ \text{boundary and shock structures,} \\ \text{inhibited further densification,} \\ \text{relief axes or outflows,} \\ \text{nonlinear lensing or rotational response.} \end{cases}$$

These signatures do not always occur together. A system can be dynamically young, turbulent, disturbed, or only partially coupled. The UQSH therefore does not expect a simple equation such as "more magnetic field equals saturation." What matters is the combination of density, coupling, directional order, inhibition, and relief. In this view, magnetism is a particularly important indicator, but not saturation itself.

Formally, this can be written as a macroscopic saturation factor:

$$S_{\text{macro}} = S_{\text{macro}}\left(\rho_{\text{bar}}, \rho_{\text{plasma}}, \mathcal{B}_{\text{coh}}, \Omega, \nabla v, K_{\text{field}}\right).$$

Here, ρ_{bar} denotes the baryonic density, ρ_{plasma} the plasma density, $\mathcal{B}_{\text{coh}}$ the coherent magnetic order, Ω the rotation, ∇v shear or velocity gradients, and K_{field} the already existing field curvature or field tension. The saturation degree does not arise from any one of these contributions alone, but from their joint organization.

A simple qualitative form is:

$$S_{\text{macro}} \uparrow \quad \Longleftrightarrow \quad \text{density, coupling, magnetic order, and field tension become coherent together.}$$

As S_{macro} increases, the system's ability to absorb additional tension locally decreases. Redistribution, boundary formation, relief, and directional order become more important instead:

$$S_{\text{macro}} \uparrow \Rightarrow \Delta\mathcal{R}_{\text{local}} \downarrow, \qquad \Delta\mathcal{R}_{\text{boundary/relief}} \uparrow .$$

Here, $\Delta\mathcal{R}_{\text{local}}$ denotes the free local reorganization capacity, while $\Delta\mathcal{R}_{\text{boundary/relief}}$ denotes the tendency toward redistribution at boundaries, axes, or larger field regions.

This clarifies the magnetic conjecture: large-scale magnetic fields are not automatically the cause of saturation, but they are one of the most important observable forms in which macroscopic saturation becomes visible. Where a filament, cluster, or galactic node no longer densifies its field tension further in a local way, this appears as directional order, plasma guidance, boundary formation, or relief structures.

Magnetic order is not saturation itself, but a visible trace of directed macroscopic field saturation.

17.11. Regime-Bound Saturation and Continued Boundary Dynamics

The scale-relative logic of saturation leads to a necessary distinction. A field state is maximally saturated within a regime without the field dynamics as such being ended. Saturation denotes the boundary of local deepening and readability of a field state, not the cancellation of all internal or boundary-near reorganization.

This resolves the apparent tension between maximal saturation and continued activity. What appears to a macroscopic observational regime as a fully saturated boundary state remains dynamic at its edges, in its microscale fine structure, and in adjacent regime readings. Saturation is therefore not identical with field rest. It means that additional field motion is no longer absorbed as further local deepening of the same regime form.

For an Anun, this distinction is central. An Anun is maximally bound within its regime and cannot be locally deepened further. Precisely for this reason, additional field motion no longer appears as deepening, but as boundary structure, inertia, horizon formation, relief, weak emission, internal fine structure, or regime connection. The maximal saturation of the Anun therefore describes the boundary of a macroscopic reading, not the end of scale-transcending field motion.

In this view, an Anun is not a dead endpoint, but a boundary node. It closes the local deepening capacity of a regime, while field motion at the boundary of this saturation shifts into other forms.

These shifted forms are only indirectly visible to the original regime. They no longer appear as ordinary free motion, but as weak relief, thermal boundary dynamics, quantum fluctuation, boundary tension, or transition into another coupling level.

Hawking radiation therefore receives a special meaning. Within the UQSH framework, it shows that a maximally saturated macroscopic field node is not absolutely devoid of dynamics. Even if the local depth of a black hole appears saturated for our regime, a microscale side of relief and reorganization remains present at the boundary surface. Hawking-like emission therefore does not contradict saturation, but shows precisely that saturation does not mean absolute field rest.

$$\text{macroscopic saturation} \neq \text{absolute field rest.}$$

More precisely,

$$\boxed{\text{saturation}^{(n)} = \text{boundary of local readability in } R_n, \quad \text{not the boundary of field dynamics itself.}}$$

This makes it understandable why the same saturation logic has different forms of appearance on different scales. In the microscopic domain, saturation appears as a stable coupling portion, a bound field mode, or quantized transition capability. In the macroscopic domain, it appears as inertia, horizon formation, magnetic order, inhibited densification, jet formation, boundary structure, or large-scale relief. The forms differ, but their function is the same: a field region does not absorb additional motion locally and indefinitely in the same way.

This view also explains why a maximally saturated macroscopic state remains microscopically active. Saturation concerns the macroscopic absorption capacity, not every fine structure of the field. What appears on the large scale as a boundary state continues, on the smaller scale, to consist of reorganization, rhythm, and relief processes. The macroscopic form is saturated; its boundary and fine structure remain dynamic.

Schematically,

$$\text{saturation in } R_n \Rightarrow \begin{cases} \text{local boundary binding for } R_n, \\ \text{continued fine structure in } R_{n-1}, \\ \text{coupling form for } R_{n+1}. \end{cases}$$

This threefold reading is especially important for the Anun. For its own regime, it is the maximal saturation form. From the perspective of deeper fine structures, it is a region of extremely dense reorganization. From the perspective of a higher regime, the same total form appears, once it becomes sufficiently coherent, as a new minimal coupling unit. The Anun function of one regime is thereby connected with the Anektron function of the next higher regime.

$$\text{Anun}^{(n)} \Rightarrow [\text{macrosaturation in } R_n + \text{boundary dynamics in } R_{n-1} + \text{Anektron function in } R_{n+1}] \,.$$

This structure presupposes a scale-open field reality. If field organization were absolutely closed in

extent or depth, a final saturation would arise. Such a final saturation would lead either to complete singularity or to complete standstill. Both contradict the UQSH ground assumption of continued reorganization.

The UQSH therefore does not require a simple spatial infinity in the sense of a merely ever larger outer space. What is decisive is scale-relative openness: field organization expands, differentiates, dedensifies, densifies, and passes into other coupling forms under local saturation. The field does not end where a regime loses its local readability. It changes its form of appearance.

saturation is locally finite, but not field-logically final.

Or, formulated differently,

each saturation limits a reading of the field, not the field itself.

The UQSH thereby obtains a consistent connection between microscopic and macroscopic saturation. A quantum, a bound field mode, an atom, a star, a galactic node, a cluster, or an Anun do not appear similar because they are externally identical. They appear similar because the same functional logic recurs in them: absorption, binding, boundary, redistribution, and transition.

The observable traces of saturation therefore depend on scale. In small bound systems, they appear as discrete transitions, stability, and coupling thresholds. In large systems, they appear as pressure, magnetization, boundary formation, shock fronts, jets, rotational limits, lensing structures, or inhibited further densification. These differences are not opposites, but different regime readings of the same field logic.

In this order, an Anun is the clearest limiting case. It is maximally saturated for its regime, but not maximally saturated for the field as a whole. Its macroscopic inertia, its horizon formation, its Hawking-like boundary dynamics, and its role as a regime transition are therefore not separate properties, but different expressions of the same scale-relative saturation.

An Anun is maximally saturated for its regime, but not maximally saturated for the field as a whole.

18. Entanglement and the Continuity of the Field Medium

From the UQSH perspective, the so-called spooky action at a distance of entangled states indicates that space should not be understood as an absolute stage. If two entangled states show shared correlations across a large spatial distance, this does not have to be interpreted as signal motion through space. It may rather indicate that both states remain jointly anchored in a deeper field organization.

In the standard picture, entanglement allows no usable transmission of classical information faster than light. This restriction is also preserved within the UQSH framework. The interpretation, however, shifts: it is not a signal that moves instantaneously from one place to another, but spatially separated measurement events appear as local readings of a shared deeper field order. Distance belongs to the spacetime reading of our regime, while the underlying field relation does not necessarily have to be organized in the same spatial form.

Space thereby becomes relative to coupling. For our observational regime, the field appears spatially extended, temporally ordered, and causally limited by the speed of light. At a deeper level of the Qu-foam field medium, however, the same organization may act pre-spatially or with less spatiality: not in the sense that it violates space, but that space is first brought forth as a regime-dependent form of appearance.

Within the UQSH framework, it follows from this that no point in space and time exists completely isolated. What appears separated, delayed, or connected only through light paths for our observational regime remains, at the level of the underlying field medium, part of a coherent organization. This connection does not allow arbitrarily fast classical information transfer; it only means that field states are not ontologically cut off from one another in their possibility of coupling.

The apparent separation of places arises through the spacetime reading of our regime. The deeper field order remains continuous. Every place is therefore both local expression and part of the whole: locally limited in its observable coupling, but not separated from the rest of the field.

Every point is locally distinguishable, but not separated from the field whole.

19. Observer, Temporal Linearity, and Scale-Relative Reality

The previous sections described space, time, proper time, and entanglement as readings of the physically effective field reality Φ. The extension developed here assumes the already introduced basic UQSH order

$$Q \rightarrow \Phi \rightarrow B(\Phi)$$

and applies it to the question of why an observer experiences reality as local, causal, and temporally linear. The distinction between the antecedent Qu-ground Q, the physically readable field reality Φ, and its motion $B(\Phi)$ is therefore not derived again here, but used as the foundation for the following interpretation of observation.

The decisive point is no longer only that time appears as the local rhythm of bound field reorganization. In addition, every such rhythm becomes resolvable only for a particular coupling point. An observer does not experience the totality of possible field states, but only that sequence of state changes to which its own structure can couple in a scale-relative way. From this limitation arises the impression of a linear reality.

The question of the observer touches a fundamental point in the UQSH. A field reality may contain a rich range of possible state forms; it becomes physically experienceable only where local coupling takes place. An observer is therefore not merely an external spectator looking at an already fully completed world. It is a coupling point within Φ, at which certain field states become distinguishable, measurable, and readable as reality.

In this context, the term observer need not be restricted to human consciousness. An atom, a molecule, a detector, a biological sense organ, or a technical measuring system can also act as a coupling point. The difference lies in the degree of internal organization. A simple detector registers a limited state change. A biological nervous system integrates many such couplings into a stable order of experience. Consciousness would, in this reading, be a particularly complex and self-referential form of such field

coupling.

This also explains why experienced reality cannot be the totality of all possible field states. Every coupling has a limit. An observer can resolve only those differences that are connectable within its regime, its material structure, its internal rhythm, and its stability. What does not couple is not part of this observer's experienceable reality, even if it is not excluded within the more comprehensive field possibility.

This limitation can be described formally by a scale-relative distinguishability function,

$$\mathcal{D}_O(\ell) = \frac{dN^{(O)}_{\text{resolvable}}}{d\ln \ell}.$$

Here, O denotes the observer or coupling point, ℓ the scale under consideration, and $N^{(O)}_{\text{resolvable}}$ the number of field state changes distinguishable for this observer. The function $\mathcal{D}_O(\ell)$ therefore does not measure the totality of the field, nor the depth of the Qu-ground. It describes only how many differences within Φ a particular observer can actually resolve on a particular scale.

It is precisely this finiteness of resolution that generates the stable experience of a world. An observer with unlimited resolution would not experience a normal reality, but every possible state branching, every scale shift, and every regime coupling at once. Such total access would have no unambiguous present, no clear direction of action, and no simple identity. Experience therefore presupposes limitation. Reality appears to an observer not because it grasps everything, but because it couples only to an ordered and connectable subset.

From this limited coupling arises the impression of linear time. The observer does not experience all possible field states at once. It couples to a sequence of states that must be compatible with one another. The next state must not be arbitrary, but must connect to the internal structure of the preceding state. Only in this way can memory, identity, causality, and action appear as a stable order.

Formally, this connection condition can be written schematically as

$$O_{n+1} \sim C(O_n, \Phi),$$

where O_n denotes the observer-related coupling state at step n, and C denotes the compatibility condition for the next possible coupling. The next experienced state is not a movement along an absolutely stored time axis, but a connectable continuation of the previous coupling order.

Linearity is therefore not simply an illusion. It is real as a form of experience for the observer. But it is not necessarily the fundamental structure of the field itself. Linearity lies in the limited connectivity of the observing system. The observer experiences an ordered sequence because its own coupling can stabilize only a particular chain of compatible states.

Memory also receives a different meaning in this framework. Memory is not access to a past that is stored somewhere as a complete external archive. Memory is a present field organization of the observer, in which earlier coupling orders are preserved as internal structure. The past therefore continues to act within the present state, but not necessarily as a separate place on an objective timeline.

The future is correspondingly not an already completed path that only has to be traversed. It is the set of connectable possibilities that can emerge from the present coupling state. For the observer, only one of these possibilities becomes the experienced continuation. This is not because other possibilities

would be logically impossible, but because they cannot be resolved and lived simultaneously by this observer within this coupling path.

This leads to an important reinterpretation of the concept of reality. Reality for an observer is not the totality of all field possibilities, but the stable, scale-relatively coupled order that the observer can actually resolve. The observer does not generate the field. But its coupling structure determines which subset of the field appears to it as world.

Consciousness can therefore be described as a stabilized coupling path. A conscious moment is not simply an identical point that is pushed forward along an external timeline. It is a concrete self-coupling of the field, which continues in the next moment as the same identity only if the next state is sufficiently compatible. The self thus appears as continuity, although in field-mechanical terms it is a sequence of connectable coupling states.

In this view, every observed moment exists only once. It is not continued as the same moment, but replaced by a new compatible state. What persists is not the moment itself, but the connection order between moments. It is precisely this connection order that generates the impression of a personal timeline.

This also makes clear why a classical linear time travel is not meaningful within the UQSH framework. Such a time travel would presuppose that past and future are stored as stations on a common, modifiable axis. If time, however, is the observer-bound order of compatible field couplings, then there is no single external timeline along which one could simply travel back and alter one's own cause.

What would be conceivable instead is a state transition. An observer could theoretically couple to another locally observable field configuration that corresponds to an earlier or alternative order. This would not be a journey along the same timeline, but a change of coupling within the field reality Φ. The observer would not alter its own past, but enter or read another possibility of field organization.

In such a case, no classical paradox would arise. If an observer encountered a version of itself, this would not be an encounter with its own earlier state on the same linear time axis. It would be the encounter of two different coupling configurations within a more comprehensive field possibility. The original connection chain of the observer would not be retroactively erased, because it is not assumed to be the only modifiable timeline.

This interpretation fundamentally shifts the meaning of time travel. What is usually understood as a journey into the past would, in the UQSH framework, rather be a field-state displacement or a change of observer-related coupling. The past is then not a place to which one returns, but a particular order of field states to which a coupling could connect under special conditions.

The central statement is therefore:

$$\text{time} = \text{order of limited field coupling},$$

not

$$\text{time} = \text{absolute storage axis of reality}.$$

Linearity is the perspective of a limited observer on a nonlinearly structured field possibility.

This does not devalue the experienced world. It is not a mere illusion. It is the real, locally stable appearance of that field order to which an observer can couple. But it is not necessarily the totality of what is possible in the Qu-ground. The physical reality of the observer is scale-relative, finitely resolvable, and dependent on connection. Precisely for this reason it appears as space, time, causality, and history.

This interpretation of the observer is relevant not only for consciousness and time. It also determines how large-scale structures are read. A galaxy does not appear to us as a complete field reality, but as that spectral, dynamical, and gravitational coupling layer which our regime can resolve from its more comprehensive field organization. Thus the question of the observer leads directly to the question of how galaxies appear as spectral field nodes.

20. Galaxies as Spectral Field Knots

The previous sections have shown that bound field organizations appear not only locally as matter, radiation, or reemission, but also become readable through their spectral and gravitational effect. The same logic can now be transferred to larger scales. If atoms, stars, or plasma zones can be understood as field knots with their own tension and modulation order, then this also applies to galaxies within the UQSH framework.

In the usual picture, galaxies are systems of stars, gas, dust, plasma, magnetic fields, and a dark-matter halo. The UQSH supplements this view: galaxies additionally appear as large-scale field knots with their own tension organization. Their electromagnetic spectra thereby show not the entire field reality, but an observable layer of this organization.

Stellar populations, gas clouds, dust, star-forming zones, central cores, supernovae, pulsars, magnetars, jets, and active black holes jointly contribute to the total radiation and to the gravitational field response of a galaxy. The galaxy as a whole can therefore be regarded as a spectrally modulating field knot: not as a single luminous object, but as a superposed field organization whose internal states express themselves in light, dynamics, and gravitational effect.

The analogy to atoms is structural. Atoms possess characteristic spectral lines because their internal bound states allow only certain resonances. Galaxies could, analogously in the superregime, possess characteristic organizational signatures. These, however, would not consist of individual atomic lines, but of the coarse form of their spectral, dynamic, and gravitational field organization.

The following table summarizes this analogy as a preliminary UQSH classification. It orders galactic systems not according to new particle types, but according to their coarse modulation, reprocessing, and activity character. The classification therefore does not claim a chemical periodic table of galaxies. It serves as an analogy for an ordering principle: stable spectral signature, internal organization, coupling capacity, and reaction behavior can be comparable across scales without the objects being physically equated.

UQSH class	Galactic field knot	Modulation character
G-I	Elliptical galaxies / bulges	smooth, old, broadly organized spectra; stabilized ground-state knots
G-II	S0 / early spirals	medium modulation; transition structure
G-III	Sb–Sc spirals	high local fine structure; active modulation knots
G-IV	Starburst galaxies	strong line and reprocessing imprint; excited knots
G-V	dust-rich IR galaxies	global spectral redistribution; obscured reprocessing
G-VI	AGN / Seyfert	core-dominated high-energy imprint
G-VII	Quasars	extreme central activity; overcritical field knots
G-VIII	supernova-dominated systems	impulse-like tension discharge
G-IX	merging galaxies	unstable, strongly superposed reorganization

Table 1: Preliminary UQSH classification of galactic field knots as a scale-relative analogy to spectral organizational classes.

20.1. What Was Measured in the Tests

The previous tests within the framework of this work and the preparatory analyses do not measure the complete field configuration of a galaxy. Such a configuration is currently not directly observable. Instead, projections within our coupling regime were measured: rotation curves, baryonic structural quantities, photometric colors, UV and IR proxies, as well as galaxy spectral templates. These quantities are not the field organization itself, but its readable traces in dynamics, light distribution, and spectrum.

The data used were not generated as new observational data, but derived from publicly accessible astronomical datasets and catalogs. For the rotation-curve analysis, SPARC-based quantities in particular were used; supplementary photometric and spectral comparisons were carried out using available WISE, GALEX, SDSS, and template data. The evaluation essentially consisted of the calculation of derived quantities, the normalization of profiles, the comparison of dynamic regimes, and the testing of simple correlations with observable proxy parameters.

A complete provision of all intermediate tables, test runs, and derived auxiliary files is deliberately omitted in this work in order to limit the scope of the paper. The central statements do not depend on proprietary or hard-to-access data. They can be checked from the named public datasets through an independent reproduction of the described processing steps. The results presented here are therefore to be understood as a methodological and conceptual summary, not as a substitute for a later complete data and code publication.

It must therefore be clearly distinguished: the electromagnetic spectrum is not itself the postulated superregime spectrum. It is the observable modulation layer of a galactic field knot in our regime. The dynamic field response, by contrast, becomes visible indirectly through rotation curves, D_{DM}, lensing, or other gravitational observables. Only the combination of both levels approaches the actual field organization.

20.2. SPARC, q, and Baryonic Organization

An important test concerns the dynamic switching parameter q, which was obtained from rotation-curve analyses. In earlier UQSH works, q was interpreted as a dynamically activated transition scale. The central question was whether q is merely a fit artifact or whether this parameter is connected with real properties of the galaxies.

The comparison with SPARC properties shows that q is not simply determined by morphology, distance, or disk inclination. Instead, systematic differences appear in structural quantities such as disk surface brightness, baryonic compactness, gas-fraction proxies, 3.6-μm luminosity, and D_{DM}. This difference becomes especially clear between the peak regime and the diffuse regime.

The peak regime shows, in the median, a higher dynamic excess and higher gas-fraction proxies. The diffuse q-regime, by contrast, is baryonically more compact, more luminous, and poorer in gas. The conceptual distinction is important: *diffuse* here does not describe the visible baryonic distribution, but the form of the additional dynamic field response. A baryonically compact galaxy can therefore possess a diffuse q-response.

Within the UQSH framework, this suggests that the dark-matter effect depends not only on the total mass, but on the way in which baryonic matter organizes the field medium. q then describes not a simple morphological type, but a dynamic form of reorganization.

The rotation-curve basis used is oriented toward the SPARC database [1]; the comparison with baryonically expected dynamics is also closely related to the Radial Acceleration Relation [2].

20.3. WISE, GALEX, and SDSS

The tests with WISE, GALEX, and SDSS proxies show a different picture. Simple broadband colors do not robustly separate the q-regimes. WISE colors such as $W1 - W2$, $W2 - W3$, and $W3 - W4$ show no strong correlation with q. GALEX NUV likewise shows no significant relationship. Available SDSS colors are also not sufficient to clearly distinguish the dynamic regimes.

This is not a failure, but an important limitation. q is apparently not a simple color, dust, or star-formation indicator. The dynamic field response is more strongly connected with spatial baryonic organization than with global broadband colors. For a deeper spectral test, integrated spectra, line ratios, D_{4000}, star-formation rates, metallicities, and spatially resolved spectral data would be required.

20.4. Multiscale Galaxy Spectral Templates

The template tests were considerably more interesting. Here, not only individual colors were considered, but continuous galaxy spectra. The spectra were brought onto a common wavelength range, normalized, and smoothed on several scales. This made it possible to test whether the spectral identity of a galaxy depends only on narrow lines or whether a coarse large-scale spectral form is also preserved.

The central result is: the template classes remain distinguishable even after strong smoothing. Elliptical and S0 templates remain red-weighted and smooth. Spiral templates form an intermediate class. Starburst templates remain bluer, activated, and more strongly modulated systems. The sequence

$$\text{elliptical/S0} \rightarrow \text{spiral} \rightarrow \text{starburst}$$

remains visible even when a large part of the fine structure is suppressed.

For the UQSH, this finding is important. A possible superregime spectrum of a galaxy would not have to resolve the atomic lines of our regime. It could rather read the broad field-tension organization of the entire galactic knot. The smoothing analysis shows that such a coarse organizational form is indeed preserved in the electromagnetic templates.

The data basis of this template test was small and is not sufficient for a definitive galaxy classification. Nevertheless, the test is methodologically useful. It shows that galactic spectral identity does not lie only in narrow lines, but also in robust large-scale forms. Precisely such forms could serve as proxies for a scale-relative field-knot signature.

21. Matter Formation

Within the UQSH framework, matter formation denotes the transition of free light bubbles into bound, stable, and regime-internally readable field organization. Matter does not arise from a predefined substance, but from the densification, coupling, and geometric closure of field motion within the physically readable field projection Φ.

$$\begin{aligned}\text{free light bubble} &\to \text{densification}\\ &\to \text{selective coupling}\\ &\to \text{geometric closure}\\ &\to \text{bound field organization.}\end{aligned}$$

In the previous UQSH works, matter was therefore understood as stabilized field motion. A free light bubble can propagate, interfere, be modulated, carry energy, and relax again. It becomes matter-capable only when, under suitable local conditions, it does not simply continue onward or become reemitted, but passes into a closed, sustainable, and stable coupling form.

$$M = B_{\text{bound}}(\Phi).$$

More precisely, matter is bound field motion with stabilized tension order:

$$M = B_{\text{bound}}(\Phi) + K_{\text{stabilized}}(B(\Phi)).$$

This binding does not arise arbitrarily. Matter formation presupposes a selective coupling geometry. Not every radiation, not every densification, and not every field tension leads to matter. What is required is the coincidence of specific prerequisites: phase, energy, orientation, geometry, local field tension, coupling direction, and carrying capacity of the surrounding field region must fit together. Only then can free field tension pass into a stable bound organization.

$$\text{Matter formation} \Rightarrow \{\text{phase}, \text{energy}, \text{orientation}, \text{geometry}, \text{coupling direction}, \text{carrying capacity}\}\,.$$

Matter formation is thus distinct from mere reemission. In reemission, absorbed field motion is

released again as free radiation. In matter formation, by contrast, part of the field motion is densified, coupled, and closed in such a way that an independent bound field knot arises.

Reemission = relief of absorbed field motion,

Matter formation = closure and binding of densified field motion.

The most elementary expression of this transition is the formation of electron-positron pairs. Within the UQSH framework, this corresponds to the first stable splitting of free, highly densified field tension into two complementary matter-capable and regime-readable counter-modes. Electron and positron thereby do not appear as particles created from nothing, but as oppositely coupled stable field forms that can arise from a sufficiently dense tension state.

$$B^{\gamma}_{\text{free}}(\Phi) \rightarrow B^{(+)}_{\text{bound}}(\Phi) + B^{(-)}_{\text{bound}}(\Phi).$$

High-frequency radiation has a special significance here. Gamma radiation is not simply radiation of high energy, but an extremely densely timed free field motion. It lies close to that domain in which free tension fronts can reach local saturation conditions. As a result, it can play a special role both in matter formation and in the unbinding of matter.

In matter formation, gamma radiation can act as a field-saturating impulse. Its high timing density, its short reorganization scale, and its strong local tension content can drive a field front into the domain of geometric closure. If, under suitable geometry, it encounters other radiation modes, plasma fields, magnetized zones, or already bound field structures, a first matter-capable coupling form can arise from free field motion.

$$B^{\gamma}_{\text{free}}(\Phi) \rightarrow \text{densification} \rightarrow \text{geometric closure} \rightarrow B_{\text{bound}}(\Phi).$$

Conversely, gamma radiation appears during the unbinding of bound field organization as a high-frequency relief form. When complementary field modes mutually cancel their bound organization, the field motion itself does not disappear. The binding is dissolved, and the motion and tension content emerges as free high-frequency radiation.

$$B^{(+)}_{\text{bound}}(\Phi) + B^{(-)}_{\text{bound}}(\Phi) \rightarrow B^{\gamma}_{\text{free}}(\Phi).$$

This explains why gamma radiation is conspicuous in connection with matter formation and matter unbinding. It marks a boundary domain between free and bound field organization: densely timed enough to favor local saturation and closure, and at the same time free enough to emerge as relief radiation from bound field tension.

Matter formation is therefore not exclusively bound to maximally saturated Anun states. It can, in principle, occur in all sufficiently tensioned field regions: in red dwarfs, normal stars, hot stars, accretion zones, neutron stars, magnetars, jets, quasar environments, and in the relief fields of an Anun. What is decisive is not the absolute size of the field knot alone, but whether field-saturating radiation selectively encounters other radiation modes or bound field structures.

The difference between ordinarily tensioned fields and maximally saturated Anun environments lies not in the principle, but in the intensity, frequency, and spatial extent of the conditions. In less extreme stellar or plasma fields, such processes can occur rarely, locally, or briefly. In Anun, quasar, or jet environments, by contrast, the radiation densities, frequencies, tension gradients, and coupling geometries are so strong that pair formation and subsequent matter organization are markedly favored.

Anun states are therefore not the only locations of matter formation, but the strongest natural amplifiers of this process. Their maximum field saturation generates relief zones in which high-frequency radiation, strong magnetization, plasma, shock fronts, and dense coupling fields interact. Quasar environments are therefore especially suitable spaces in which pair formation and early matter organization can occur more frequently, more densely, and across larger spatial and temporal domains.

Strongly saturated field knots such as neutron stars and magnetars also belong in this context. They are not Anun states in the strict sense, but they possess extreme field tensions, strong magnetization, and high reemission capacity. Their environment can therefore generate local conditions in which high-energy radiation, plasma, and magnetic coupling geometries favor the emergence of short-lived or stabilizable material modes.

$$\begin{aligned}\text{strongly tensioned field knot} &\rightarrow \text{high-frequency reemission}\\ &\rightarrow \text{local pair formation}\\ &\rightarrow \text{possible binding.}\end{aligned}$$

The UQSH therefore distinguishes three levels of matter formation. First, there is local pair formation as the splitting of highly densified field motion into complementary counter-modes. Second, there is stabilization, in which part of these modes passes into bound field organization through suitable coupling. Third, there is embedding, in which such bound states are integrated into atomic, stellar, galactic, or still larger field orders.

$$\text{Pair formation} \neq \text{permanent matter organization.}$$

$$\text{permanent matter} = \text{pair formation} + \text{stabilization} + \text{embedding in a coupling regime.}$$

Observationally, frequent high-energy processes, ionized gas regions, hydrogen and helium lines, as well as indications of pair plasmas in quasar and jet environments fit this picture. These findings are not understood here as conclusive proof of new matter generation, but as compatible indications that highly tensioned relief fields can favor minimal matter-capable modes.

Matter formation thereby becomes an ongoing process of field organization. It does not occur once at the beginning of a universe, but wherever free light bubbles are sufficiently densified, selectively coupled, and geometrically closed. In this view, the universe is not a finished container full of matter, but a field process in which bound organizations continually arise, decay, reemit, and become newly embedded.

Matter does not arise from a substance, but from the binding and geometric closure of densified light bubbles.

This logic can be transferred scale-relatively to larger field knots. What appears in the atomic regime as matter formation can recur in the galactic regime as the formation of large-scale bound field organization. Galaxies possess internal structure, spectral signature, gravitational coupling, boundary zones, reemission, relief channels, and dynamic stability. They can therefore appear as large-scale bound field knots of a higher organizational domain.

$$\text{Matter} \sim \text{bound field knot in the local regime,}$$

$$\text{Galaxy} \sim \text{bound field knot in the large-scale regime.}$$

Just as matter emerges from the binding and stabilization of tension fronts, galactic field knots can arise from the large-scale binding, redistribution, and stabilization of field tension. Black holes, jets, star-forming zones, magnetic fields, gas flows, and baryonic densifications do not act in isolation, but as a jointly coupled reorganization structure.

In this sense, matter and galaxies follow a related organizational logic. Both appear as stable coupling forms of the field: matter on the minimal regime-internal scale, galaxies on the large-scale cosmic scale.

Matter and galaxies can be understood as bound field knots of different regimes of the same organizational logic.

Testable Prediction for Targeted Matter Formation. From the UQSH interpretation follows a concrete experimental expectation: if matter emerges from the densification, selective coupling, and geometric closure of free light bubbles, then matter formation should not only be possible in astrophysical extreme environments, but should in principle also be technically favorably induced. Particularly suitable would be configurations in which directed high-frequency gamma radiation meets one or more differently weighted radiation components under controlled angles.

What would be decisive here is not the energy of the radiation alone, but the coupling geometry. Two or more light bubbles would have to meet in such a way that their phase relation, timing density, orientation, and local field tension allow geometric closure. An asymmetric weighting of the participating radiation modes could thereby favor the splitting into complementary coupling directions. The angle between the fronts would then not be an external parameter, but part of the closure condition.

$$B^{\gamma}_{\text{free}}(\Phi) + B^{\lambda}_{\text{free}}(\Phi) \xrightarrow[\text{angle, phase, coupling}]{\text{selective densification}} B^{(+)}_{\text{bound}}(\Phi) + B^{(-)}_{\text{bound}}(\Phi).$$

This yields the following UQSH prediction: directed gamma radiation should, under suitable superposition with further radiation modes and fitting angles of incidence, facilitate the formation of material counter-modes. Matter formation would then not be random particle generation from energy, but the result of a controlled field-mechanical closure geometry.

Matter formation should be favored when high-frequency light bubbles with suitable phase, weighting, and geometry are forced into a local field closure.

In particular, frequency density, radiation weighting, angle, phase relation, local field tension, and coupling environment should be investigated. If the UQSH is correct, matter formation should become more probable where these quantities allow a stable geometric closure of free field motion.

21.1. Regime Coupling and Subregime-Like Matter Modes

The formation of visible matter does not presuppose that all participating field configurations couple fully to our observational regime. Within the UQSH framework, a series of subregime-like intermediate modes may exist between a free light bubble and visibly classifiable matter. These modes are not less real and not merely empty auxiliary states. They themselves belong to matter in the broader field-mechanical sense, but they do not couple to our regime in the same way as electrons, atoms, molecules, or macroscopic substances.

It is therefore more precise not to distinguish between matter and non-matter, but between visibly regime-coupled matter and subregime-bound matter modes.

$$\text{visible matter} = \text{matter above the coupling threshold of our regime},$$

$$\text{subregime-like matter modes} = \text{matter below or beside this coupling threshold}.$$

Such intermediate modes can distribute internal tension, facilitate geometric closure, stabilize resonances, or prepare transitions between free and bound field motion. They do not necessarily appear as additional visible particles or as independent chemical substances. Nevertheless, they can help carry the stability, inertia, coupling form, and load-bearing capacity of a visible matter structure.

$$\begin{aligned}\text{free light bubble} &\rightarrow \text{subregime-like matter modes}\\ &\rightarrow \text{visibly regime-coupled matter}.\end{aligned}$$

The coupling threshold thereby describes not whether a state exists, but whether it becomes directly visible, measurable, or chemically classifiable for our regime.

$$E_{\text{org}} < E_{\text{coupling}} \Rightarrow \text{subregime-like matter mode},$$

$$E_{\text{org}} \geq E_{\text{coupling}} \Rightarrow \text{visible coupling in the regime}.$$

Matter is thus not only the visible knot that our regime recognizes as an electron, atom, molecule, crystal, or material. It possesses an internal coupling architecture in which even such modes can be effective that do not immediately appear as visible components. The visible substance would then be the regime-coupled surface of a deeper field organization.

$$\text{visible substance} = \text{regime-coupled surface of a deeper matter order}.$$

This view corresponds to macroscopic field organization. In galaxies, halos, jets, or relief fields as well, a field configuration can be dynamically effective without becoming fully electromagnetically

visible. Analogously, a material substance can also possess internal subregime-bound stabilization components that do not appear as their own visible material phase, but help determine its properties.

This becomes especially illustrative in materials. A material such as iron, steel, or cast iron is usually described by chemical composition, crystal structure, carbon content, phases, defects, grain boundaries, and heat treatment. Even small changes can generate enormous differences in properties. Steel typically contains less than about two percent carbon, whereas higher-carbon iron-carbon materials are more commonly assigned to the domain of cast iron. Nevertheless, hardness, toughness, elasticity, brittleness, formability, and load-bearing capacity already differ considerably through very small changes in internal order.

Within the UQSH framework, this sensitivity can additionally be read field-mechanically. A material then consists not only of the visibly chemically detected atoms and phases, but also of subregime-like matter modes that stabilize, tension, relieve, or occupy coupling spaces between these visible structures. A seemingly pure material would therefore not be field-mechanically completely empty between its visible components. Between the regime-coupled atoms, subregime-like stabilization fields could be present that contribute to strength, toughness, conductivity, magnetic order, or fracture dynamics.

$$\text{material} = \text{visible atomic arrangement} + \text{subregime-like stabilization modes}.$$

This gives rise to a far-reaching possibility. If such intermediate modes could actually be deliberately influenced, displaced, ordered, or redistributed, materials engineering would not only alter chemical composition and crystal structure, but also the subregime-like stabilization architecture of a substance. A material could thereby gain new internal coupling spaces or distribute existing tension modes differently.

This would mean: the properties of a material depend not only on which visible atoms are present, but also on which invisible subregime-like matter modes stabilize between them. If this internal field architecture is altered, a substance could arise that has markedly different properties while having the same or a similar chemical composition.

$$\text{new material} \sim \text{visible composition} + \text{altered subregime-like coupling order}.$$

For steel, this would mean within the UQSH framework: not only the carbon content and the visible microstructure would be decisive, but also the question of which subregime-like stabilization modes exist between iron, carbon, and alloy structures. If it were possible to reduce, reorder, or deliberately replace such modes, new coupling spaces could arise. These could either take up more stable visible structure or carry the existing structure differently. This would yield the possibility of materials with new or improved properties.

21.2. From Electron-Positron Pairs to Hydrogen and Helium

The first matter-capable splitting of free, highly densified field tension appears as the formation of complementary electron-positron modes. This, however, does not yet yield complex matter in the chemical sense. Electron and positron initially mark the simplest form in which free field motion can pass into bound, oppositely oriented coupling states.

$$B^{\gamma}_{\text{free}}(\Phi) \rightarrow B^{(+)}_{\text{bound}}(\Phi) + B^{(-)}_{\text{bound}}(\Phi).$$

The next step does not consist in finished particles being mechanically assembled like small bodies. Within the UQSH framework, field modes configure themselves jointly. Their tension directions, coupling orientations, phases, energies, and geometric closures must fit together in such a way that a larger stable composite arises. Matter therefore does not grow through the mere juxtaposition of particles, but through the joint stabilization of several field modes.

$$\text{field mode + field mode} \rightarrow \text{shared coupling geometry} \rightarrow \text{larger field composite}.$$

From the simplest stable coupling forms, the first atomic matter modes can thus emerge. Hydrogen and helium assume a special role here because they represent the simplest and most widespread stable atomic composites. They form the first major threshold between minimal matter formation and later chemistry.

Hydrogen can be read as the simplest stable atomic coupling form: a minimal bound composite of positively and negatively oriented field components that appears to our regime as an atom. Helium already represents a higher level of stability: more densely bound, more strongly symmetrized, and field-mechanically more robust. Both are therefore not only chemical elements, but early stable organizational forms of the field projection.

$$\text{electron-/positron modes} \rightarrow \text{first atomic field composites} \rightarrow H,\ He.$$

This makes understandable why hydrogen and helium appear so fundamental and widespread in the universe. Within the UQSH framework, they are not random initial products of a one-time event, but the simplest stable matter modes that can emerge from the further coupling and closure of elementary field modes. Wherever sufficiently high-frequency field tension, coupling geometry, and stabilization are present, such simple composites can preferentially arise or remain preserved.

The further development toward heavier elements and chemical structures builds on this threshold. Only when simple atomic field composites such as hydrogen and helium are stably present can denser nuclei, more complex electron orders, molecules, solids, organic chemistry, and finally life-capable structures arise. Chemistry therefore does not begin with isolated particles, but with stable field composites that enable repeatable coupling forms.

$$H,\ He \rightarrow \text{stars} \rightarrow \text{heavier elements} \rightarrow \text{chemistry} \rightarrow \text{complex structures}.$$

For the later question of life, this point is essential. Life presupposes not only matter, but matter that already possesses stable, repeatable, and combinable coupling forms. Hydrogen and helium form the first universal basis here. From them, heavier elements arise in stars and highly tensioned field environments; from these in turn chemical bonds, molecules, and finally those complex field organizations that can carry biological order.

Within the UQSH framework, hydrogen and helium are
the first simple stable matter modes
from which later chemistry and complex structure can emerge.

21.3. Dwarf Galaxies as Secondary Matter and Field Knots

A natural consequence of the previous logic of matter formation concerns young, gas-rich dwarf galaxies in the environment of larger galactic systems. If strongly tensioned relief fields can favor the formation of minimal matter-capable modes, then such dwarf galaxies do not have to be understood exclusively as original small galaxies or as old remnants of early structure formation. Within the UQSH framework, they can also arise as secondary field organizations that emerge from the tension, jet, or relief zones of larger galactic knots.

Central black holes, active galactic nuclei, accretion zones, jets, and galactic tension axes generate large-scale domains of increased field tension. There, high-frequency radiation, plasma, magnetic fields, shock fronts, and existing field structures can selectively couple with one another. Under suitable conditions, minimal matter-capable coupling forms and early gas structures first arise. If these are further stabilized, gas clouds, star formation, and finally independent dwarf galaxies can emerge from them.

Hydrogen- or helium-rich dwarf galaxies near large galaxies thereby do not necessarily appear as isolated foreign bodies. They can be read as local products of an ongoing field-mechanical relief. The host galaxy acts here not only through gravitational attraction, but also structurally: its central and boundary tension fields create coupling spaces in which small galactic field knots arise, bind gas, and gradually develop their own stability.

It is to be expected that not all dwarf galaxies possess the same degree of organization. Some may lie farther away from the host galaxy and thereby gain enough dynamic margin to form their own internal structure, their own core, more stable stellar populations, and a clearer gravitational self-order. Others remain more closely bound to the host galaxy. Their field organization is then more strongly under the influence of the superordinate galactic tension space. They can appear more diffuse, less strongly densified, and remain longer in an early or incompletely stabilized state.

$$\text{greater distance} \rightarrow \text{more self-organization} \rightarrow \text{more stable dwarf knot},$$

$$\text{smaller distance} \rightarrow \text{stronger anchoring} \rightarrow \text{more diffuse field organization}.$$

Precisely their often diffuse, gas-rich, and developmentally poor-looking form fits this interpretation. Such a dwarf galaxy would then not necessarily have passed through the same long field history as its host galaxy. It can appear as a younger reorganization knot that already possesses enough coupling to hold gas and enable star formation, but whose internal saturation, metal enrichment, and dynamic closure are still much less developed.

In very close dwarf systems, this self-organization can additionally be inhibited. Proximity to the host galaxy means not only stronger gravitation, but also stronger embedding in its field tension, boundary dynamics, shear, and relief structure. A young dwarf knot can thereby be gravitationally

held, stretched, kept diffuse, or even reabsorbed into the larger field organization before it forms its own stable core. Its matter would then be present, but its field organization not yet sufficiently closed to appear as an independent compact galaxy.

$$\text{young dwarf knot + strong host coupling} \rightarrow \text{diffuse bound state}$$

or

$$\text{young dwarf knot + strong host coupling} \rightarrow \text{reintegration into the host galaxy.}$$

A high fraction of hydrogen and helium would in this context indicate that the organization is still in an early or little-processed stage. The dwarf galaxy is then not a mere clump of matter, but a newly stabilized or still forming galactic field knot: materially present, gravitationally effective, but field-mechanically still young. Its proximity to the main galaxy can be read as an indication that it emerged from its large-scale tension, jet, or relief environment.

The contrast with the host galaxy thereby becomes especially significant. The large galaxy carries a long, strongly structured, and partially saturated field history. The dwarf galaxy, by contrast, appears as a fresher coupling knot in its boundary or relief field. Its diffuseness would then not be a mere deficit of structure, but the visible trace of an early binding phase in which simple hydrogen and helium structures are already present, while heavier elements, dense stellar populations, and more complex internal organization arise only gradually.

From this perspective, different types of dwarf galaxies can represent different stages of field organization. More compact and more distant dwarf galaxies would already have achieved more self-stability. Diffuse, gas-rich, or very close dwarf systems, by contrast, could be more strongly held, shaped, or limited by the host galaxy. Their behavior would then be determined not by their own mass alone, but by the relation between their own field closure and external anchoring.

$$\text{dwarf-galaxy state} = \text{self-organization} + \text{anchoring to the host galaxy} + \text{local field tension.}$$

This connects with the scale-relative analogy to matter formation on a smaller level. Just as an electron-like mode becomes stable only when free field motion passes into a bound coupling form, a dwarf galaxy can also arise as a larger coupling unit. It forms where free or diffuse field tension is transferred, through suitable geometry, density, duration, and coupling, into a long-lived organization. Whether this organization forms its own core, remains diffuse, or re-enters the host galaxy depends on whether its self-binding becomes stronger than the external field anchoring.

$$\begin{aligned}\text{galactic relief field} &\rightarrow \text{matter-capable modes} \rightarrow \text{gas binding} \\ &\rightarrow \text{star formation} \rightarrow \text{dwarf galaxy as secondary field knot.}\end{aligned}$$

$$\text{secondary dwarf knot} \rightarrow \begin{cases}\text{own core formation,} \\ \text{diffuse bound state,} \\ \text{reintegration into the host galaxy.}\end{cases}$$

Within the UQSH framework, the diffuseness of young dwarf galaxies can
be read as early, still weakly saturated field organization
whose self-development depends on anchoring to the host galaxy.

21.4. Quantum Numbers as Stable Field Markers

Within the UQSH framework, quantum numbers are not understood as externally attached properties of isolated particles, but as stable markers of bound field modes. They describe how a field organization carries mass, couples outward, is internally oriented, and can assume only certain stable states. In this view, quantization means that the field cannot permanently carry every arbitrary intermediate form, but only those organizational states whose tension, geometry, phase, and coupling fit together stably.

Mass appears here as bound field tension of a stabilized knot. It is a particularly stable form of regime-effective field curvature, but not the only source of gravitational field response. A mass-carrying state is a field motion that does not propagate freely, but is locally held, organized in circulation, and stabilized against external reorganization:

Mass = bound field tension.

Charge describes the directed coupling asymmetry of such a field knot. A charged state does not couple neutrally in all directions, but possesses a sign, orientation, or counter-coupling structure through which it responds selectively to electromagnetic field organizations:

Charge = coupling asymmetry.

Spin is not understood as the classical rotation of a small body, but as the internal orientation, chirality, or circulation order of a bound field mode. It describes how the internal field motion is nested in such a way that it permits only certain stable orientations and coupling responses:

Spin = internal orientation or circulation order.

Further quantum numbers can correspondingly be read as markers of stable coupling roles. They indicate which internal tension directions, parities, symmetries, binding modes, or regime connections a field state possesses. Within the UQSH framework, quantum numbers are therefore not abstract labels, but the readable signatures that a field knot can exist only in certain stable organizational classes.

Quantum numbers are stable markers of bound field organization.

22. Chemistry as Flexible Field Coupling

Within the UQSH framework, chemistry describes not only the interaction of finished particles, but the stable coupling of bound field knots within an intermediate saturation domain. Atomic nuclei represent highly densified, strongly bound field organizations. Electrons or electron-like coupling

forms, by contrast, form the more flexible outer coupling layer through which atoms can enter into relation with one another. Only this outer, electromagnetically ordered shell makes it possible for molecules, materials, and more complex structures to arise from individual atoms.

The electromagnetic interaction occupies a special intermediate position here. It is not as locally locked as the strong interaction in the nucleus and not as freely large-scale as gravitation. It forms a domain of intermediate saturation: bound enough to carry stable forms, but open enough to enable coupling, exchange, polarization, reaction, and reorganization. Precisely in this intermediate domain, the diversity of chemistry arises.

$$\text{nuclear densification} \rightarrow \text{electromagnetic shell} \rightarrow \text{chemical coupling capacity}.$$

Chemical bonds thereby appear as stabilized field bridges between several bound knots. A molecule is not merely an accumulation of individual atoms, but a shared field organization in which charge distribution, electron density, geometry, polarity, bond angles, and vibrational modes come together into a new stable overall form. The chemical identity of a substance therefore lies not only in the atoms involved, but in the way their field shells are jointly organized.

$$\text{chemical bond} = \text{stable coupling of atomic field shells}.$$

A chemical reaction is correspondingly not the generation of new substance from nothing. It is a reorganization of existing field bonds. Certain couplings are dissolved, transition states arise, energy is absorbed or released, and new, more stable or differently adapted field configurations form. What appears in chemistry as a reaction pathway is, within the UQSH framework, the path along which a system passes from one field organization into another.

$$\text{old bonding order} \rightarrow \text{transition state} \rightarrow \text{new bonding order}.$$

The energy absorption or energy release observed in this process describes the difference between the participating organizational states. A system can absorb field motion in order to break an old bond or reach an unstable intermediate form. It can release field motion when a new configuration is more stable and requires less free reorganization. In established language, this appears as binding energy, heat, radiation, motion, or internal vibration.

Molecular vibration thereby also receives a field-mechanical meaning. Vibrational and rotational states are not merely mechanical motions of small bodies, but periodic modes of reorganization of the entire molecular field. They show how the coupled field structure distributes, stores, and releases its tension again. Spectral lines of molecules are therefore readable modulations of this internal coupling order.

Water has a special role in this context. It combines high stability with pronounced polarity, flexible hydrogen-bond formation, large heat capacity, and versatile solvent function. Within the UQSH framework, this means: water forms an especially broad coupling window for chemical reorganization. It can absorb, distribute, buffer, and mediate field motion between different molecular structures. Water thereby becomes a preferred medium in which more complex chemical order can arise and remain stable.

Chemistry is therefore the bridge between matter formation and biological organization. Matter

formation generates the first stable atomic coupling forms; chemistry connects these forms into flexible reaction networks. The conditions under which such networks can pass into biological organization are treated later in the section on the recurring mode of field operation.

$$\text{Matter} \rightarrow \text{chemical coupling} \rightarrow \text{reaction networks} \rightarrow \text{biological organization.}$$

Scale-relatively, chemistry corresponds to the general principle that bound field knots can form new superordinate coupling forms. In the atomic regime, this occurs as molecule formation. In the galactic regime, several field knots can form larger systems, filament relations, or coupled galaxy groups through gravitational, dynamic, and magnetized connections. The forces are not literally the same, but the mode of operation is comparable: field organizations connect, reorganize their binding, exchange energy, and form new stable overall states.

$$\text{Chemical bond} \leftrightarrow \text{scale-relative field coupling.}$$

Chemistry is thus not reduced within the UQSH framework, but expanded. It remains the concrete science of atomic and molecular bonds, but at the same time is understood as the expression of a more general field logic: stability arises through fitting coupling, reaction through reorganization, and complexity through the repeated formation of new shared field states.

Chemistry is the flexible coupling of bound field knots into new stable organizational states.

22.1. Periodic Table and Scale-Relative Organizational Analogy

In established chemistry, the periodic table of the elements describes the order of atomic systems according to nuclear charge, electron structure, bonding capacity, and recurring chemical properties. This order is not replaced within the UQSH framework. It remains a precise and experimentally confirmed description of atomic matter. The UQSH merely asks, in addition, whether the recurrence of stability, coupling, and saturation visible in it could be the expression of a more general mode of field operation.

No literal equation of atoms and galaxies is claimed here. The analogy concerns functional roles within different scale regimes. What is decisive is not external equality, but the recurrence of certain organizational forms: a central field center, a binding space, peripheral coupling zones, stable and unstable intermediate states, saturation, neutralization, and reorganization.

In the atomic regime, the nucleus corresponds to the strongly bound center of the field organization. The electron shell forms the domain in which coupling, shielding, binding, and chemical reactivity are organized. The position of an element in the periodic table shows which bonding roles are stably possible: closed shells, open coupling sites, high reactivity, transition behavior, or especially stable noble-gas states.

Scale-relatively, a similar role logic can also be transferred to larger systems. A galactic core, especially a massive central field knot, can assume a nucleus-like function in the superregime. Disk, halo, satellite systems, gas structures, and magnetic orders then do not form an electron replacement in the literal sense, but a peripheral coupling space. In this space it is decided which structures remain

bound, which are expelled, which form transition states, and which reorganizations are possible.

The periodic order of atomic elements can thereby be read as an example of a more general UQSH question: do stable coupling roles repeat across scales as soon as a field system forms a center, a binding space, and a limited number of stable external couplings? If so, the periodic table would not be only a chemical order, but also an indication that field organization prefers certain stability patterns.

On the galactic level, this does not yield identity, but a comparable role logic. The galactic core can assume the function of a central field knot. Disk and halo form a large-scale coupling space. Satellites, gas structures, and dark or weakly visible companions can be read as bound or only partially coupled external modes. Jets, outflows, and strong radiation phases correspond to relief and transition forms, while the cluster environment forms the superordinate binding space.

Chemical and cosmic order can thus be described in a shared language without being equated. Chemical reactivity corresponds in the atomic regime to the capacity to occupy open coupling sites, rebuild bonds, or form stable molecules. In the galactic or cosmic regime, comparable roles can occur through accretion, satellite binding, gas inflow, star formation, jet activity, mergers, or cluster coupling. In both cases, what is decisive is not the existing mass alone, but the way in which a field system carries, limits, shields, or redistributes coupling.

What is decisive here is the coupling capacity to the respective regime. Not every existing field configuration must immediately appear as a stable, observable structure. In the subatomic domain, electron-like field configurations could be present without already coupling as electrons of our regime. They would appear as observable states only when their field organization reaches the necessary coupling conditions. Before that, they would not be nothing, but not yet sufficiently coupled to our regime.

The same applies scale-relatively to galactic external modes. Dwarf galaxies, gas clouds, dark satellite candidates, or weakly coupled substructures must therefore not simply be numerically equated with the electrons of an atom. For an analogous counting, they would have to be distinguished according to their coupling capacity: stably bound external modes, weakly coupled companion structures, transition modes, unstable reorganization remnants, and hardly visible field configurations. Only the number of actually coupling-capable roles would be relevant for a scale-relative analogy.

A testable consequence would therefore be that the mere number of satellite structures is not sufficient. What is decisive would be a coupling-adjusted count: which companions are stably bound, which only temporarily coupled, which visible through gas or star formation, which dynamically present but electromagnetically weak or invisible? Only after such a separation could it be investigated whether a galaxy fulfills a specific element-analogous coupling role in the superregime.

Chemical structure and chemical behavior can also be read scale-relatively as indications of recurring field roles. A chemical element is determined not only by its mass or nuclear charge, but by its bonding capacity, reactivity, stability, saturation, and tendency toward transformation. Analogously, a galaxy can also be described not only by mass or size, but by its capacity to bind satellites, absorb gas, form jets, expel matter, activate star formation, or pass into other organizational states.

The comparison with heavy and radioactive elements is especially instructive. A radioactive element such as uranium is, in the atomic regime, not a simply stable endpoint of the matter order, but a strongly loaded field state that reduces internal tension through decay, radiation, and transformation. Scale-relatively, very massive, overloaded, or highly active galaxies could possess a similar functional role. They would then not simply be larger galaxies, but oversaturated field organizations that relieve

their internal tension through jets, outflows, strong radiation, satellite formation, gas redistribution, or activation events.

In this analogy, radioactive decay does not literally correspond to a galactic explosion. What is meant is a shared organizational role: a field knot that is too strongly saturated cannot hold its tension locally without limit and therefore releases energy, matter, or structuring field motion to its environment. In the atomic regime, this appears as decay and radiation; in the galactic regime, it could appear as jet activity, quasar phase, starburst, gas outflow, formation of dwarf galaxies, activation of filaments, or transition into a calmer organizational form.

Conversely, stable chemical elements can serve as an analogy for stable galactic organizational forms. Systems with closed or especially favorable coupling roles behave relatively inertly: they bind their external modes stably, show low reactivity, and change their structure only under stronger external influences. Reactive elements, by contrast, correspond to open field organizations that easily couple, enter into bonds, or change their external structure. Scale-relatively, strongly gas-rich, interacting, or satellite-rich galaxies could assume such reactive roles.

Molecule formation also possesses a possible galactic correspondence. Chemical bonds arise when two or more atomic field organizations coordinate their open coupling roles in such a way that a shared stable structure emerges. On the cosmic scale, galaxy groups, interacting galaxy pairs, cluster cores, or filament knots could form similar roles: not every encounter leads to a stable bond, but under suitable velocities, tension states, gas fractions, and coupling conditions, a superordinate organization can arise from several individual systems.

The scale-relative analogy is therefore not:

$$\text{specific element} = \text{specific galaxy}.$$

Rather, it is:

$$\text{chemical behavior} \sim \text{galactic organizational role}.$$

What is decisive are stability, reactivity, saturation, open coupling sites, transition states, and relief forms. A very massive active system can therefore assume, in the superregime, more the role of a heavy, unstable element, while a quiet, closed galaxy corresponds more to a saturated or weakly reactive coupling order.

23. Large-Scale Structures and light bubbles

If galaxies are field knots, then their environment also belongs to the field configuration. A galaxy does not end sharply at its visible boundary. Halo, satellites, filament connection, and dynamic excess are parts of its overall tension organization. On cosmological scales, the cosmic web can therefore be understood as the visible projection of large-scale field tension.

In this view, filaments, walls, and knots would not only be matter distributions, but possible paths, superposition zones, or remnants of propagating light bubbles of higher regimes. Such a front would not have to appear from our perspective as a perfect spherical shell. Since we observe along our light cone and through redshift spaces, a spherical or partly spherical front can become visible as an arc, wall, ring, filament chain, or large-scale concentration.

Especially large and partly controversially discussed structures, such as the Hercules–Corona Borealis Great Wall, are interesting in this context as hypothetical test examples. If an extremely large-scale concentration of discharge events or galaxies at similar redshift is present, this could be read within the UQSH framework as the projection of a superordinate activation or light bubble. This interpretation remains hypothetical, but provides a search direction: large-scale structures should be investigated not only according to matter density, but also according to dynamic activation, lensing potential, hot gas, galaxy knots, and reprocessing indicators.

The knots of the cosmic web would be especially important here. They could be superposition zones of several fronts. Stronger lensing signals, higher galaxy densities, dynamically active systems, hot gas, and systematic deviations in the relation between baryonic organization and gravitational effect should therefore accumulate there. This connects the large-scale structure of the universe with the same field-tension logic that appears on galactic scales in the dark-matter effect and in q-regimes.

23.1. Imperfect Symmetry as a Prerequisite for Structure

In the idealized limiting case, the spherical light bubble possesses a geometrically clear form. Its surface grows with

$$A(r) = 4\pi r^2.$$

This formula describes the ideal symmetry of a spherical front. Physically, however, such a front does not arise in an empty, completely homogeneous space, but in an already excited, structured, and coupling-capable field medium.

The circle itself already shows an important peculiarity in its mathematical description: the ratio of circumference to diameter is determined by π, and π is not a finite, closed number. This does not mean that the mathematical circle would be uneven. Rather, it shows that even the simplest ideal round form possesses a not completely finite numerical description. Within the UQSH framework, this can be read as an image for the fact that ideal symmetry and real field organization are not identical.

A real light bubble runs through a field that already contains motion, gradients, couplings, local tensions, and earlier reorganizations. Its spherical basic form can therefore remain preserved without its surface appearing completely homogeneous. Local deviations, phase shifts, density differences, and coupling unevenness are not merely disturbances of a perfect order, but the conditions under which new order can arise.

If a field front were completely smooth, homogeneous, and without internal difference, it could hardly form local couplings, resonances, or bound structures. Only small differences in tension, phase, density, and geometry make it possible for certain regions to couple more strongly, while others relax or continue propagating. Structure therefore arises not despite imperfection, but through it.

This makes understandable why large-scale light bubbles in the universe do not have to appear as perfect spherical shells. Their ideal propagation form can be spherical, while their observable realization becomes visible only as arcs, shell remnants, filament edges, rings, or partial fronts. The deviation from the perfect form is not a defect, but an expression of the field-mechanical reality: a dynamic medium forms order not through absolute smoothness, but through controlled unevenness.

The imperfect is not a disturbance of order, but its condition of emergence.

24. Gravitational Wave as Gravitational Light

The term light bubble initially denotes, in general, any free propagation of field tension. Not every light bubble, however, should immediately be called a gravitational wave. A gravitational wave is the macroscopic curvature wave of a stable field knot. It arises where a star, a neutron star, a magnetar, a black hole, or another dense field knot not only holds its field tension locally, but imprints it as a geometrical wave into the surrounding field order.

light bubble = general free field propagation,

gravitational wave = macroscopic curvature wave of a field knot.

Such a gravitational wave does not belong only to a single point of the source. It encompasses the entire macroscopic field organization of the knot. In a star, it is the large-scale wave form of its bound field tension. In a neutron star or magnetar, it carries the higher density, stronger curvature, and stronger internal order of these objects. In an Anun, that is, a maximally saturated field knot, it becomes the boundary wave of an extremely densified regime.

star $\rightarrow$ stellar gravitational wave,

neutron star / magnetar $\rightarrow$ highly densified gravitational wave,

Anun $\rightarrow$ gravitational wave of a maximally saturated field knot.

For us, such waves are not visible like ordinary light. In our regime, they appear as changes in the spatial and length order, that is, as what is called gravitational waves in established physics. This view remains compatible with observation: gravitational waves propagate at the speed of light, carry energy, possess frequency, phase, direction, and curvature, and act as real motion of field geometry.

$$v_{\mathrm{GW}} = c.$$

visibility in our regime = curvature modulation.

Scale-relatively, however, the same wave can appear differently in another regime. What becomes visible to us as a gravitational wave can be readable in a superregime as radiation. In this extended sense, one can speak of gravitational light. Gravitational light is then not a second kind of light beside electromagnetic light, but the regime-dependent visibility of a macroscopic gravitational wave.

gravitational light = regime-dependently visible reading of a gravitational wave.

The separation between radiation and gravitation thereby becomes softer without being blurred. The source generates a gravitational wave. This wave carries modulation. Depending on the regime, this modulation appears as spatial curvature, as electromagnetically readable radiation, as X-ray or gamma order, or remains below the respective coupling threshold.

$$\text{gravitational wave} \rightarrow \begin{cases} \text{curvature modulation in our regime,} \\ \text{radiation modulation in another regime,} \\ \text{non-coupled internal modulation.} \end{cases}$$

The Sun therefore generates not only electromagnetic light. It also carries a stellar gravitational wave into space. This wave is the macroscopic propagation of its bound field tension. Visible sunlight is the electromagnetically coupled modulation that we can receive. The solar gravitational wave is the geometrical field order that carries the same source on large scales.

sunlight = electromagnetically coupled modulation,

solar gravitational wave = macroscopic curvature wave of the solar knot.

The fact that we do not measure the gravitational wave of the Sun as an ordinary gravitational wave, as in the merger of black holes, is not because it does not exist. It is too uniform, too stationary, and too strongly embedded in the general field order. Our measuring instruments respond above all to rapid, coherent changes of curvature. Stable stars, by contrast, generate a persistent, quiet field wave whose effect appears to us rather as gravitation, orbital order, and local spatial curvature.

stable source $\rightarrow$ quiet, persistent curvature wave,

dynamic event $\rightarrow$ measurable gravitational-wave pulse.

It follows that gravitational waves should not occur only in extreme events. Every stable field knot imprints its environment tension-geometrically. Every star, every compact mass, every galactic core, and every larger field organization generates a macroscopic curvature wave. However, only those components become measurable that are strong enough or coherent enough to stand out from the background of the general field order.

every stable field knot $\rightarrow$ gravitational wave,

measurable gravitational wave = amplified or rapidly changing curvature modulation.

In extreme sources, the readability shifts. Black holes as maximally saturated field knots generate the densest and strongest gravitational waves of our regime. In a superregime, these waves could appear as

gamma-like radiation. Neutron stars, magnetars, or other highly densified stars could there carry rather X-ray-like modulations. Normal stars would, in the same reading, form lower-frequency modulations.

$$\text{Anun} \rightarrow \text{gamma-like superregime modulation},$$

$$\text{neutron star / magnetar} \rightarrow \text{X-ray-like superregime modulation},$$

$$\text{normal star} \rightarrow \text{lower superregime modulation}.$$

This does not mean that gravitational waves in the same regime would simply be ordinary radiation. Our Sun emits electromagnetic radiation that couples directly to our matter and therefore becomes visible or measurable. Its gravitational curvature wave is another coupling layer of the same field knot. Both belong to the overall organization of the Sun, but they are read differently in our regime.

$$\text{one source} \rightarrow \begin{cases} \text{electromagnetically coupled modulation}, \\ \text{geometrically coupled curvature wave}. \end{cases}$$

Conversely, light bubbles from a superregime can dedensify into our regime. We then do not see the complete original field motion, but only the modulation that couples to our matter, our measuring devices, and our spacetime reading. Such a front can carry several levels of internal order: an origin modulation, a regime-transition modulation, a visible coupling modulation, and a geometrical curvature modulation.

$$\text{one front} \rightarrow \begin{cases} \text{origin modulation}, \\ \text{regime-transition modulation}, \\ \text{visible coupling modulation}, \\ \text{geometrical curvature modulation}. \end{cases}$$

This also makes understandable why we may detect only a small section of such waves. Some modulations couple directly to our matter and appear as light, heat, X-ray, or gamma radiation. Others couple to our geometrical field order and appear as gravitational waves. Still others remain hidden from our regime because their modulation lies below our coupling threshold or is carried in a scale order that is not directly readable for us.

$$\text{measured} = \text{regime-coupled modulation},$$

$$\text{not measured} \neq \text{not present}.$$

Gravitational waves thereby become a key to regime coupling. They show that field motion does not have to appear only as visible radiation. A macroscopic curvature wave can appear geometrically in our regime, but in another regime become readable as radiation, energy outburst, or high-frequency relief form. The decisive question is not whether a front is “really” light or gravitation, but which level of its modulation couples to a regime.

gravitational wave ≠ counterpart to light,

gravitational wave = macroscopic field wave whose modulation is read scale-relatively.

In this view, gravitational light is the more general term for the scale-relative visibility of such field waves. Electromagnetic light is then a regime-coupled visible modulation. A gravitational wave is the geometrical reading of the same basic motion at the macroscopic level. Both belong not to separate worlds, but to different visibilities of one field motion.

We do not see the whole field wave, but only the modulation level that couples to our regime.

24.1. Gravitational Waves as an Additional Component of Free Field Tension

From the interpretation developed so far, a natural extension follows: gravitational waves are not only rare, measurable pulses from extreme events such as the merger of black holes. They are a form of motion of the Qu-foam itself. They carry curvature, tension, direction, frequency, and energy. For this reason, they can contribute to the gravitational field effect even when they do not become visible to our measuring instruments as an individual gravitational-wave event.

The important distinction is between a measurable gravitational wave and a gravitationally effective curvature modulation. A signal can be too weak, too slow, too uniform, too broadly distributed, or too strongly embedded in the general field order to be detected directly. Nevertheless, it can remain part of the total field tension that co-organizes a galaxy, a stellar system, or a halo.

measurable gravitational wave $\subset$ gravitationally effective curvature modulation.

This expands the view of stars, planets, and galaxies. A star is not a resting point. It rotates, oscillates, radiates, carries plasma, magnetic fields, internal shear, flares, and mass motions. A planet is also not a passive body, but possesses rotation, orbital dynamics, internal tension, and in many cases magnetic or geological activity. A galaxy contains billions of such moving field knots, which move around a common center, possess their own axes of rotation, are embedded in spiral, halo, and filament structures, and continuously generate radiation, plasma, magnetic fields, and curvature tension.

This leads to a strong conjecture: not only black holes and neutron stars generate relevant curvature waves. Ordinary stars, planets, stellar systems, galactic cores, rotating disks, satellite systems, and galaxies as whole systems can also generate weak curvature modulations. These do not have to be individually measurable. What matters is their sum, their spatial order, their repetition, and their embedding in the total galactic field.

stars + planets + galactic rotation + filament binding $\rightarrow$ weak free curvature modulations.

At least three gravitationally effective forms of motion within one regime can therefore be distinguished:

1. **Light and radiation fronts** = free field motion with frequency and tension structure
2. **Gravitational waves** = free or partially bound curvature motion
3. **Galactic field potentials** = large-scale tension of disk, halo, and environment.

The dark-matter effect would then not be only a consequence of the visible baryonic mass. It would arise from the interplay of several field components: baryonic matter forms the stable anchors, radiation carries free field motion, rotation and magnetic fields order this motion, and weak curvature modulations add to the large-scale field tension.

$$K_{\text{free}} = K_\gamma + K_{\text{GW}} + K_{\text{plasma}} + K_{\text{rot}} + K_{\text{gal}} + \dots$$

For the observed gravitational excess, this gives the following working form:

$$g_{\text{obs}} = g_{\text{bar}} + \mathcal{F}_{\text{gal}} \left[S_n \left(K_{\text{bind}} + K_\gamma + K_{\text{GW}} + K_{\text{rot}} + K_{\text{gal}} \right) \right] .$$

Here, K_{bind} describes the bound baryonic field organization. K_γ stands for free radiation fronts, K_{GW} for curvature modulations through gravitational waves, K_{rot} for rotational and magnetically twisted field order, and K_{gal} for the large-scale galactic potential. These components do not have to be visible as separate signals. They can nevertheless jointly shape the measured field effect.

This leads to a concrete prediction: dark-matter-like excesses should be especially strong where many of these components come together: many baryonic anchors, strong rotation, a high number of stars, active star formation, radiation, magnetized gas structures, satellite binding, filament connection, and signs of past or ongoing reorganization.

baryonic anchors + radiation + rotation + gravitational-wave components $\longrightarrow$ dark-matter effect.

This does not mean that every stellar or planetary gravitational wave must be directly measurable. It is much more likely that many individually unmeasurable curvature modulations act together as a weak background. Just as not every field motion appears as an individual photon, particle, or pulse, not every gravitationally effective field motion has to become visible as a classically measurable gravitational wave.

not individually measurable $\neq$ gravitationally ineffective.

This conjecture would be testable above all statistically. The gravitational excess of a galaxy should not only be related to its visible mass, but also to its dynamic field activity: star-formation rate, radiative power, gas and plasma distribution, magnetic field order, rotation structure, satellite dynamics, filament connection, and reorganization history. A gravitational-wave background that is difficult to resolve directly would then not merely be a side effect, but a possible component of free field tension.

Not every gravitationally effective field motion has to be measurable as an individual gravitational-wave event. Many weak curvature modulations can jointly contribute to the additional galactic field effect.

25. Planck Limit, Vacuum Energy, and Coupling Filter

Within the UQSH framework, the Planck limit receives a different meaning. It is not a physical infinity and not necessarily the final structure of the field. It primarily describes the lower resolution and coupling limit of our regime. Below this limit, field motion does not disappear. It continues to reorganize in subregimes that are no longer directly resolvable for our measuring instruments and our material structure.

This must be distinguished from the upper saturation limit. It appears where local curvature can no longer be deepened further, for example in compact limiting states such as black holes. In short:

The Planck limit bounds our resolution downward; field saturation bounds curvature upward.

This distinction helps with the vacuum-energy problem. In standard physics, an enormous discrepancy arises when zero-point energies are summed up to the Planck scale and compared with the observed cosmological energy density. The difference is often given as about 10^{120} to 10^{122} orders of magnitude. Within the UQSH framework, this number does not have to mean that the calculation captures every real field energy incorrectly. It may also mean that an extremely large amount of field motion lies in subregimes that are not directly coupled. The known discrepancy between the formally expected vacuum energy and the observed cosmological effect is known as the cosmological constant problem [13, 14].

The decisive distinction is:

$$\rho_{\text{field}}^{\text{total}} \neq \rho_{\text{grav}}^{\text{observed}}.$$

The total field energy comprises all nested reorganization and tension components of the Qu-foam. The observed gravitational energy density is only the portion that becomes effective and readable as field curvature within our regime. The large discrepancy would then not be proof that the field energy does not exist, but an indication that only an extremely small portion of the total organization falls into our gravitational readability.

The vacuum-energy problem thereby becomes a coupling problem:

Not: Why does so much field energy exist?

But: Why does only a small fraction of it couple into our regime?

One possible structural formula is:

$$\rho_{\text{obs}} = \mathcal{F}_{\text{regime}} \left(S[\rho_{\text{field}}^{\text{total}}] \right),$$

where $\mathcal{F}_{\text{regime}}$ describes the coupling filter of our regime and S the saturation. This equation too is not meant here as a finished derivation, but as a working form: observed energy density is a filtered and saturated projection of deeper field energy.

25.1. Quantum Fluctuations as Short-Lived Potential Spaces

Within the UQSH framework, quantum fluctuations do not appear as the random appearance of particles from nothing. They arise from the dynamics of the field medium itself. Even below or outside our direct resonance capacity, field motions, residual tensions, and non-coupling-capable organizational forms remain present.

When such non-coupling-capable field structures encounter one another, overlap, orbit one another, or briefly enter into a shared geometry, a local potential space can arise. For a short time, this potential space possesses more coherence, more tension, and a higher coupling capacity than its individual components. What was previously not measurable becomes briefly connectable to our regime in the shared state.

A quantum fluctuation is therefore not an emergence from nothing, but a temporary coupling opening:

$$\begin{aligned} \text{non-coupling-capable field structures} &\rightarrow \text{superposition} \rightarrow \text{short-lived potential space} \\ &\rightarrow \text{measurable fluctuation.} \end{aligned}$$

Scale-relatively, this process can be illustrated by galactic processes. When two galaxies encounter one another, interpenetrate, and move apart again, they temporarily generate a larger shared potential space. The individual galaxies remain their own field knots, but during the interaction an additional overall organization arises: shared halo tension, shear surfaces, tidal bridges, star-forming zones, shock fronts, and large-scale field distortions. For a hypothetical superregime, this entire process could appear like a short-lived fluctuation: a potential space that emerges, couples, and decays again.

The same principle can apply in the subregime. There, field modes that are not directly measurable encounter one another, briefly form a shared order, and then fall apart again. For our regime, this brief moment appears as a quantum fluctuation. The fluctuation is then not inexplicable, but the expression of a temporary crossing of the coupling threshold.

What is decisive is that the participating structures do not newly emerge from nothing. What is new is only their temporary shared coupling capacity. The fluctuation is the brief becoming-visible of a deeper field motion that was not readable individually, but jointly forms a measurable potential space.

Quantum fluctuations are short-lived coupling windows of deeper field organization.

26. Electromagnetism as Twisting Order of the Field

Within the UQSH framework, electromagnetism is understood as a directed twisting order of the field medium. The starting point is the assumption that, in its undisturbed state, the field possesses no

preferred direction. Free field motion can propagate spherically; bound field motion can stabilize itself in matter, mass, and gravitation. Direction arises only where field motion is locally superposed, bound, twisted, polarized, or coherently aligned.

Within the UQSH framework, gravitation is not reduced to bound matter or mass. It arises wherever field tension forms an effective curvature within a regime. Planets, stars, galaxies, halos, filaments, light bubbles, and saturated knots are therefore different carriers of gravitational field response, insofar as their curvature couples into the respective observational regime.

Within this view, magnetism describes no second fundamental substance beside gravitation. It arises when a field curvature is not only effective, but coherently twisted, polarized, and directed. While the gravitational response describes the regime-compatible curvature effect of a field state, magnetism appears as the short-range, axially ordered near projection of this curvature.

Gravitation = regime-effective field curvature,

Magnetism = coherently twisted near order of this field curvature.

This interpretation explains why magnetic effects can be strong, local, and direction-dependent. In a magnet, it is not the entire mass that couples outward uniformly, but rather a part of the internal field organization that emerges as a coherent twisting order. This order can be read as directed near curvature of the field. Its strength depends on how many internal field modes are aligned, how stable this alignment remains, and how well it couples into the surrounding field medium.

A magnet is therefore not a perfectly aligned body. It contains many subregime twisting modes in different directions. Macroscopic magnetization arises when a particular twisting direction statistically and coherently predominates. Other twisting components remain present. They can generate internal tension fields, residual couplings, hysteresis, heat losses, or electrical voltage differences.

macroscopic magnetization = dominant coherent twisting direction,

residual tensions = not fully aligned twisting components.

This makes understandable why different materials react magnetically in different ways. In some substances, the internal field twistings largely cancel one another. They possess only weak or short-lived magnetic response. In other substances, internal twisting modes can indeed be aligned, but fall back again after the external field is removed. In ferromagnetic and permanent-magnetic materials, by contrast, many of these modes can align collectively and remain stably preserved. The magnetic strength of a material therefore describes its capacity to project subregime field twisting coherently into our regime.

The behavior of a divided magnet also follows from this. When it is broken, north and south pole are not separated from one another as distinct substances. What is divided is an internal twisting order. Each fragment forms again a complete polarization axis from its own field organization. This is why each fragment again possesses north and south pole. Polarity belongs not only to the ends, but to the continuous internal alignment of the field knot.

In this view, electricity appears as the open tension form of the same twisting logic. While magnetism

describes a bound or closed twisting order, electric effect denotes an open coupling asymmetry or tension difference. If the magnetic order is displaced, altered, or relieved, electrical voltages can arise from the not fully balanced twisting components. In this sense, electricity is not separate from magnetism, but the moved or released tension side of the same field twisting.

magnetic order = stored directed field twisting,

electricity = open or moving tension difference of this twisting.

Electromagnetism thus denotes the dynamic interplay between bound twisting and open tension. A change in the magnetic twisting order can generate electrical field work; an electrical tension motion can in turn build up magnetic twisting. What appears in the standard picture as the coupling of electric and magnetic fields is read within the UQSH framework as mutual transformation between open coupling tension and bound twisting order.

This view can be extended scale-relatively. Planets, stars, magnetars, accretion disks, galaxies, galaxy clusters, and filaments are not only accumulations of mass, but rotating, layered, or flowing field knots. Where such systems possess sufficient internal motion, plasma, charge separation, shear, or coherent twisting, they can form large-scale magnetic orders. Magnetism then acts not only as a small material property, but as an axis, channel, spiral, and stabilization function of the larger field.

Within the UQSH framework, this means: gravitation collects, curves, and binds field tension; magnetism aligns a part of this field tension. It orders not primarily through additional mass, but through coherent twisting. This gives rise to preferred axes, rotation directions, plasma flows, relief channels, spiral paths, and stable coupling zones.

Gravitation $\rightarrow$ binding and curvature,

Magnetism $\rightarrow$ direction, twisting, and order.

This ordering function is already illustrative on the planetary and stellar levels. Earth possesses not only gravitational field binding, but also a magnetic axis order. A compass does not align itself with an abstract point, but couples its own internal twisting axis to the larger magnetic order of the planet. Solar magnetic fields also order plasma, solar wind, flares, coronal structures, and large-scale relief paths. Magnetars show an extreme form of such saturated magnetic twisting.

On the galactic level, the same logic appears on a larger scale. Spiral galaxies are not only rotating stellar systems. Their spiral arms can be read as large-scale relief and coupling paths in which rotation, gas flow, star formation, plasma, magnetic fields, and gravitational field tension are organized together. The spiral form is not accidental here. It connects radial extension with tangential twisting and allows the field to distribute tension not abruptly, but in an ordered manner across a growing surface.

radial extension + tangential twisting $\rightarrow$ spiral order.

Within the UQSH framework, spiral arms thereby become visible paths of ordered field relief. They

show how a galactic field knot does not simply distribute its internal tension spherically, but organizes it in twisted, stabilized, and repeatedly star-forming zones. Magnetic fields can help carry this order by stabilizing plasma flows, gas densifications, shear regions, and star-forming zones along preferred paths.

It is striking that many spiral forms appear approximately logarithmic across large domains. The golden ratio should not be understood here as a mandatory law of every spiral galaxy, but as a possible ideal value of a stable growth proportion. In such an order, the relation between expansion, rotation, and coupling remains similar across several scales. The field then does not grow arbitrarily, but in a form that can stably carry its twisting and relief onward.

stable spiral order ~ preservation of the relation between growth, rotation, and coupling.

In this reading, the golden ratio can appear as a limiting image of a field-mechanically favorable proportion: not as a mystical number, but as a possible form of order in which a system can grow without losing its internal twisting structure. Magnetism would then be one of the field effects that favors such proportional orders, because it can stabilize axial and spiral couplings.

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

can therefore, within the UQSH framework, be read as an idealized proportional quantity of self-similar field relief, insofar as rotation, extension, and coupling remain in a stable relation across scales.

This logic also continues within the spiral arms. Stars, planetary systems, gas clouds, and local magnetic fields are not merely embedded individual objects. They form smaller field knots within the larger spiral order. A star possesses its own magnetic activity, planets can form their own magnetospheres, and gas or dust structures can be densified along larger field lines. The spiral arm itself thereby appears as a nested magnetic ordering space: guided on large scales by the galaxy, and locally further differentiated by stars and planets.

spiral arm → stellar fields → planetary systems → local magnetospheres.

Scale-relatively, this yields a continuous role logic. On small scales, magnetism aligns atomic, molecular, or material-internal twisting modes. On the planetary level, it orders magnetospheres and charge flows. On the stellar level, it structures plasma, flares, and corona. On the galactic level, it can help shape spiral arms, gas flows, and star-forming zones. On the cluster level, magnetized filaments, hot gas, shock fronts, and large-scale plasma flows could assume comparable ordering functions.

material magnetism → planetary magnetosphere → stellar magnetism → galactic spiral order → cluster and filament

Magnetism thereby becomes readable in the UQSH as an ordering force of the field. It does not generate mass and does not replace gravitation. It gives field tension direction, axis, twisting, and coupling path. Gravitation binds the field state; magnetism organizes the direction in which tension preferentially flows, rotates, relieves, or stabilizes.

Within the UQSH framework, magnetism is the ordering twisting of the field;
it aligns field tension across scales into axes, channels
and spiral orders.

Earth, for example, possesses not only an overall gravitational tension, but also a directed planetary field order. A compass does not align abstractly with the North Pole, but couples its own internal twisting axis to the larger magnetic axis order of Earth. In the same way, solar magnetic fields can order plasmas, solar wind, flares, and coronal structures. Magnetars show an extreme form of saturated magnetic twisting. In spiral galaxies, magnetic field orders can help shape gas flows, star formation, plasma streams, and spiral-arm stability.

This yields two complementary readings of the same field foundation. Gravitation describes the regime-effective curvature tension of a field state. Magnetism describes that portion of this field order which is coherently twisted, axially polarized, and thereby becomes locally directed in its effect. The general saturation limits of gravitation were treated above; what is decisive here is that magnetism does not deepen this field effectiveness, but directs it.

This gives rise to poles, channels, relief directions, field lines, plasma flows, and preferred coupling axes. Gravitation gives the field state its curving effectiveness; magnetism makes a portion of this effectiveness locally directed and coupling-selective.

$$\text{field curvature} \rightarrow \text{regime-effective gravitation},$$

$$\text{coherently twisted field curvature} \rightarrow \text{magnetic near order}.$$

This interpretation also has a speculative technological consequence. If magnetism is a directed twisting form of field tension, then a partial decoupling from the field gravitation of a larger body should not be expected through switching off gravitation. Rather, it would have to arise through a controlled reorganization of one's own coupling twisting. A body would have to alter its local field anchoring in such a way that it does not merely lie passively in the larger gravitational field, but redirects its embedding in a directed manner.

An ordinary body possesses passive field binding. A magnet possesses directed field binding. An electromagnet possesses controllable directed field binding. A further-developed field technology would have to extend this control to the entire body: not merely generate a magnetic field, but a coherent, dynamically controlled twisting geometry of the entire field knot.

$$\text{passive mass} \rightarrow \text{bound field tension},$$

$$\text{magnet} \rightarrow \text{directed field tension},$$

$$\text{field-coupling system} \rightarrow \text{controlled twisting geometry}.$$

In this context, reports of rotating or disk-shaped objects are to be understood only as speculative

illustration, not as proof. If a technical system could modulate its coupling to the surrounding field medium, the entire body would not necessarily have to rotate mechanically. An internal rotation would also be possible: a circulating ring, a plasma layer, a field rotor, a superconducting current, or a phase-stable coupling zone. What would be decisive is not the visible rotation of a casing, but the rotation of coherent field tension inside.

Such a circulating twisting order could generate a directed counter-order or transverse order to the normal gravitational embedding. The object would then not simply be accelerated against its inertia, but would alter its coupling to the larger field space. In this view, motion would not only be force against mass, but a change of field anchoring.

This prospect remains highly speculative. It shows, however, what role magnetism could in principle play in the UQSH: it is not merely a side effect of electric charges, but the first observable indication that matter can not only carry its field tension passively, but organize it in a directed way. The next stage would not be stronger magnetization alone, but the controlled, dynamic, and scale-compatible steering of this twisting order.

Every regime-effective field curvature forms gravitation;
coherent twisting forms its magnetic near order.

Field decoupling would not be switching off gravitation,
but a controlled redirection of coupling twisting.

27. The Four Interactions within the UQSH Framework

In this view, the four known interactions do not appear as four ontologically separate forces. They are four coupling modes of the same field medium. The difference lies in the degree of organization, in range, in saturation, and in the freedom of reorganization.

27.1. Gravitation

Gravitation is the universal field response to organized tension curvature. Every stable or metastable field organization changes the tension state of the medium. For bound observers, this change appears as gravitation. Gravitation is not the maximally locally bound saturation mode, but the large-scale continuation of that field curvature which is not completely locally locked. It is therefore locally weak, but universal and not screenable.

27.2. Electromagnetism

Electromagnetism was previously described as the interplay between open coupling tension and bound twisting order. In the systematics of the four interactions, it forms the directed, regime-internal coupling layer between free radiation, bound matter, charge asymmetry, magnetism, and chemical organization.

27.3. Strong Interaction

The strong interaction is the local saturation mode of maximally densified field knots. In a nuclear regime, the field organization is so densely nested that deviations cannot freely run out. They are immediately returned into the existing structure and bound. Its strength arises from this local locking. Its short range is the reverse side of the same property: because the tension remains almost completely locally bound, it can hardly form a far-reaching signature outward.

Scale-relatively, this can be compared with the difference between the atomic nucleus and the galactic center. In the atomic nucleus, the binding is so dense that protons and neutrons do not appear as freely separating field knots, but remain locked in a shared nuclear order. On the galactic level, the central field knot, for example a dense core region or an Anun-near state, assumes a comparable organizational role. There too, field motion is strongly locally bound, stabilizes the surrounding structure, and allows only limited relief outward. Jets, accretion zones, or boundary-field reliefs would then not be a direct correspondence of the strong interaction, but macroscopic evasive forms that arise when the local locking reaches its saturation limit.

27.4. Weak Interaction

The weak interaction is not a permanently binding mode, but a selective reorganization channel of marginally stable field patterns. It emerges where an existing configuration can no longer stably carry its old order and passes into another admissible form. Decay and transformation thereby appear not as the annihilation of particles, but as transitions between field patterns.

Strong interaction	=	locally saturated binding of dense field knots
Electromagnetism	=	oriented modulation coupling of bound knots
Weak interaction	=	selective reorganization of marginally stable patterns
Gravitation	=	large-scale field response to tension curvature

The fine-structure constant α is not yet treated formally here. It remains a separate next step, because it presumably expresses the dimensionless relation between electromagnetic modulation coupling, bound orientation asymmetry, and maximum reorganization rate.

Scale-relatively, this can be compared with transition states between stable orders. In the atomic nucleus, the weak interaction appears where a neutron-like or proton-like state can no longer stably maintain its previous coupling role and passes into another admissible role. On the galactic level, analogous processes would not be direct particle decays, but slow reorganization processes of marginally stable systems: an unstable satellite knot can be incorporated into a larger galaxy, a gas or star-forming region can tip into a new stable structure, or a small field knot can lose its former order through tidal forces, jets, shock fronts, or relief zones and pass into another coupling form.

The analogy therefore lies not in a literal equation of beta decay and galactic development, but in the shared functional role: a marginally stable pattern changes its state because the old order is no longer supportable. In the microphysical regime, the weak interaction is the precise, quantized channel of such transformation; on larger scales, the same logic appears as slow, structure-dependent reorganization of field knots.

28. Recurring Mode of Field Operation and the Possibility of Enduring Life Formation

A central motif of the UQSH is the assumption that the field medium works according to the same basic principles on all dimensional scales. Radiation, matter, gravitation, dedensification, densification, saturation, relaxation, and reorganization are not separate processes, but different forms of expression of the same mode of field operation. The field moves, tensions itself, binds itself, relieves itself, and continually reorganizes its states.

This yields a different picture of cosmic development. Life is then not the result of a one-time, unrepeatable accident in an otherwise passive universe, but a possible consequence of recurring organizational chains. Because neither complete saturation nor complete relaxation is really reached, the field remains in residual dynamics, tension, relief, and reorganization. This repeatedly gives rise to new star-forming regions, new forms of matter, new chemical environments, and new coupling windows.

The prerequisite for this is not that the same conditions occur identically everywhere. What is decisive is that the field repeats the same operational logic: free field motion becomes bound, bound organization generates gravitation, gravitation collects matter, matter forms stars, stars generate heavier elements, elements enable chemistry, chemistry generates complex coupling networks, and coupling networks can, under suitable conditions, form biological organization.

$$\begin{aligned}\text{field motion} &\rightarrow \text{matter formation} \rightarrow \text{stars} \rightarrow \text{elements} \\ &\rightarrow \text{chemistry} \rightarrow \text{complex organization} \rightarrow \text{life}.\end{aligned}$$

In this view, life is not a foreign body in the universe. It is an especially complex, locally stabilized, and self-reorganizing field organization. Biological systems do not stand outside field mechanics, but are late coupling forms of the same dynamics that also carries radiation, matter, gravitation, and chemical bonding. Life is therefore not separated from physics, but a higher readability form of organized field motion.

The boundary domains are especially important here. In filaments, halos, accretion zones, star-forming regions, planetary transition zones, and Anun-relief domains, densification and relaxation meet. Such zones are not only places of high dynamics, but also places of selective stabilization. There, field motion can pass into new bound forms; there, matter and radiation can be modulated again; and there, the conditions for chemical and biological complexity arise.

The role of destruction also changes thereby. Supernovae, collisions, jets, radiation outbursts, accretion, or galactic reorganization are not only destructive events. They distribute field motion, heavy elements, dust, plasma, and new coupling conditions. What appears locally as an end can elsewhere become the prerequisite for new order. Relief and decay are therefore not merely opponents of life, but part of the cosmic redistribution through which later habitats become possible.

$$\text{decay} \rightarrow \text{redistribution} \rightarrow \text{new coupling conditions} \rightarrow \text{new organization}.$$

From this view follows a natural prediction: if water can form and remain stable wherever hydrogen, oxygen, suitable temperatures, pressure conditions, and chemical coupling spaces come together,

then life-capable environments are not in principle restricted to a single location. Life is thereby not automatically expected everywhere, but its formation is understood as a repeatable field process: where water remains permanently or cyclically stable, where chemical gradients exist, where energy flow is present, and where sufficient time for self-organization exists, life-like or for us recognizable life forms can arise.

In established astrobiology, this possibility is often described through the life-friendly zone or habitable zone. What is meant is the distance range around a star in which liquid water can remain stable on a planetary surface for longer periods of time. This zone is an important approximate concept, because water, temperature, radiation flux, and planetary atmosphere jointly decide whether chemical self-organization becomes durably possible.

Within the UQSH framework, the habitable zone is not rejected, but expanded. It describes not only a geometrical distance from the star, but a local coupling window of the field. A planet is life-friendly if its position, atmosphere, internal heat, magnetosphere, water chemistry, geological activity, and radiation environment allow stable reorganization of complex matter. The classical habitable zone would thereby be the visible astronomical surface of a deeper field condition: water must not only be present, but be held in a stable, cyclic, and chemically active coupling space.

$$\text{habitable zone} = \text{distance range of stable liquid water},$$

$$\text{UQSH coupling window} = \text{water} + \text{energy flow} + \text{chemical gradients} + \text{field stability}.$$

Thus, a place can be life-friendly even if it is not simply at the right distance from its star. Subsurface oceans, icy worlds with internal heat, hydrothermal systems, or moons in strong tidal fields can possess their own coupling windows. Conversely, a planet within the classical habitable zone can remain hostile to life if atmosphere, magnetic field, water balance, geology, or chemical stability do not cooperate sufficiently.

Such life forms develop differently depending on their environment. On water-rich planets, in subsurface oceans, in icy worlds, in hydrothermal systems, in dense atmospheres, or at interfaces between rock, water, ice, plasma, and radiation, different coupling conditions can prevail. Life is therefore not a fixed form, but purpose-bound self-organization within a local coupling window: its shape follows the environment, because every living being obtains its stability from the field, energy, water, temperature, and matter conditions present there.

$$\text{water} + \text{energy flow} + \text{chemical coupling} + \text{temporal stability} \rightarrow \text{life-capable self-organization}.$$

The UQSH thereby expands the question of life in the universe: what is decisive is not only whether individual planets possess favorable conditions, but whether the field medium repeatedly generates coupling windows in which order, energy flow, water chemistry, and stability act together.

In the end, a further question remains open. If life is understood within the UQSH framework as self-stabilizing field organization, then it must be asked whether water is the only possible carrier condition of such organization. For life as we know it, water is especially suitable because it mediates chemical reactions, buffers temperature ranges, transports molecules, and stabilizes complex coupling

networks. There could, however, be life forms that get by with very little water or use other chemical and physical coupling media, as long as these fulfill a comparable function: stabilization, exchange, energy flow, self-organization, and preservation of internal order.

$$\text{life} \neq \text{necessarily identical water biology},$$

$$\text{life} = \text{durable self-reorganizing coupling order}.$$

Water thereby remains the most important known indicator for the search for life, but not necessarily the final boundary of the concept of life. The UQSH therefore allows water-based habitats to be considered preferred and especially recognizable coupling windows, without fundamentally ruling out that other field and chemical orders can carry life-like self-organization.

Where water, energy flow, and stable coupling windows exist,
the field can repeatedly bring forth life-capable organization.

Water is the strongest known indicator of life-capability,
but the more general condition is durable
self-reorganizing coupling order.

29. c as the First Constant of Field Motion.

For the formalization, it is decisive that the speed of light is not treated as a local random variable of an individual tension space. Within the UQSH framework, c denotes the first constant of physically readable field motion: the absolute limiting rate of free reorganization within the field projection Φ. It is not the velocity of a material object, but the limiting form of $B_{\text{free}}(\Phi)$ itself.

$$c = \text{first constant of free field motion}.$$

Bound field states can appear differently timed through gravitation, saturation, dedensification, densification, or regime coupling. A strongly tensioned field region can slow clocks, shift frequencies, curve paths, or alter coupling. It does not, however, make c smaller. What changes is the bound reorganization order of the respective system; what remains constant is the free limiting motion of the field.

For a local observer within the same tension space, the speed of light therefore remains unchanged. Their clocks, measuring rods, matter, and light signals belong jointly to the local field organization. They continue to measure

$$c = \text{constant}.$$

An external observer, however, can read the same region as slowed. For that observer, emission, motion, frequency, timing, and internal reorganization appear stretched. This slowing does not mean that c has become locally smaller, but that the bound field organization of the tensioned region is

translated less freely into the external comparison order.

$$\text{local measurement} \Rightarrow c = \text{constant},$$

$$\text{external observation} \Rightarrow \text{stretched timing of bound processes}.$$

Time dilation thereby becomes possible without violating the constancy of the speed of light. The time form of a system can change because its bound field organization times differently under tension. The free limiting rate c, however, remains the unchanged reference constant by which this timing change becomes measurable at all.

If, in strongly tensioned fields, the free limiting rate c itself were to decrease in addition to the bound timing, a double slowing of dynamics would arise. In the limiting case of maximum saturation, this could lead to a genuine standstill of field reorganization. The UQSH avoids this contradiction by understanding saturation not as an abolition of c, but as binding and stretching of regime-internal readability.

$$\text{bound timing} = \text{regime- and tension-dependent},$$

$$c = \text{cross-regime constant of free field motion}.$$

c is the first constant of field motion; time dilation changes the readability of bound processes, not the free limiting motion of the field.

Besides c, two further quantities emerge within the UQSH framework as especially fundamental: π and φ. They do not, however, denote the same kind of constant. c describes the first constant of field motion itself. π describes the geometrical basic form of free propagation. φ is regarded as a strong candidate for a scale-spanning growth and stability proportion.

$$\pi = \text{constant of the spherical propagation form},$$

$$\varphi = \text{candidate of a stable growth and redistribution proportion}.$$

The meaning of π follows from the free propagation of the field. If a light bubble is not locally bound or directed, it propagates spherically in the idealized limiting case. The sphere is therefore the simplest free motion form of the field. Its surface grows with

$$A(r) = 4\pi r^2.$$

Thus, within the UQSH framework, π appears not only as a geometrical number, but as an expression of the only completely free basic form of field motion: spherical unfolding. Where no preferred

direction, no binding, and no coherent twisting dominate, free field motion distributes itself radially in all directions.

$$\text{free field motion} \rightarrow \text{spherical propagation} \rightarrow \pi.$$

π thereby describes the rounding and propagation order of the field. It is the constant of free field unfolding, not in the sense of an additional force, but as the geometrical form condition of every spherical light bubble.

The situation is different with φ, the golden ratio. This quantity is not posited here as an already proven absolute field constant. It appears, however, as a strong candidate for a preferred growth proportion of stable field organization. While π describes free spherical unfolding, φ could describe ordered redistribution once field motion no longer merely expands freely, but at the same time rotates, couples, twists, and stabilizes.

$$\varphi = \frac{1 + \sqrt{5}}{2}.$$

In this view, φ does not simply describe an aesthetic proportion, but a possible stability condition for growth while preserving internal order. A field system can grow without losing its form if expansion, rotation, relief, and coupling remain in a favorable relation to one another. This is precisely where the connection to spiral orders, magnetic twisting, and scale-spanning stabilization arises.

$$\text{growth} + \text{rotation} + \text{coupling} + \text{relief} \rightarrow \text{stable proportion}.$$

The strong assumption is therefore: if magnetism is the ordering twisting of the field, then φ could mark the proportion in which this twisting is distributed especially stably across scales. Magnetism directs field tension, forms axes, channels, and spiral orders. φ could indicate the growth proportion in which such orders preserve their form during ongoing redistribution.

$$\text{magnetic twisting} + \text{scale-relative redistribution} \rightarrow \varphi\text{-like stability order}.$$

This yields a first ordering of the three constants:

c = constant of free field motion,
π = constant of the spherical propagation form,
φ = candidate of stable growth and redistribution order.

In this order, c is the dynamic fundamental constant, π the geometrical fundamental constant of free propagation, and φ a strong working hypothesis for stable scale-spanning organization. The first two quantities are directly connected, within the UQSH framework, with free field motion and spherical unfolding. The third quantity must be examined further, but possesses a natural role where growth, magnetism, rotation, spiral formation, and stabilization interact.

30. Formalization Program

The present work remains predominantly conceptual in its ontological interpretation. It does not, however, stand outside established mathematical formalizations. Many central relations of physics, such as $E = mc^2$, the Newtonian approximation of gravitation, the relation between frequency and energy, or the invariance of the speed of light, are not rejected within the UQSH framework, but read as already measured limiting and coupling forms of a deeper field order.

The formalization program therefore does not consist in reinventing all known equations, but in working out their shared field-mechanical reading. Established formulas can be understood as limiting cases, regime projections, or effective descriptions of Q, Φ, and $B(\Phi)$. For a further mathematical elaboration, several fundamental quantities must be clearly distinguished:

Q Qu-ground; the deepest ontological foundation of the UQSH.

Φ Physically readable field projection of Q.

$B(\Phi)$ Motion of this field projection.

$K^{(n)}(x,t)$ Field curvature or tension curvature of a regime n.

$M^{(n)}(x,t)$ Modulation structure within a carrier front.

$S_n(K)$ Saturation function of the regime.

$\mathcal{F}_n$ Coupling filter between field organization and observational regime.

R_n Reorganization rate whose limiting value appears in our regime as c.

The term *projection* in the case of Φ does not mean virtuality or mere apparentness. What is meant is the real field form in which Q becomes physically accessible for a regime.

The UQSH basic order is thereby formally set as an ontological and dynamic stratification:

$$\boxed{Q \rightarrow \Phi \rightarrow B(\Phi)} \tag{15}$$

Here, Q denotes the Qu-ground, Φ its physically readable field projection, and $B(\Phi)$ the motion of this field projection. From this motion arise the first physical readings:

$$\boxed{B(\Phi) \Rightarrow L, , E, , \mathfrak{G}} \tag{16}$$

Here, L denotes the freely propagating motion form of the field projection, E its motion content or its timing and frequency order, and $\mathfrak{G}$ its regime-effective curvature and tension side. Light and gravitation are thereby not equated. They appear as different readings of the same underlying field motion.

A simple structural form for observable modulations would be:

$$\text{observed modulation} = \mathcal{F}_n \left[M^{(n)} \left(K^{(n)}, K^{(n-1)}, K^{(n+1)} \right) \right]. \tag{17}$$

For redshift, a dedensification function would have to be formulated:

$$\nu_{\text{obs}} = \nu_{\text{em}}, E_{\text{path}}^{-1}, \tag{18}$$

where E_{path} describes the cumulative field-mechanical dedensification along the propagation path.

For gravitation, it would accordingly have to be examined whether the observed excess can be written as a saturated projection of bound and free field tension:

$$g_{\text{obs}} = g_{\text{bar}} + \mathcal{F} * \text{gal}\left[S_n\left(K * \text{bind} + K_{\text{free}}\right)\right]. \tag{19}$$

Here, K_{bind} describes the tension curvature of bound field organization, such as baryonic matter, while K_{free} describes the tension curvature of freely propagating light bubbles. Free light tension fronts therefore do not stand outside field reality. They are free motion forms of Φ and carry timing, tension, and curvature relation. Their gravitational effectiveness, however, is not identical with classical gravitation, but the regime-dependently readable curvature side of the same field motion.

The classical Newtonian limiting case is preserved if the filtered field tension in the weak, quasi-static regime is read as effective energy or mass:

$$g(r) = G_N \frac{E_{\text{eff}}}{c^2 r^2}. \tag{20}$$

Here, G_N is not gravitation itself, but the empirical coupling scale by which our regime measures field tension as acceleration.

These equations are not meant as a finished theory. They show which building blocks are necessary: Qu-ground, field projection, field motion, saturation, coupling, modulation, dedensification, free and bound field tension, and regime projection.

A concrete application of this program is the energetic regime width between Anektron and Anun.

30.1. Initial Derivation of the Regime Width between Anektron and Anun

The concepts *Anektron* and *Anun* allow a first formalizable description of the energetic span of a scale-relative regime. A regime (R_n) is thereby not defined by an arbitrary size scale, but by two saturation limits: a minimal saturation form from which an independent coupling unit becomes measurable, and a maximum saturation form from which local deepening can no longer be continued.

As defined above, the *Anektron* denotes the lower saturation limit of a regime, while the *Anun* describes the upper saturation limit. For the formalization, it is decisive that both limiting forms can be treated as characteristic energies of the same regime.

For a regime R_n, we therefore define two characteristic energies:

$$E_{\text{A}}^{(n)}$$

as the energy of the minimally saturated Anektron form and

$$E_{\mathrm{U}}^{(n)}$$

as the energy of the maximally saturated Anun form.

This yields the energetic span of a regime as

$$\Lambda_n = \frac{E_{\mathrm{U}}^{(n)}}{E_{\mathrm{A}}^{(n)}}.$$

This quantity Λ_n describes not simply a mass ratio, but the linear saturation span of a regime: the distance between first measurable saturated coupling and maximum local field saturation.

For our observational regime, the electron rest energy can be used as a first calibration point for the lower coupling limit:

$$E_{\mathrm{A}}^{(n)} \approx E_e = m_e c^2.$$

This does not mean that electron and Anektron are completely equated, but that the electron energy serves as an observable approximation of the minimal coupling form.

As the upper calibration, the minimal black-hole limit can be used. In this view, a black hole is the strongest candidate for an Anun of our regime:

$$E_{\mathrm{U}}^{(n)} \approx M_{\mathrm{BH,min}} c^2.$$

It follows for the regime width:

$$\Lambda_n = \frac{M_{\mathrm{BH,min}} c^2}{m_e c^2} = \frac{M_{\mathrm{BH,min}}}{m_e}.$$

The speed of light cancels out. In this first approach, the regime width is therefore a pure mass or energy ratio between lower minimal coupling and upper maximum saturation.

If one inserts approximately a few solar masses for the minimal black-hole limit,

$$M_{\mathrm{BH,min}} \sim 2 - 5\, M_\odot,$$

then Λ_n lies, as an order of magnitude, at about 10^{61}. For the frequently used calibration value

$$M_{\mathrm{BH,min}} \approx 3\, M_\odot$$

one obtains

$$\Lambda_n \approx \frac{3 M_\odot}{m_e} \approx 6.5 \times 10^{60}.$$

The quadratic span is especially interesting:

$$\Lambda_n^2 \sim 10^{122}.$$

This order of magnitude lies in the range of the known discrepancy between formally expected vacuum energy and observed gravitational effect and motivates the following working hypothesis:

$$\frac{\rho_{\text{vacuum}}^{\text{formal}}}{\rho_{\text{grav}}^{\text{observed}}} \sim \Lambda_n^2.$$

Here, $\rho_{\text{vacuum}}^{\text{formal}}$ denotes the vacuum energy density formally expected from field fluctuations, and $\rho_{\text{grav}}^{\text{observed}}$ the actually observable gravitational effect. The quadratic dependence is not claimed here as a final derivation, but as an approach: if coupling between regimes acts not linearly, but projectively, surface-like, or through two-sided filtering, the observable discrepancy can appear as the quadratic trace of the regime width.

This gives rise to a UQSH interpretation of the vacuum-energy discrepancy: not the entire formal field energy has to become gravitationally effective in our regime. Only a filtered portion of the nested field organization couples as observable gravitation. The known discrepancy would then not be the expression of too much energy, but of too little direct coupling between total field motion and our observational regime.

The regime relation can therefore be summarized as

$$\rho_{\text{field}}^{\text{total}} \gg \rho_{\text{vacuum}}^{\text{formal}} \gg \rho_{\text{grav}}^{\text{observed}}.$$

The total field motion can be substantially larger than the vacuum energy calculated formally up to the Planck limit. This in turn can be substantially larger than the actually observed gravitational projection. The reason lies in saturation, coupling filtering, and regime limitation.

The regime transition itself can be written as a recursive relation:

$$E_{\text{A}}^{(n+1)} \approx E_{\text{U}}^{(n)}.$$

This means: the Anun of one regime can assume the function of an Anektron in the next higher regime. The end of maximum saturation in one regime thereby becomes the beginning of minimal coupling capacity in the next regime.

This yields the scale-relative chain:

$$E_{\text{A}}^{(n)} \rightarrow E_{\text{U}}^{(n)} = E_{\text{A}}^{(n+1)} \rightarrow E_{\text{U}}^{(n+1)}.$$

In words: a regime begins with minimal saturation, unfolds its matter and structural forms within these limits, reaches maximum saturation in its Anun, and precisely thereby can be translated into the next higher regime.

The relation developed here is not a completed mathematical derivation of the electron, the black hole, or the cosmological constant. It does, however, represent a grounded initial derivation. It connects three previously separate quantities within a shared saturation framework:

electron/Anektron, black hole/Anun, vacuum-energy discrepancy.

The central UQSH approach is therefore:

$$\boxed{\Lambda_n = \frac{E^{(n)}_{\text{Anun}}}{E^{(n)}_{\text{Anektron}}} \sim 10^{61}}$$

and

$$\boxed{\Lambda_n^2 \sim 10^{122}.}$$

The first quantity describes the linear energetic width of our regime. The second quantity could represent the quadratic coupling or projection trace of this regime width in the observed vacuum-energy discrepancy.

31. Astrophysical Mass Gap as a Saturation Zone

In the astrophysical context, the so-called mass gap, that is, an observed mass gap between compact stellar remnants and black holes, can be understood within the UQSH framework as a transition domain between maximally supportable matter organization and minimally stable Anun formation. A compact star can carry its field organization only up to a certain limit. If this limit is exceeded, the previous matter structure is no longer sufficient to hold the local field tension stably. The field must reorganize. However, a fully coherent Anun form does not necessarily arise immediately. Between the two states, an unstable saturation zone can therefore lie.

Formally, this domain can be written as

$$M^{(n)}_{\text{max,matter}} < M < M^{(n)}_{\text{Anun,min}}.$$

Here, $M^{(n)}_{\text{max,matter}}$ denotes the maximum mass that can still be carried as compact matter organization in the regime R_n, while $M^{(n)}_{\text{Anun,min}}$ denotes the minimum mass from which a durable Anun function becomes possible. In this view, the mass gap would not be an empty domain in the absolute sense, but an unstable reorganization domain: too strongly densified for ordinary matter stability, but not yet fully coherently formed as an upper saturation form.

The astrophysical mass gap is the unstable zone between matter supportability and Anun coupling.

With growing mass, range, and boundary structure, such a state can enter the Anun transition band. The minimum black-hole mass then does not mark the end of all dynamics, but the first clear threshold from which maximum saturation can occur as an upper regime form.

31.1. Yang–Mills Mass Gap in Light of the UQSH

The Yang–Mills mass-gap problem is not mathematically solved here, but only interpreted from the perspective of the UQSH. In the usual formulation, the question is that a quantized non-Abelian gauge theory possesses, despite massless fundamental fields, a positive distance between the vacuum and the first physical massive excitation state. Schematically, this can be written as an energy gap:

$$\Delta E_{\mathrm{YM}} = E_1 - E_0 > 0.$$

Within the UQSH framework, this gap can be read as the expression of a minimal saturation threshold. Free field motion or massless fundamental fluctuation does not automatically become observable as an independent inertial state. Only when the field organization crosses a minimal binding and saturation threshold does a stable, mass-carrying and coupling-capable mode arise.

The simplest UQSH reading initially formulates this threshold through the saturation condition:

$$E_{\mathrm{A}}^{(n)} = \min \left\{ E \;\middle|\; S_n(K) \geq S_{\mathrm{min}}^{(n)} \right\}.$$

The later supplemented coherence condition further specifies this threshold. Here, $E_{\mathrm{A}}^{(n)}$ denotes the minimal Anektron energy of a regime, $S_n(K)$ the saturation function of the field curvature or field organization K, and $S_{\mathrm{min}}^{(n)}$ the minimal saturation threshold from which an independent inertial coupling mode can arise. In this reading, the Yang–Mills mass gap does not appear as an additional property beside field motion, but as the formal trace of that threshold at which free field organization first passes into measurable inertia.

Within the UQSH, the Yang–Mills mass gap corresponds to the minimal saturation threshold between free field motion and inertial coupling form.

This interpretation does not replace a mathematical Yang–Mills derivation. It merely places the problem within the language of the UQSH: mass appears there not as an isolated substance, but as the result of minimal saturation, binding, and regime-capable stabilization.

31.2. Coherence, Hysteresis, and Observation Filter

For the formation of stable field forms, high energy or field tension alone is not sufficient. The field organization must be sufficiently coherent spatially, temporally, and geometrically in order to couple as an independent mode. At the same time, saturation possesses hysteresis: a saturated state once reached can persist even when the original excitation partly decreases. Finally, every observational regime reads such states only filtered through its energy, time, and structure resolution. These three conditions are formally supplemented in the following.

31.3. Formal Supplement: Coherence, Hysteresis, and Observation Filter

For the coupling capacity of a field state, high energy or field tension alone is not sufficient. A state must at the same time be sufficiently coherent in order to appear as an independent field form within a

regime. We therefore introduce a coherence function

$$C_n(x,t) \in [0,1]$$

for a regime R_n. It describes how strongly a field state is ordered spatially, temporally, and geometrically. Formally, it can be written as the product of three contributions:

$$C_n = C_{\text{space}}^{(n)}\, C_{\text{time}}^{(n)}\, C_{\text{phase}}^{(n)}.$$

Here, $C_{\text{space}}^{(n)}$ describes the spatial stability of the field form, $C_{\text{time}}^{(n)}$ its duration relative to the temporal resolution of the regime, and $C_{\text{phase}}^{(n)}$ its internal phase or geometry order.

A field state becomes readable as its own coupling form only when both its saturation and its coherence exceed a threshold:

$$S_n(K) \geq S_{\text{min}}^{(n)} \qquad \text{and} \qquad C_n \geq C_{\text{crit}}^{(n)}.$$

This yields for minimal Anektron formation:

$$E_{\text{A}}^{(n)} = \min\left\{ E \;\middle|\; S_n(K) \geq S_{\text{min}}^{(n)} \;\wedge\; C_n \geq C_{\text{crit}}^{(n)} \right\}.$$

For upper Anun formation, correspondingly:

$$E_{\text{U}}^{(n)} = \min\left\{ E \;\middle|\; S_n(K) \geq S_{\text{max}}^{(n)} \;\wedge\; C_n \geq C_{\text{U}}^{(n)} \right\}.$$

Here, $S_n(K)$ denotes the saturation function of the field curvature or field organization K, while $C_{\text{U}}^{(n)}$ describes the coherence necessary for a maximally saturated state not only to arise transiently, but to be carried as a stable Anun form.

Saturation also possesses hysteresis. Entry into a saturated state and exit from it do not necessarily occur at the same threshold. We therefore distinguish an activation threshold $S_{\text{on}}^{(n)}$ and a maintenance threshold $S_{\text{off}}^{(n)}$:

$$S_{\text{on}}^{(n)} > S_{\text{off}}^{(n)}.$$

A state can thus remain stable even if the original saturation excitation has already somewhat decreased:

$$S_n(K) \geq S_{\text{off}}^{(n)} \quad \Rightarrow \quad \text{saturated form remains preserved.}$$

This hysteresis explains the longevity of stable field forms. Once formed, inertia does not immediately dissolve as soon as the external excitation decreases. The field organization possesses a memory of its saturation.

For observation by a regime R_n, an additional coupling filter can be introduced:

$$\mathcal{F}_n = \mathcal{F}_E^{(n)}\, \mathcal{F}_t^{(n)}\, \mathcal{F}_L^{(n)}.$$

Here, $\mathcal{F}_E^{(n)}$ describes the energy resolution, $\mathcal{F}_t^{(n)}$ the temporal resolution, and $\mathcal{F}_L^{(n)}$ the structure or length resolution of the observational regime.

The observable field effect of a state then does not result from the total field energy, but from the filtered, coherent, and saturation-weighted projection:

$$W_{\text{obs}}^{(n)} = \mathcal{F}_n \left[C_n \, S_n(K) \, E_{\text{field}} \right] .$$

A field state appears as its own object when

$$W_{\text{obs}}^{(n)} \geq W_{\text{crit}}^{(n)} .$$

This condition connects energy, coherence, saturation, and observability. It shows why not every fluctuation forms matter, why not every large mass immediately possesses Anun function, and why not every existing field organization is measured as its own object.

In summary:

$$\boxed{\text{Coupling = filtered effect from energy, coherence, and saturation.}}$$

Or formally:

$$\boxed{W_{\text{obs}}^{(n)} = \mathcal{F}_n \left[C_n \, S_n(K) \, E_{\text{field}} \right] \geq W_{\text{crit}}^{(n)} .}$$

In simple language, this says:

A regime does not see the whole field energy. It sees only the portion that is saturated enough, coherent enough, and readable through the observation filter of the regime.

32. Falsifiable Core Points

The UQSH does not stand or fall by whether every detail of its interpretation is immediately and completely formalized. What is decisive is whether its basic assumptions can in principle be refuted. The following points mark those places where the theory cannot evade arbitrarily.

32.1. No Ontologically Real Particles

A central point concerns the particle model. The UQSH does not deny that measurements show discrete particle events. It denies, however, that these events go back to ontologically independent small bodies. In this view, an electron, a positron, or a photon is not an isolated thing, but a stable or free coupling form of the field.

The UQSH is therefore falsifiable: if it could be shown that elementary particles exist as ontologically isolated objects independently of any underlying field organization, the core approach would be false. Likewise, the UQSH would be weakened if stable discrete states were fully explainable without structural stability conditions of the field.

$$\boxed{\text{Real isolated particle substance would refute the UQSH.}}$$

32.2. No Gravitation without Field Organization

In this view, gravitation is not an additional force, but the geometrical effect of bound or persistent field organization. It follows that gravitational effect must not occur completely independently of field structure, coherence, stability, and organizational history.

The UQSH would be falsified if gravitational excesses occurred durably and systematically without relation to baryonic structure, lensing geometry, field coherence, radiation history, or large-scale organization.

A gravitational excess without any trace of field organization $\Rightarrow$ contradiction to the UQSH.

32.3. No Arbitrary Decoupling of Matter and Field

Matter is bound field motion. Matter must therefore not appear as a substance that is completely detachable from its field binding. Its mass, inertia, stability, and coupling must remain expressions of its internal organization.

The UQSH would be falsified if matter were demonstrated as a completely field-independent substance: stable, inertial, gravitationally effective, and interacting, without any underlying field structure or coupling order being required.

$$M = B_{\text{bound}}(\Phi) + K_{\text{stabilized}}(B(\Phi)).$$

Field-free matter would be incompatible with the UQSH.

32.4. No Variable Speed of Light as a Local Fundamental Quantity

The speed of light c is the first constant of free field motion. Bound processes can appear stretched, clocks can run more slowly, frequencies can be shifted. But locally, c must not appear as a variable material property of the field.

The UQSH would be falsified if, in a local inertial system, a genuine change of c as the fundamental speed of free field reorganization were demonstrated, not merely a change of path, medium, measurement procedure, refraction, time dilation, or observer perspective.

$$c = \text{first constant of free field motion.}$$

A local variability of c as the fundamental field rate would refute the UQSH.

32.5. No Infinite Saturation

The UQSH assumes that field tension cannot be concentrated locally without limit. Extreme states must pass into saturation, redistribution, boundary formation, or relief. Mathematical singularities must therefore not occur as real physical states.

The UQSH would be falsified if true infinite density, true infinite curvature, or a physical point without extension and without regime transition were demonstrated.

$$K_{\text{local}}(B(\Phi)) \to K_{\text{sat}},$$

$$K_{\text{local}} \nrightarrow \infty.$$

A real physical singularity would be a hard contradiction.

32.6. No Purely Random Regime Formation

The UQSH claims that stable structures do not arise arbitrarily, but through coupling, thresholds, saturation, and geometrical stabilization conditions. Transitions between states should therefore not be distributed purely randomly, but show recurring organizational patterns.

The UQSH would be falsified if stable matter, galaxy, or field states showed no thresholds, no preferred stability forms, no regime formation, and no recurring coupling orders.

$$\text{Stability} = \text{coupling} + \text{threshold} + \text{organization}.$$

Completely patternless stability contradicts the UQSH.

32.7. Concise Falsification Core

The UQSH can therefore be reduced to a few hard test stones:

1. Are there truly field-free particles?
2. Is there field-free matter?
3. Is there gravitation without any organizational trace?
4. Is c locally variable as a fundamental rate?
5. Are there real physical singularities?
6. Do stable structures arise without thresholds and patterns?

The UQSH is false if reality consists of isolated particles,
field-free matter, boundless singularity
or patternless stability.

33. Experimental Test of Wave-Particle Dualism

Within the UQSH framework, wave-particle dualism can be tested by an extended double-slit or single-photon experiment. The central question is not whether light shows wave-like and point-like

properties. This observation is established. Rather, what is to be tested is whether the behavior can be understood as a combination of extended field motion and local coupling.

Within the UQSH framework, a light excitation is not understood as a small object flying through a slit. It is interpreted as free, spherically propagating field motion of Φ. Detection appears point-like because the extended field motion couples only locally to a bound field knot of the detector. The wave-like character thus belongs to the propagation of the front, while the particle-like character belongs to the local measurement coupling.

A suitable experiment would therefore have to examine three levels simultaneously:

- the spatial propagation of the light front,
- the internal phase, frequency, and coherence structure of the front,
- the local coupling to detector matter.

An idealized setup would consist of a single-photon source as point-like as possible or a source of ultrashort light pulses, a double-slit or multi-slit arrangement, and a detector plane with high spatial and temporal resolution. In addition, a weak reference or test pulse could be used to test whether the passage of the light front leaves behind a short-lived change in the local coupling conditions.

Schematically:

$$\boxed{\text{emission} \rightarrow \text{spherical field front} \rightarrow \text{interference structure} \rightarrow \text{local detection}} \tag{21}$$

In the standard picture, the interference pattern is described as a probability distribution for individual photon events. The UQSH adopts this observable structure, but reinterprets it field-mechanically: it is not a point-like particle that interferes with itself, but an extended field front that passes through both slits as a coherent motion structure. The single point on the screen is not the light itself, but the local coupling event between the front and a bound state of the detector.

A first test quantity is therefore the stability of the interference pattern at strongly reduced intensity. If the source is attenuated so that the detection events occur individually, the pattern should continue to build up statistically:

$$N(x) \propto I_{\text{front}}(x), \tag{22}$$

where $N(x)$ describes the frequency of local detection events and $I_{\text{front}}(x)$ the intensity structure of the extended field front at the location x. Within the UQSH framework, this means: the point structure of the measurement arises locally, while the underlying propagation remains front-like.

A second test quantity concerns the question of whether the detection itself depends on the coupling capacity of the screen. Different detector materials, temperatures, field environments, or polarization filters should not only change the number of hits, but also provide indications of how the extended field front couples locally into bound states. Formally, this can be written as a coupling filter:

$$N(x) \propto \mathcal{F}_{\text{det}}\left[I_{\text{front}}(x), M_{\text{front}}(x), K_{\text{det}}(x)\right]. \tag{23}$$

Here, $\mathcal{F}_{\mathrm{det}}$ denotes the detector coupling filter, $I_{\mathrm{front}}(x)$ the local intensity of the front, $M_{\mathrm{front}}(x)$ its modulation structure, and $K_{\mathrm{det}}(x)$ the local coupling state of the detector material.

A third and more UQSH-specific test quantity would be a possible aftereffect of the front on a subsequent test pulse. For this purpose, a strong, short pump pulse could be sent through the arrangement, followed by a weak probe pulse. One would search for a very small, short-lived change in phase, travel time, polarization, or coherence of the probe pulse:

$$\Delta\varphi_{\mathrm{probe}} = \varphi_{\mathrm{probe}}^{\mathrm{after}} - \varphi_{\mathrm{probe}}^{\mathrm{before}}. \tag{24}$$

Such a measurement would be relevant to the UQSH only if it cannot be explained by known effects such as heating, scattering, material response, nonlinear optics, or electromagnetic noise. The decisive test would consist in whether a propagating light front generates a measurable, spatially ordered, and temporally short-lived change in the coupling conditions.

The experimental core can therefore be summarized as follows:

$$\boxed{\begin{aligned} \text{wave} &= \text{extended field motion} \\ \text{particle} &= \text{local coupling event} \\ \text{dualism} &= \text{regime transition between propagation and detection} \end{aligned}} \tag{25}$$

The UQSH therefore does not predict that the known double-slit experiment must produce different hit patterns. Rather, it predicts that the known hit patterns can be understood as local coupling of a really extended field front. This interpretation becomes testable above all where propagation, interference, detector coupling, and possible aftereffects of the same front are measured together.

34. Prediction on the Emergence of Life in the Universe

The UQSH yields a clear qualitative prediction: life should not be a unique special case in the universe. It becomes repeatedly possible wherever stable coupling windows of matter, energy flow, chemical diversity, and suitable environmental conditions arise over sufficiently long periods of time. This does not mean that life must necessarily be present on every suitable celestial body. It does mean, however, that similar conditions should repeatedly favor similar organizational chains.

The reason lies in the recurring mode of operation of the field. Free field motion can bind itself, bound states enable matter, matter enables chemistry, and chemistry can, under suitable conditions, pass into self-stabilizing structures. Life thereby appears not as a principle added from outside, but as a possible continuation of the same order: stability, coupling, energy flow, uncertainty, reorganization, and adaptation.

Water plays a special role. It provides a broad chemical coupling space, can dissolve and transport substances, buffer temperature, enable charge separation, and favor molecular self-organization. In this view, water is not magical, but an especially effective medium in which flexible chemical and field-bound couplings can work durably.

The search for life-friendly environments should therefore not remain restricted to Earth-like planets. Subsurface oceans, icy moons, hydrothermal systems, water-rich dwarf planets, dense atmospheres,

interfaces between ice, rock, and water, as well as chemically active boundary zones of planets and moons, can also form suitable spaces. What is decisive is not the external similarity to Earth, but the durable stability of a chemical reorganization space.

$$\text{water} + \text{energy flow} + \text{chemical gradients} + \text{temporal stability} \rightarrow \text{life-capable self-organization}.$$

From this follows a second expectation: life forms should take different forms depending on their environment. In water-rich oceans, different structures would be expected than in icy worlds, hydrothermal boundary spaces, dense atmospheres, or strongly radiation-loaded domains. Life would then not be a fixed form, but a local solution: it assumes the form that can remain stable under the existing energy, water, temperature, matter, and coupling conditions.

This view becomes testable through the frequency and diversity of prebiotic chemistry. If the UQSH interpretation is correct, complex organic molecules, water chemistry, stable chemical gradients, and precursors of biological self-organization should occur in many places: in protoplanetary disks, comets, ocean moons, the Martian subsurface, exoplanet atmospheres, or hydrothermal systems. Recurring chemical patterns in very different environments would be especially meaningful.

The UQSH expects life not as a one-time accident, but as a recurring possibility of stable field, water, energy, and chemical coupling.

If the universe is not reduced to a short, one-time emergence window, the possible time window for life also expands. Life forms can repeatedly arise, persist, vanish, and appear again under new conditions. Some lines of life could be very young, others already past, and still others adapted over long periods of time to stable coupling windows.

Consciousness would not be a foreign body in this order. It can be understood as a higher degree of self-preserving organization. A living system not only stabilizes its structure, but can form models of its environment, its boundaries, and its possible actions. Consciousness would then be the self-reference of an organization that co-regulates its own preservation and its future couplings.

This view does not claim that every life form possesses consciousness. It orders consciousness as a possible higher stage: life stabilizes itself; conscious life recognizes and steers parts of this self-stabilization. Consciousness thereby remains anchored in the same chain of development:

$$\text{field motion} \rightarrow \text{matter} \rightarrow \text{chemistry} \rightarrow \text{life} \rightarrow \text{self-reference}.$$

More highly developed life forms could then be systems that not only exist in the field, but increasingly recognize and use its coupling conditions. Such a development would not necessarily have to become louder, larger, or more energetically conspicuous. The better a civilization understands the mode of operation of the field, the more likely it could work with it rather than against it.

Technological symbiosis means in this sense: motion would not only be thrust against an environment, but controlled coupling to local and nonlocal field states. Energy generation would not merely be combustion or coarse release of stored matter, but targeted relief, densification, or reorganization of existing field tension. Matter generation would not be creation from nothing, but the transfer of free

field motion into bound stable field knots.

A very highly developed life form could therefore appear quiet but effective on a cosmic scale. Its technology would not have to generate strong waste traces, massive thermal signatures, or violent energy excesses, if it couples, redirects, and rebinds field tension efficiently. It would appear less like a consumer of the universe, and more like a local co-organizer of the field: stabilizing, relieving, densifying, shifting, or connecting.

From this perspective, the search for extraterrestrial life also expands. One would search not only for biological traces or strong technical emissions, but also for signs of unusually coherent field organization: low-loss motion, controlled relief zones, abrupt changes of direction without classical propulsion logic, altered inertial responses, artificially stabilized matter forms, or structures in which energy flow, matter organization, and motion interact with striking precision.

Such considerations remain speculative, but follow from the assumption that matter, motion, gravitation, energy, and life are different organizational forms of the same field. A highly developed life form might then not be the loudest appearance in the universe, but the one that works most quietly and precisely with its basic motion.

Advanced life would not have to stand out through maximum energy output,
but could reveal itself through especially precise
coupling to the field order.

35. Outlook: Field Coupling, Location, and Nonclassical Motion

If space and time are not absolute containers, but readings of the field order, then the concept of motion also changes. A body would not merely move through a ready-made space. It would itself be a stable field knot whose location is determined by its coupling to the surrounding field structure.

In this view, a location is not an empty coordinate. It is a local coupling state: a specific order of field tension, phase, curvature, timing, and connection to neighboring field domains. Motion therefore means not only change of location, but also change of this field binding.

$$\text{classical motion} = \text{traversing a distance},$$

$$\text{field-mechanical motion} = \text{change of coupling to field states}.$$

This gives rise to a speculative possibility. If a system could change its own coupling deeply enough, it would not have to traverse every distance classically. It could attempt to modulate the local field order in such a way that it becomes connectable to the coupling signature of a distant domain. The target location would not thereby be brought closer; rather, the system's own field binding would be switched to another spatial relation.

$$\text{target location} \sim \text{coupling signature of a field domain},$$

$$\text{travel} \sim \text{transition between coupling signatures.}$$

Such motion would not be classical faster-than-light flight. The local reorganization of the field would remain bound to c. What would be changed is not the speed of light, but the assignment of the system to a spatial field reading. Distance would then not merely be length, but coupling difference.

$$\text{faster-than-light flight} \neq \text{field-coupling transition,}$$

$$\text{location} = \text{stabilized field relation.}$$

For this, several conditions would have to come together. The local field tension would have to be controllably alterable. The matter form of the system would have to remain stable even though its external field binding is switched. The intended target signature would have to be generated in a phase- and structure-compatible way. And the process would not be allowed to violate the local limiting rate c.

The decisive point is inertia. In the normal regime, a body cannot be accelerated arbitrarily because its bound field organization resists reorganization. If, however, the coupling to the surrounding field geometry could be changed, the required reorganization work could also change. The body would then not simply be accelerated against its inertia, but brought into a state in which its inertial response is organized differently.

$$\text{inertia} = \text{resistance of bound field organization against reorganization,}$$

$$\text{inertia modulation} = \text{change of the coupling conditions of this organization.}$$

From this perspective, reports of abrupt motions, apparently absent inertia, or very rapid changes of direction would not be explainable by classical propulsion, but at most by coupling, field geometry, and inertia modulation. This is not a claim that such phenomena are technically or really already understood. It only shows where a field-mechanical explanation would have to begin.

The popular idea that one would have to copy the "vibration" of a location can also be formulated more soberly. What would be meant is not a mystical vibration, but a specific bundle of frequency structure, field tension, phase, curvature, saturation, local timing, and coupling geometry. A location would not only be a coordinate, but a field state. Whoever could make this state controllably connectable would not copy the matter of a target location, but alter the relation to its field order.

It would not be the distance that has to be overcome, but the coupling difference between two field locations.

Such an outlook lies far beyond today's technical and experimental state. It nevertheless remains logically connectable: if location and time emerge from field relations, then the deepest form of locomotion would not be propulsion through space, but controlled change of the field readability of location and time.

36. Comparison with Established Models

The UQSH does not replace general relativity. General relativity remains the precise geometrical description of gravitational effects. The UQSH sets in one level below it and asks what could physically carry this geometry. Spacetime curvature is therefore not rejected, but read as the expression of a real field tension.

The cosmological standard model is also not simply pushed aside here. CMB, redshift, lensing, structure formation, and galaxy distribution remain real findings. The UQSH reads them differently, however: not necessarily as the consequence of an expansion of passive space from a singularity, but as possible consequences of ongoing field reorganization, dedensification, saturation, and coupling thresholds.

Compared with particle models of dark matter, the UQSH avoids the assumption of an additional invisible substance. The dark-matter effect is interpreted as organized, saturation-dependent field tension. Compared with MOND [23], the UQSH shares the insight that baryonic structure is central for the dynamics. It interprets this connection, however, not as a mere change of an acceleration law, but as a field response to stable matter organization.

Related approaches include thermodynamic and entropic interpretations of gravitation [20–22]. With them, the UQSH shares the idea that gravitation may not be an isolated fundamental force. The difference lies in the carrier interpretation: gravitation, light, spectrum, CMB, black holes, the Planck limit, and the dark-matter effect are not treated separately here, but as different appearances of a dynamic field medium with tension, saturation, coupling, and free field motion.

37. Possible Misunderstandings and Limits

The UQSH should not be understood as ignoring established measurements or successful mathematical models. Its strength must lie precisely in not explaining away known findings, but in reading them differently. The question is therefore not whether spectra, redshift, lensing, rotation curves, or the CMB exist. The question is what field organization could stand behind these observations.

Nor is it claimed that an electromagnetic spectrum already contains the complete field configuration of a galaxy. A spectrum is only one layer. A robust reconstruction would have to bring together several levels: spectrum, rotation curve, baryonic structure, gas distribution, lensing, large-scale environment, and dynamic activation.

The analogy between galaxies and atoms is also not an equation. It concerns not size, substance, or concrete components, but a role logic. Bound organization can characteristically modulate free field motion. Atoms do this in the electromagnetic regime; galaxies could do so on a higher scale through gravitational and electromagnetic field organization.

The data analyses so far should therefore be read as first test traces. The SPARC-q-regimes show structural differences, while simple WISE, GALEX, and SDSS colors have so far not yielded a robust separation. The template smoothing analysis supports the idea of coarse spectral organizational forms, but remains limited by the small number of templates. For a robust test, larger, homogeneously prepared datasets and a joint evaluation of several observational levels are needed.

38. Conclusion

The present work develops *gravitational light* as a scale-relative interpretation of radiation, matter, gravitation, spectrum, and observation. The starting point is the assumption that the Qu-foam is not the background or stage of physics, but the common dynamic fundamental structure in which states arise, couple, relax, and reorganize themselves.

The central idea is: radiation, matter, and gravitation are not ontologically separate things. They are different organizational forms of the same field motion. Radiation is the free, propagating form. Matter is the bound and stabilized form. Gravitation is the field response to curvature, tension, and bound organization.

radiation = free field motion,

matter = bound field motion,

gravitation = geometrical response of the field to organization.

This shifts the physical question. Individual objects are not in the foreground, but the transitions between free motion, binding, dedensification, densification, modulation, and observability. Light appears as a free field front. Its spectrum is the readable modulation of this front. What is measured is never the complete field reality, but the layer that couples to matter, measuring device, and the observer's regime.

observation = regime-coupled readability of a field order.

In this view, the dark effects also receive a shared language. The dark-matter effect appears as baryonically anchored, radiation-excited, and saturation-limited field tension. Dark energy appears as the relaxation tendency of the field, especially clearly in voids. Cosmological redshift is not denied, but interpreted as dedensification of propagating field fronts.

This gives rise to a unified reading of light, spectrum, dedensification, CMB, galaxies, q-regimes, the dark-matter effect, the dark-energy effect, black holes, the Planck limit, quantum numbers, and interactions. The UQSH does not claim that all details are already formally completed. It does, however, offer a shared language in which many separate problems appear as different states of the same field organization.

Anektron and Anun mark the lower and upper saturation limits of a regime. The Anektron stands for minimal stable coupling capacity, the Anun for maximum local field saturation. The recursive idea is: the Anun of one regime can assume Anektron function in the next higher regime. Saturation is therefore not an end of field motion, but a transition.

Anektron = lower stable coupling limit,

Anun = upper saturation limit of a regime.

The associated uncertainty is not a weakness, but part of stability. At the lower boundary, it prevents absolute fixation of the minimal coupling form. At the upper boundary, it prevents singularity and enables redistribution, growth, and connection to a further regime.

The UQSH thus understands itself as a field-ontological interpretive level. It does not seek to replace known phenomena, but to read them in their shared carrier logic: as different visibilities of a single dynamic field medium.

Explanation does not take away the magic of the universe. It only changes its place. Wonder no longer lies only in the misunderstood, but in the depth of connection. The more order becomes visible, the greater the wonder becomes: that motion, tension, light, matter, and gravitation do not have to be separate riddles, but different expressions of the same fundamental motion.

Energy is field motion;
radiation, matter, and gravitation
are its different organizational forms.

The magic does not become smaller through knowledge, but greater.

38.1. Concluding Remark on the Qu-Ground

At the end of this work, one question remains open that cannot be answered completely with formulas: what actually is the Qu-ground Q?

This work has primarily described physical processes. It dealt with light, energy, gravitation, matter, inertia, magnetism, time, redshift, and dark effects. All these phenomena were interpreted as readings of a deeper field motion. With Q, however, what is meant is not another field within this physics. Q denotes the ground from which a physical field reality can arise in the first place.

This makes the concept difficult to grasp. We are used to connecting reality with something tangible: with bodies, locations, measured values, motion in space, or change in time. But Q lies precisely before this level. For this reason, Q has in this work no definable zero, no definable unit, and no location in space and time. A zero, a one, a scale, or a measured value already belongs to a world in which something has become distinguishable and measurable. Q is not this measurable state, but that from which measurability first becomes possible.

The physically accessible level is therefore designated as Φ. With Φ, what is meant is not an illusory world, not a virtual environment, and not a projection in the sense of a projector image. Φ is the real field reality in which Q appears for a regime as space, motion, tension, curvature, coupling, and organization.

A direct measurement connection between Q and Φ is not claimed here. According to the present state of this work, it is not even completely explainable how Q and Φ can be distinguished according to the physical standards of our universe. For every measurement already presupposes space, time, coupling, and observation. It therefore already takes place within Φ.

Precisely for this reason, Q should not be prematurely understood as a physical object. It is also not a

substance, not a force, and not empty space. Q is a boundary concept. It marks the point at which physical description points to its own origin without being able to measure it itself completely.

Perhaps a simple image may help, without taking it literally. Thoughts can assume infinitely many forms. In thought, something can arise, change, disappear, return, and be ordered anew, without the thought itself being a spatial object. One can imagine worlds within it without this imagination being bound to a location, a distance, or a measurable boundary. In a similar way, Q can be thought of as the intangible ground from which physical forms first emerge.

This does not mean that Q is consciousness in the human sense. It only means that our ordinary images of space, body, and measurability are no longer sufficient for this ground.

At this point, the UQSH also touches concepts that are traditionally used more philosophically or religiously: existence, origin, wholeness, consciousness, God, or power. These concepts are not introduced here as physical definitions. At most, they can indicate why Q should not be understood as a single something within the world. If everything that appears physically emerges from this ground, then Q is not a part of the whole, but the prerequisite for a whole to be able to appear at all.

The question of the beginning and end of the universe also remains open in this way. If cosmological redshift should not arise mainly through an expansion of space, but through the dedensification of propagating field motion, then the visible boundary of the universe would not be an absolute boundary of reality. It would only be the boundary of what our regime can still couple and measure. The universe could then be older, larger and more open.

Acknowledgements

The author gratefully acknowledges the intellectual heritage of the many physicists, mathematicians, astronomers and philosophers whose work has shaped our present understanding of space, time, gravitation, light, matter and field organization. This acknowledgement is not limited to the most famous names or the greatest individual breakthroughs. It also extends to the many researchers, observers, experimentalists, engineers, teachers and critical thinkers whose incremental contributions, measurements, corrections and questions have made modern physics possible.

The author also acknowledges the developers and researchers behind modern AI-based tools. In this work, such tools were used as imagination-extending and editorial aids: for manuscript structuring, linguistic refinement, translation support, code debugging, consistency checks and the clarification of conceptual formulations. They did not replace the author's responsibility for the scientific content.

All interpretations, hypotheses, conclusions, methodological decisions and the final form of the argument remain solely the responsibility of the author.

Citation

Rexhepi, U. Q. (2026).
Gravity-Light.
Independent Researcher, Tübingen, Germany.
Researchgate.net:https://doi.org/10.13140/RG.2.2.33455.91042

References

[1] Lelli, F.; McGaugh, S. S.; Schombert, J. M. SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. *The Astronomical Journal* **2016**, *152*, 157. doi:10.3847/0004-6256/152/6/157.

[2] McGaugh, S. S.; Lelli, F.; Schombert, J. M. Radial Acceleration Relation in Rotationally Supported Galaxies. *Physical Review Letters* **2016**, *117*, 201101. doi:10.1103/PhysRevLett.117.201101.

[3] Planck Collaboration; Aghanim, N.; Akrami, Y.; Ashdown, M.; et al. Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics* **2020**, *641*, A6. doi:10.1051/0004-6361/201833910.

[4] Hawking, S. W. Particle Creation by Black Holes. *Communications in Mathematical Physics* **1975**, *43*, 199–220. doi:10.1007/BF02345020.

[5] Breit, G.; Wheeler, J. A. Collision of Two Light Quanta. *Physical Review* **1934**, *46*, 1087–1091. doi:10.1103/PhysRev.46.1087.

[6] Schwinger, J. On Gauge Invariance and Vacuum Polarization. *Physical Review* **1951**, *82*, 664–679. doi:10.1103/PhysRev.82.664.

[7] Peterson, B. M. *An Introduction to Active Galactic Nuclei*. Cambridge University Press, Cambridge, **1997**.

[8] Cackett, E. M.; Bentz, M. C.; Kara, E. Reverberation mapping of active galactic nuclei: from X-ray corona to dusty torus. *iScience* **2021**, *24*, 102557. doi:10.1016/j.isci.2021.102557.

[9] Blandford, R. D.; Znajek, R. L. Electromagnetic extraction of energy from Kerr black holes. *Monthly Notices of the Royal Astronomical Society* **1977**, *179*, 433–456. doi:10.1093/mnras/179.3.433.

[10] Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team. NIST Atomic Spectra Database. National Institute of Standards and Technology, Gaithersburg, MD. `https://physics.nist.gov/asd`; accessed 2026-05-25.

[11] National Institute of Standards and Technology. Strong Lines of Hydrogen (H). `https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable2.htm`; accessed 2026-05-25.

[12] NASA/IPAC Extragalactic Database. Broad Emission Lines. `https://ned.ipac.caltech.edu/level5/Cambridge/Cambridge1_3_4.html`; accessed 2026-05-25.

[13] Weinberg, S. The Cosmological Constant Problem. *Reviews of Modern Physics* **1989**, *61*, 1–23. doi:10.1103/RevModPhys.61.1.

[14] Martin, J. Everything You Always Wanted To Know About The Cosmological Constant Problem (But Were Afraid To Ask). *Comptes Rendus Physique* **2012**, *13*, 566–665. doi:10.1016/j.crhy.2012.04.008.

[15] Wright, E. L.; Eisenhardt, P. R. M.; Mainzer, A. K.; et al. The Wide-field Infrared Survey Explorer (WISE): Mission Description and Initial On-orbit Performance. *The Astronomical Journal* **2010**, *140*, 1868–1881. doi:10.1088/0004-6256/140/6/1868.

[16] Martin, D. C.; Fanson, J.; Schiminovich, D.; et al. The Galaxy Evolution Explorer: A Space Ultraviolet Survey Mission. *The Astrophysical Journal Letters* **2005**, *619*, L1–L6. doi:10.1086/426387.

[17] York, D. G.; Adelman, J.; Anderson, J. E.; et al. The Sloan Digital Sky Survey: Technical Summary. *The Astronomical Journal* **2000**, *120*, 1579–1587. doi:10.1086/301513.

[18] Einstein, A. Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften (Berlin)*, **1915**, 844–847.

[19] Wheeler, J.A. On the Nature of Quantum Geometrodynamics. *Annals of Physics* **1957**, *2*, 604–614.

[20] Jacobson, T. Thermodynamics of Spacetime: The Einstein Equation of State. *Phys. Rev. Lett.* **1995**, *75*, 1260–1263.

[21] Verlinde, E. On the Origin of Gravity and the Laws of Newton. *JHEP* **2011**, *1104*, 029.

[22] Verlinde, E. Emergent Gravity and the Dark Universe. *SciPost Phys.* **2017**, *2*, 016.

[23] Milgrom, M. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophys. J.* **1983**, *270*, 365.

[24] Einstein, A. 1905, Zur Elektrodynamik bewegter Körper, *Annalen der Physik*, 17, 891–921.

[25] Einstein, A. 1916, Die Grundlage der allgemeinen Relativitätstheorie, *Annalen der Physik*, 49, 769–822.

[26] Rindler, W. 2006, *Relativity: Special, General, and Cosmological*, 2nd ed., Oxford University Press.

[27] Pound, R. V., & Rebka, G. A. 1960, Apparent Weight of Photons, *Physical Review Letters*, 4, 337–341.

[28] Hafele, J. C., & Keating, R. E. 1972, Around-the-World Atomic Clocks: Predicted Relativistic Time Gains, *Science*, 177, 166–168.

[29] Will, C. M. 2014, The Confrontation between General Relativity and Experiment, *Living Reviews in Relativity*, 17, 4.

[30] Giulini, D. 2008, The Equivalence Principle, in *The Genesis of General Relativity*, Springer.

[31] Rubin, V. C., & Ford, W. K. 1970, Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions, *The Astrophysical Journal*, 159, 379.

[32] Sofue, Y., & Rubin, V. 2001, Rotation Curves of Spiral Galaxies, *Annual Review of Astronomy and Astrophysics*, 39, 137–174.

[33] Reid, M. J., & Brunthaler, A. 2004, The Proper Motion of Sagittarius A*. II. The Mass of Sagittarius A*, *The Astrophysical Journal*, 616, 872–884.

[34] Reid, M. J., & Brunthaler, A. 2020, The Proper Motion of Sagittarius A*. III. The Case for a Supermassive Black Hole, *The Astrophysical Journal*, 892, 39.

[35] Varma, V., Biscoveanu, S., Islam, T., Shaik, F. H., Haster, C.-J., Isi, M., Farr, W. M., Field, S. E., & Vitale, S. 2022, Evidence of Large Recoil Velocity from a Black Hole Merger Signal, *Physical Review Letters*, 128, 191102.

[36] van Dokkum, P., et al. 2023, A Candidate Runaway Supermassive Black Hole Identified by Shocks and Star Formation in Its Wake, *The Astrophysical Journal Letters*, 946, L50.

[37] Gaia Collaboration, Helmi, A., van Leeuwen, F., et al. 2018, Gaia Data Release 2: Kinematics of Globular Clusters and Dwarf Galaxies around the Milky Way, *Astronomy & Astrophysics*, 616, A12.

[38] Fritz, T. K., Battaglia, G., Pawlowski, M. S., et al. 2018, Gaia DR2 Proper Motions of Dwarf Galaxies within 420 kpc, *Astronomy & Astrophysics*, 619, A103.

[39] Li, H., Hammer, F., Babusiaux, C., et al. 2021, Gaia EDR3 Proper Motions of Milky Way Dwarfs I: 3D Motions and Orbits, *The Astrophysical Journal*, 916, 8.

[40] Hammer, F., Wang, J., Pawlowski, M. S., et al. 2021, Gaia EDR3 Proper Motions of Milky Way Dwarfs II: Velocities, Total Energy and Angular Momentum, *The Astrophysical Journal*, 922, 93.

[41] Rexhepi, U.Q. Bimodal Regime Structure in Galactic Rotation Curves: Evidence for Distinct Dynamical States and a Field-Based Interpretation of the Dark Matter Effect. *Preprints* **2026**. https://www.preprints.org/manuscript/202604.0640

[42] Rexhepi, U.Q. Continuous Field Dynamics and the Dark Matter Effect: A First-Principles Approach. *ResearchGate* **2026**. https://doi.org/10.13140/RG.2.2.15717.36321

[43] Rexhepi, U. Q. *From pattern to wave — A field-ontological interpretation of light, matter and scales (UQSH)* . Zenodo, 2026. doi:10.5281/zenodo.18381978. https://github.com/ukshinrexhepi-cloud/Universal-Quantum-Foam-Hypothesis-UQSH-/blob/main/From_pattern_to_wave___A_field_ontological_interpretation_of_light__matter_and_scales__UQSH_.pdf

[44] Rexhepi, U. Q. *Light Bubbles — The mechanical solution to wave-particle duality*. Zenodo, 2026. doi:10.5281/zenodo.18476363. https://github.com/ukshinrexhepi-cloud/Universal-Quantum-Foam-Hypothesis-UQSH-/blob/main/Light_Bubbles___The_mechanical_solution_to_wave_particle_duality_Figshare_01.03.26.pdf